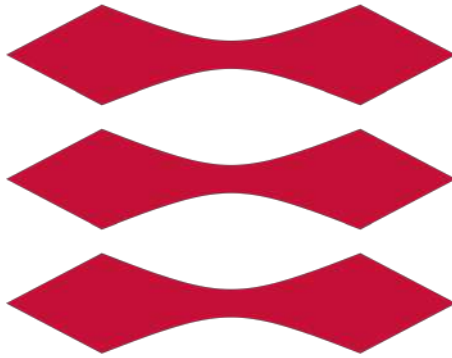


DTU



Danmarks Tekniske Universitet

A condensed summary of Computer simulation of materials 47212

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Chapter 1

Module 1

1.1 Random numbers and pseudo-random numbers

A classical example of randomness is a roulette wheel, where the probability for the numbers $0, 1, 2, \dots, 35, 36$ is equal and the result of each spin is (ideally) completely random.

Computers, however, are deterministic machines. Any number they produce is the result of an algorithm, so we cannot obtain truly random numbers, only **pseudo-random** numbers. These are generated by deterministic recursions, but are designed so that:

- their statistical properties mimic those of true random numbers (e.g. uniform in $[0, 1)$),
- correlations are minimised,
- the sequence has a very long **period** before it repeats.

In numerical simulations, one sometimes also uses **quasi-random** (or low-discrepancy) sequences. These are not random at all, but are “jittered” or “shuffled” constructions that fill space more uniformly than a purely random point set, improving convergence in some integrals.

Caveats of pseudo-random numbers

- Algorithmically created random numbers are never truly random, only pseudo-random.
- Every generator has a finite period: after some number of generated values, the sequence repeats exactly.
- Correlations can appear in multi-dimensional sampling if the generator or usage is poor.
- Shuffling or scrambling helps reduce conditional probabilities and visible patterns.

MATLAB’s default generator is the **Mersenne Twister**, based on the Mersenne prime $2^{19937} - 1$ and having a very long period,

$$2^{19937} - 1 \approx 10^{6000}.$$

This is usually sufficient for typical scientific simulations, but for very large simulations (or poorly designed usage) correlations may still appear.

1.2 Materials science: grain growth

1.2.1 Grain structure

Grains are regions of homogeneous crystallographic orientation in a polycrystal. They are separated by grain boundaries.

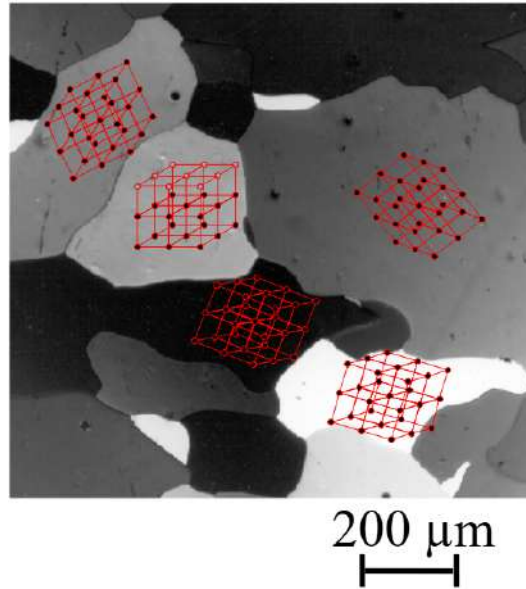


Figure 1.1: Example of grain structure in a polycrystalline material.

Orientation differences between grains are called **misorientations** (or disorientations). Deformation structures within a grain correspond to orientation gradients and local substructures.

1.2.2 Polycrystals from solidification

Grains appear when nuclei form and grow during solidification of a melt: each nucleus grows with its own crystallographic orientation; when they impinge on each other, grain boundaries form.

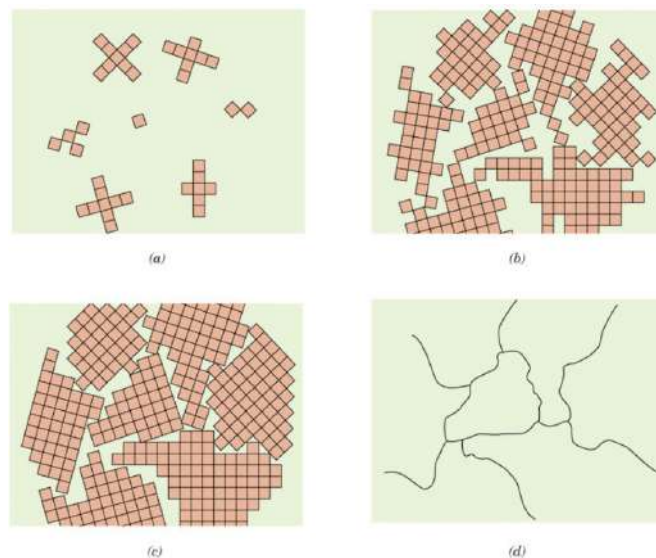


Figure 1.2: Formation of grains during solidification of a melt.

1.2.3 Grain boundaries

Grain boundaries separate two grains (same crystal structure but different orientation). They can be classified as:

- **Low-angle (subgrain) boundaries:** misorientation $< 15^\circ$.
- **High-angle (grain) boundaries:** misorientation $> 15^\circ$.

1.2.4 Grain boundary area and grain size

For an average grain with surface area S and volume V , one defines the boundary density

$$S_V = \frac{S}{V} \propto \frac{1}{D},$$

where D is a characteristic grain size. Thus, smaller grains have higher grain boundary area per volume, and vice versa.

1.2.5 Grain boundary energy

The grain boundary energy γ depends on the misorientation angle θ between neighbouring grains.

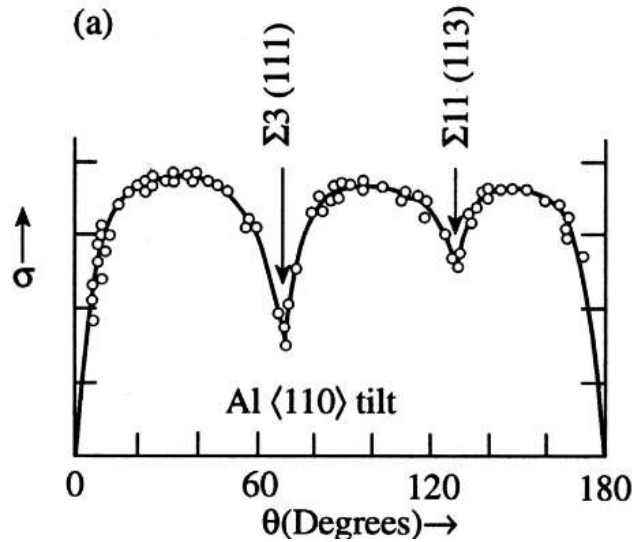


Figure 1.3: Schematic of grain boundary energy as a function of misorientation.

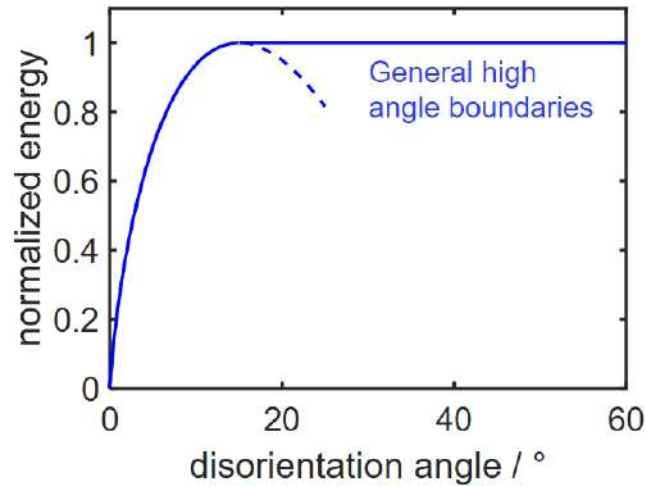


Figure 1.4: Boundary energy vs disorientation angle.

For low-angle boundaries, the **Shockley** model gives (schematically)

$$\gamma(\theta) = \gamma_{\max} \frac{\theta}{\theta_{\max}} \left(1 - \ln \frac{\theta}{\theta_{\max}} \right),$$

which shows:

- low-angle boundaries correspond to relatively low boundary energy;
- high-angle boundaries correspond to higher boundary energy.

1.2.6 Grain boundary strengthening (Hall–Petch)

Because dislocations cannot easily pass through grain boundaries, they tend to pile up at boundaries, creating stress concentrations. This additional stress can activate slip in neighbouring grains, but the overall effect is that:

- smaller grains $\Rightarrow$ more grain boundaries per volume,
- more obstacles to dislocation motion $\Rightarrow$ higher strength.

The **Hall–Petch** relation expresses the effect of grain size on yield stress:

$$\sigma_y = \sigma_0 + \frac{k_{\text{HP}}}{\sqrt{D}},$$

where:

- σ_y is the yield stress,
- σ_0 is the friction stress for dislocation motion,
- k_{HP} is the Hall–Petch constant,
- D is the grain size.

Grain refinement strengthens materials, but it can also influence other properties such as electrical conductivity.

1.2.7 Synthetic microstructures

In simulations, synthetic microstructures are constructed such that grains are:

- contiguous (space-filling, no holes),
- separated by boundaries,
- originated from discrete seeds or nuclei,
- typically modelled on a regular grid.

1.2.8 Thermal annealing of defects

Thermal annealing is a process of heating and controlled cooling used to alter microstructure and hence physical, mechanical and electrical properties.

Starting from a thermodynamically unstable state, annealing can be described in stages:

Recovery

1. Annihilation of point defects.
2. Annihilation of dislocations.
3. Rearrangement of dislocations into low-energy configurations (subgrain formation).

Recrystallisation and grain growth

1. Nucleation and growth of new, strain-free grains (recrystallisation).
2. Grain boundary migration and reduction of total grain boundary area (grain growth).

1.2.9 Grain growth driven by boundary energy

In 2D, analytical results from topology and curvature-driven motion give:

Number of edges n	Behaviour
$n = 6$	Stable (no area change on average)
$n < 6$	Shrink
$n > 6$	Grow

Table 1.1: Stability of grains with n edges in 2D curvature-driven growth.

Note: grains with 3 edges tend to disappear quickly.

The area evolution of an n -sided grain is

$$\frac{dA}{dt} = M \frac{\pi}{3} (n - 6),$$

where M is the boundary mobility. In 3D, roughly polyhedra with about 14 faces are “stable” and similar topological rules apply (though more complex).

Triple junction configurations in 2D are only stable when all dihedral angles are 120° (average is always 120°).

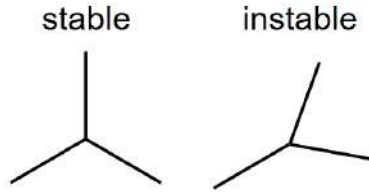


Figure 1.5: Triple junction configuration with 120° angles.

1.2.10 Relations for grain size, grain area and chord length

Circular grain (2D) For an ideal circular grain:

- Grain radius R ,
- Grain diameter $D = 2R$,
- Grain area $A = \pi R^2$.

Non-circular grains In experiments or simulations, grains are rarely circular. Often we measure the grain area A and define the **equivalent circular diameter** (ECD):

$$D = \sqrt{\frac{4A}{\pi}}.$$

For anisometric tessellations, ECD is convenient for statistics but does not directly show shape. Alternative measures:

- **Chord length** λ : distance between two boundary intersections along a test line.

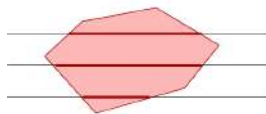


Figure 1.6: Chord length across grains.

- **Feret diameters:** calliper lengths measured in specified directions.

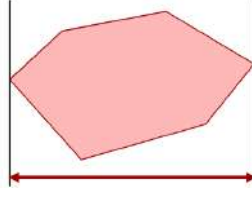


Figure 1.7: Example of Feret diameters.

Comparing Feret diameters in different directions allows assessment of anisotropy, e.g. via aspect ratio

$$A_R = \frac{D_{\text{Feret, hor}}}{D_{\text{Feret, ver}}}.$$

1.2.11 Curvature-driven grain growth law

Grain growth is driven by reduction of grain boundary area (and thus boundary energy). Boundary motion is curvature-driven.

For an approximately spherical grain of radius R :

$$\frac{dR}{dt} = v = MP,$$

where v is the boundary velocity, M is the boundary mobility, and P is the capillary driving pressure.

For a spherical grain:

$$P = \frac{2\gamma}{R},$$

where γ is the boundary energy. The average grain radius $\bar{R}$ then follows:

$$\frac{d\bar{R}}{dt} = \frac{2M\gamma}{\bar{R}} \iff \bar{R}^2 - \bar{R}_0^2 = 4M\gamma t.$$

For large grains or long times ($\bar{R} \gg \bar{R}_0$), this gives a parabolic growth law:

$$\bar{R} \approx (4M\gamma)^{1/2} t^{1/2}.$$

In general one writes

$$\bar{R} \approx ct^{1/n},$$

where the theoretical exponent is $n = 2$, while experimentally often $n \approx 2.4$.

1.2.12 Alternative grain size distributions

Observed grain size distributions can be compared to theoretical forms:

- **Rayleigh distribution** (Luoat):

$$f(R) = \frac{R}{\sigma^2} \exp\left(-\frac{R^2}{2\sigma^2}\right), \quad \sigma = \sqrt{\frac{2}{\pi}} \bar{R}.$$

- **Lognormal distribution** (Feltham, empirical):

$$f(R) = \frac{1}{R\sigma\sqrt{2\pi}} \exp\left(-\frac{(\ln R - \mu)^2}{2\sigma^2}\right),$$

where σ characterises the width and μ is related to the mean.

- **Curvature growth of individual grains** (Hillert):

$$\frac{dR_i}{dt} = MP = M \left(\frac{\bar{\gamma}}{\bar{R}} - \frac{\gamma_i}{R_i} \right),$$

where R_i is the grain radius, $\bar{R}$ the average radius, γ_i the boundary energy around grain i , and $\bar{\gamma}$ the average boundary energy.

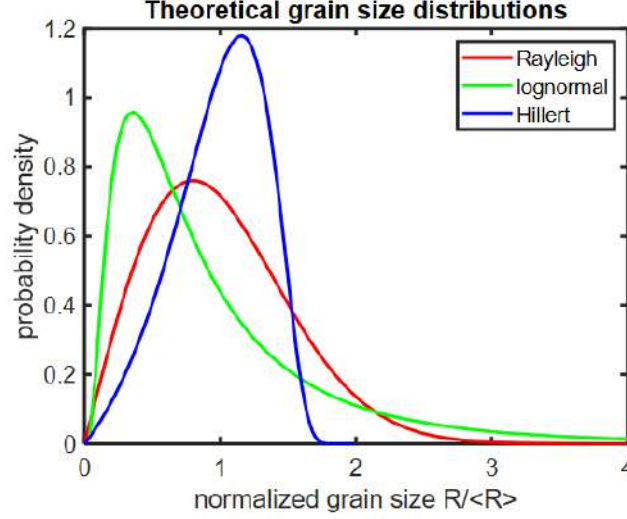


Figure 1.8: Example of theoretical grain size distributions.

1.2.13 Normal and abnormal grain growth

- **Normal grain growth:**
 - continuous coarsening;
 - self-similar (invariant) and unimodal grain size distribution.
- **Abnormal grain growth:**
 - only under certain conditions;
 - some large grains grow faster than the mean, leading to:

$$\frac{d}{dt} \left(\frac{R_i}{\bar{R}} \right) > 0;$$

- discontinuous grain growth, bimodal distributions;
- sometimes called “secondary recrystallisation” (from morphology and texture).

1.3 Tessellations

Synthetic grain structures can be generated using a **Poisson homogeneous point process** as nuclei and constructing a Voronoi tessellation.

We create a point field with N_P nuclei uniformly distributed in an $n_{\max} \times n_{\max}$ array. Then we create a tessellation (space-filling division) by assigning each pixel to the closest nucleus.

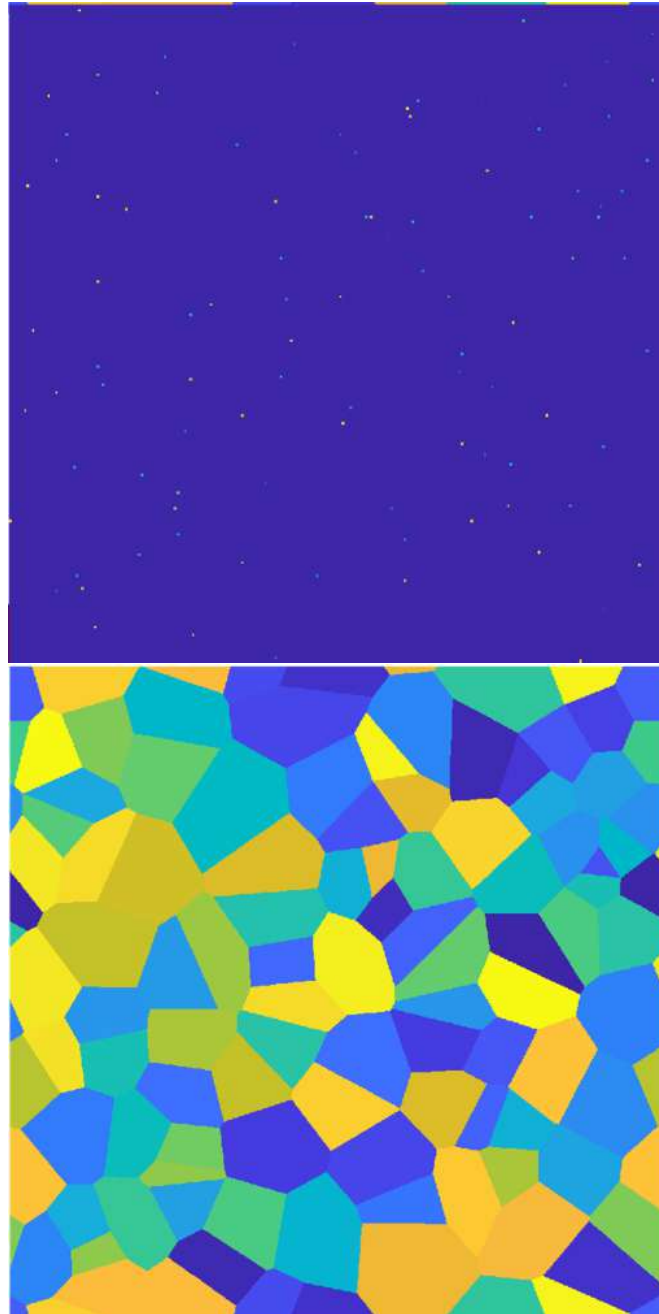


Figure 1.9: Random point field and corresponding Poisson–Voronoi mosaic.

Boundary conditions of the array are usually **periodic**, meaning the left and right edges are neighbours, as well as the top and bottom edges. The resulting construction is the well-known Poisson–Voronoi mosaic.

Why Voronoi tessellations approximate grain structures. Real polycrystals minimise interfacial energy, and grain boundaries tend to lie halfway between nuclei during solidification. The Voronoi construction reproduces this “nearest–seed” partitioning and therefore provides a good synthetic approximation of as-solidified grain structures, especially for equiaxed grains. Although real grains are not perfectly convex or space-filling, Voronoi models capture essential stereological relations used for testing statistical grain-growth theories.

1.3.1 Number of grains

MATLAB code to determine the number of non-identical nuclei (i.e. number of grains):

```

1 % function to calculate only number of grains, not sizes!
2 % counts only grains of non-vanishing size
3

```

```

4 function nog = numberofgrains(AA)
5
6 nog = numel(unique(AA)); % count number of unique entries

```

1.3.2 Area of each grain

MATLAB code to determine the number of pixels in each grain subset (i.e. grain area):

```

1 % function to calculate grain size for grains
2 % with indices from nglow to nghigh
3 % from an indexed map AA
4 % if nglow = nghigh, grain size of specific grain
5 % no explicit check, if nglow and nghigh are integers
6
7 function gsize = grainsizenew(AA,nglow,nghigh)
8
9 AAlist = AA(AA >= nglow); % too shorten further calculations
10 ngedges = nglow:(nghigh+1); % nghigh + 1 to account for grain nghigh!
11
12 gsize = histcounts(AAlist,ngedges); % list of grain sizes

```

1.3.3 Grain size distribution

We can obtain the grain size distribution by computing areas and plotting a histogram of grain area vs absolute frequency. Example MATLAB code:

```

1 %% Number of grains in tessellation
2 ngrain0 = numberofgrains(AA);
3
4 %% Calculation of grain sizes
5
6 gsize = grainsizenew(AA,1,NP);
7
8 NPsurv = nnz(gsize); % surviving nuclei
9
10 avegsizetheo = nmax*nmax/NPsurv
11 avegsizetheo = mean(nonzeros(gsize)) % do not count nonexistent grains
12 stdgsizetheo = std(nonzeros(gsize)) % do not count nonexistent grains
13
14 ngsedges = 10;
15 gsedges = 0:(4*avegsizetheo)/ngsedges:(4*avegsizetheo);
16 gscenters = zeros(size(1:ngsedges)); % otherwise artefacts when reducing ngsedges
17 gscenters(1:ngsedges) = (gsedges(1:ngsedges)+gsedges(1+(1:ngsedges)))/2;
18 gsafreq = histcounts(gsize,gsedges);
19
20 %% Grain area distribution
21 figure; hold on;
22 set(gca,'fontweight','demi','fontsize',12,'linewidth',2);
23 histogram(gsize,gsedges);
24 plot(gsceneters,gsafreq,'d-','linewidth',2);
25 text(1200,30,['{\langle}A{\rangle} = ',num2str(mean(nonzeros(gsize)),'%4.0f'),' \sigma
    = ',num2str(std(nonzeros(gsize)),'%4.0f')],'fontweight','demi','fontsize',14);
26 %text(200,3000,['{\langle}A{\rangle} = ',num2str(mean(nonzeros(gsize)),'%4.0f'),' \
    sigma = ',num2str(std(nonzeros(gsize)),'%4.0f')],'fontweight','demi','fontsize',14);
27 title(['Tessellation with ',num2str(NPsurv,'%4d'),' grains']);
28 xlabel('grain area / pixel^2');
29 ylabel('absolute frequency');
30 box on;
31 axis([gsedges(1) gsedges(end) 0 NP*6/ngsedges]);

```

This produces a histogram such as:

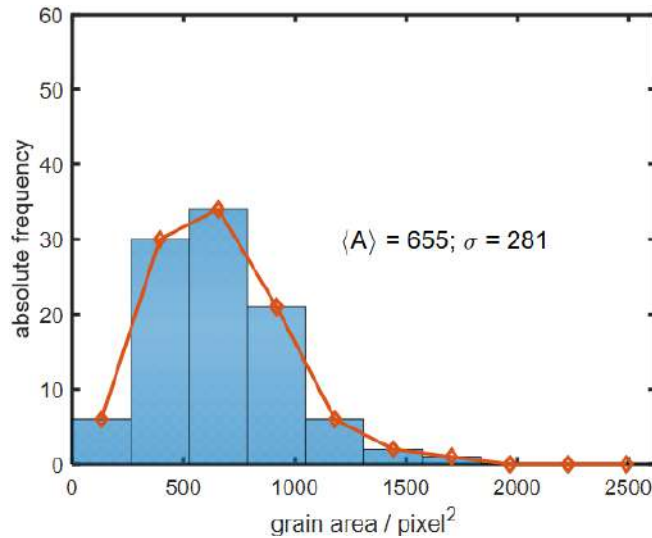


Figure 1.10: Example of grain area distribution for a Poisson–Voronoi tessellation.

1.3.4 Map of boundaries between grains

To obtain a map of grain boundaries we can compare each pixel with its neighbours:

```

1 %% Stereology
2 % Calculation of boundary length from interception points
3 % and creation of boundary map
4
5 % create shifted maps
6 AAN = circshift(AA,[1 0]); % due north in imagesc
7 AAW = circshift(AA,[0 1]); % due west in imagesc
8
9 % check the shift
10 figure;
11 imagesc(AAN);
12 axis equal;
13 axis off;
14
15 %% Create boundary map (simple solution)
16 % marking points where N is not same
17 % as well as points where W is not same
18 figure;
19 imagesc(abs(AA-AAN)+abs(AA-AAW)~=0);
20 axis equal;
21 axis off;
22
23 % stereology
24 % counting of non zero matrix elements (nnz)
25 NLN = nnz(AA-AAN)/(nmax*nmax);
26 NLW = nnz(AA-AAW)/(nmax*nmax);
27
28 % boundary density n map
29 LAemp = (NLN+NLW)*pi/4; % empirically on testing lines
30 LAc = 2*sqrt(nnz(gsize)/(nmax*nmax)); % ideally for Voronoi

```

Resulting in a binary map of boundaries:

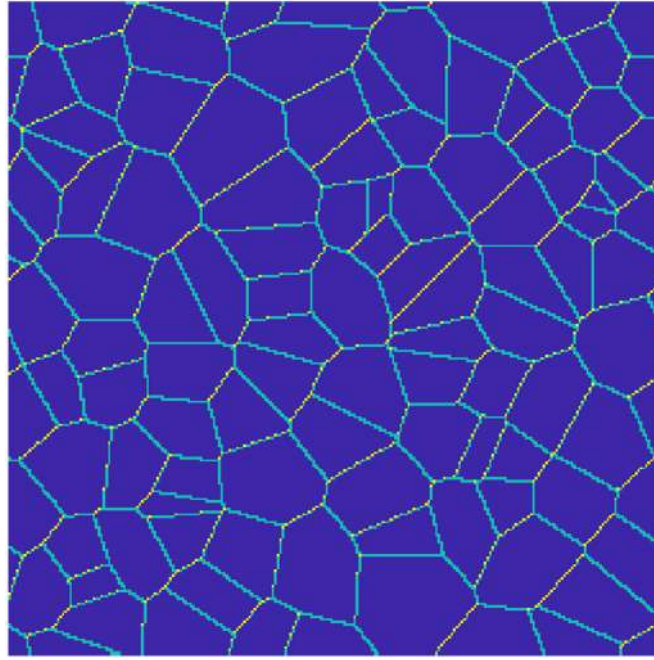


Figure 1.11: Boundary map highlighting grain interfaces.

1.3.5 Creating a connectivity matrix

We can construct a matrix M describing which grains are neighbours:

$$M(G, G^*) = \begin{cases} 1 & \text{if grains } G \text{ and } G^* \text{ are neighbours,} \\ 0 & \text{otherwise.} \end{cases}$$

MATLAB code:

```

1 %% creates a connectivity matrix between the grains
2 % script, not a function
3 % presumes existence of
4 %     an indexed map AA with size nmax x nmax
5 %     containing NP grains, does not check
6 % creates matrix of size NPxNP
7 % with 1 if grains are neighbors and 0 if not
8
9 MM = zeros(NP);
10
11 % loop runs over all points in map
12 % counting boundary segments
13
14 % create shifted maps
15 AAN = circshift(AA,[1 0]); % due north in imagesc
16 AAW = circshift(AA,[0 1]); % due west in imagesc
17
18 for yhere=1:nmax
19     for xhere=1:nmax
20
21         ghere = AA(yhere,xhere); % index of grain here
22         gN = AAN(yhere,xhere);
23         gW = AAW(yhere,xhere);
24         % calculate neighbor positions accounting for periodic boundary conditions
25         %yN = mod(yhere-2,nmax)+1; % y-1 neighbor to the north
26         %xW = mod(xhere-2,nmax)+1; % x-1 neighbor to the west
27
28         %MM(ghere,AA(yN,xhere))=1; % connection to north
29         %MM(ghere,AA(yhere,xW))=1; % connection to west
30         MM(ghere,gN)=1; % connection to north
31         MM(ghere,gW)=1; % connection to west
32

```

```

33
34     end
35 end
36
37 MM = MM+MM'; % must be symmetric but only are N and W are searched
38 MM = -floor(-MM/max(max(MM))); % trick floor(-0.123) = -1, floor(0) = 0
39
40 for i=1:NP
41     MM(i,i)=0;% grains are not neighbours of themselves
42 end

```

The number of neighbours of grain G is the sum of row G in MM . We can then obtain the distribution of neighbour numbers and e.g. the correlation between grain area and number of neighbours.

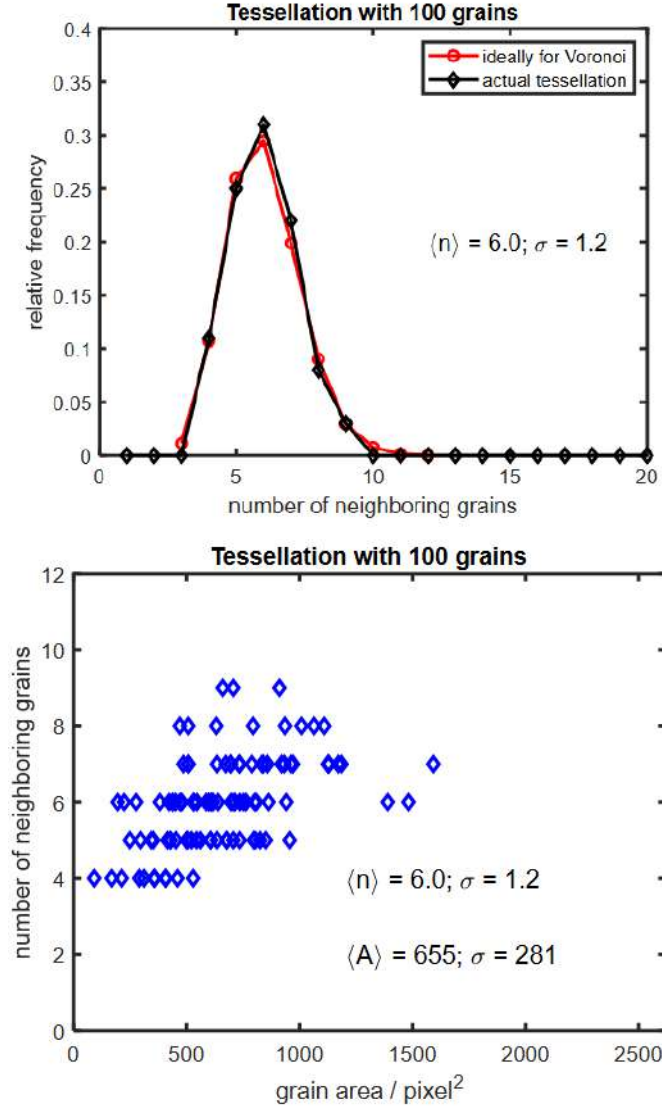


Figure 1.12: Neighbour number distribution and correlation with grain area.

1.3.6 Exotic tessellations

Voronoi tessellations depend on the distance metric used:

1. Euclidean distance

$$d_i = \sqrt{(x - x_i)^2 + (y - y_i)^2}.$$

2. Anisometric distance

$$d_i = \sqrt{(x - x_i)^2 + \eta^2(y - y_i)^2},$$

where $\eta \neq 1$ stretches one direction.

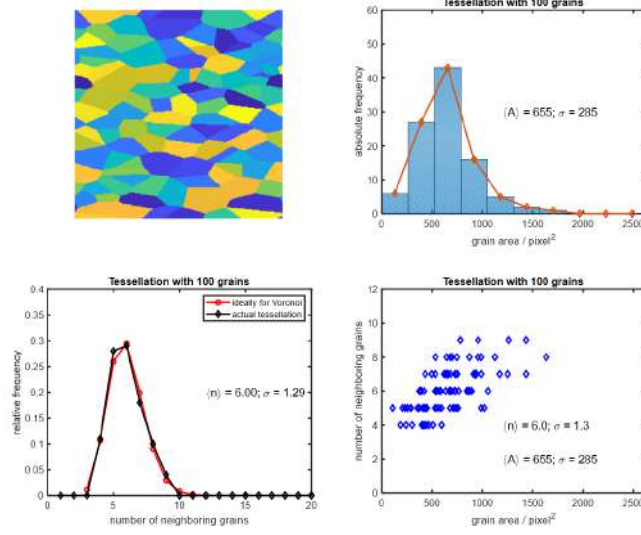


Figure 1.13: Anisometric tessellation.

3. Generalised (Minkowski) distance with exponent n

$$d_i = \sqrt[n]{(x - x_i)^n + (y - y_i)^n}.$$

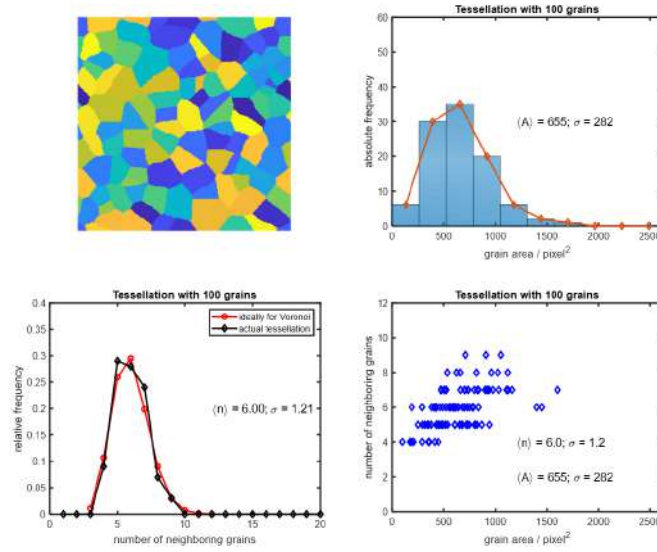


Figure 1.14: Example of tessellation with modified distance exponent.

4. Manhattan (taxicab) distance

$$d_i = |x - x_i| + |y - y_i|.$$

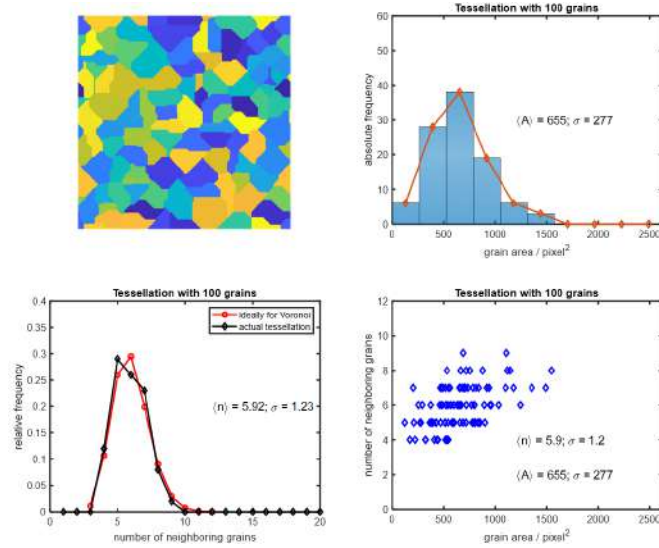


Figure 1.15: Manhattan-distance tessellation.

1.4 Cellular automata

A **cellular automaton** (CA) is an ensemble of cells on a grid; each cell has a state (here: grain identity G). At each time step, all states are updated according to local rules depending on the neighbours' states.

A typical grid structure (square lattice) is shown below:

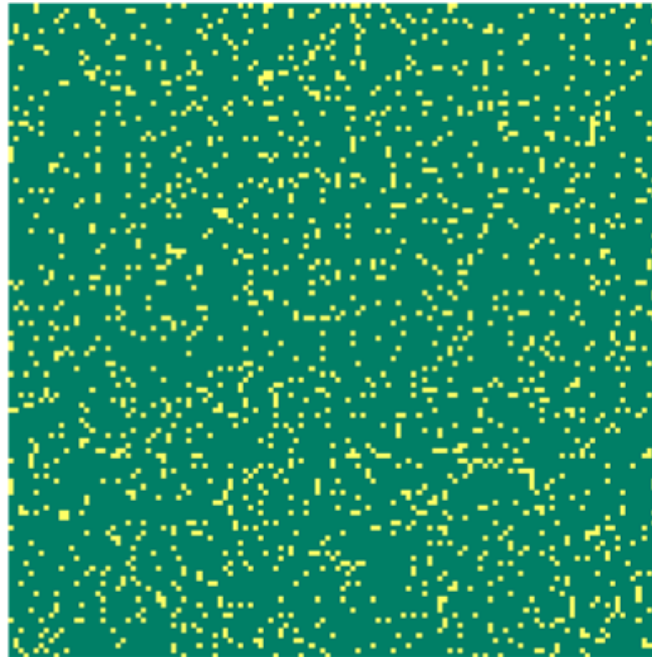


Figure 1.16: Example of a square grid for a cellular automaton.

Two main types:

- **Deterministic CA:** the future state of a cell is uniquely determined by the current state of the cell and its neighbours.
- **Totalistic CA:** the future state depends only on the *sum* (or some aggregate) of neighbour states, not their arrangement.

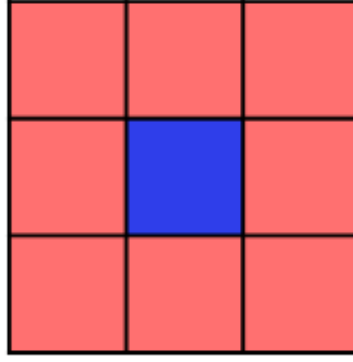


Figure 1.17: Totalistic CA: red neighbours influence the blue central cell.

1.4.1 Simulation of grain growth using Potts' model

Potts' model of grain growth describes grain boundary motion driven by boundary energy, implemented as a CA with Monte Carlo updating.

Recipe (zero temperature, $T = 0$)

- Array with N_{total} pixels (in 2D: $N_{\text{total}} = N_{\text{max}} \times N_{\text{max}}$).
- Each pixel stores grain identity G (number or orientation).
- Single Monte Carlo step (Metropolis scheme):
 1. Select randomly a pixel (j, i) with grain identity $G(j, i)$.
 2. Check neighbours: do they belong to the same grain or different grains?
 3. Select randomly one grain identity G^* from the different identities in the neighbourhood.
 4. Change the identity of pixel (j, i) to G^* (trial move).
 5. Calculate energy E^* of the new configuration and compare with old energy E .
 6. Decide based on $\Delta E = E^* - E$:
 - 6.1. if $\Delta E \leq 0$: accept the change,
 - 6.2. if $\Delta E > 0$: reject the change (keep old configuration).
 7. Repeat for N_{total} such attempts per Monte Carlo step.
- Repeat Monte Carlo steps until desired evolution is reached.

Remark (MC steps are not physical time). A Monte Carlo step (MCS) represents N_{pixels} attempted updates, but it does *not* correspond to real physical time. It only provides a kinetic ordering of microstructures under curvature-driven boundary motion. To map MCS to physical time one would need a calibrated mobility M and boundary energy γ , which is not done in the basic Potts model used in this course.

Illustration of Potts' model

Consider a neighbourhood:

NW	N	NE
W	C	E
SW	S	SE

Figure 1.18: Neighbourhood stencil used in Potts' model (8 neighbours).

Example 1 Select cell at $(4, 4)$ with $A(4, 4) = 6$:

3	8	8	8	8
3	8	8	8	8
3	3	6	6	6
3	6	6	6	6
6	6	6	6	6

Figure 1.19: All neighbours of cell $(4, 4)$ have the same identity.

All neighbours are identical, so changing the state would increase energy; at $T = 0$ no change is accepted.

Example 2 Next, select cell $(2, 2)$ with $A(2, 2) = 8$:

3	8	8
3	8	8
3	3	6

Figure 1.20: Cell with several neighbouring grains.

Five neighbours belong to different grains, so $E = 5\gamma$. Neighbouring grain identities are 3 and 6; choose one at random:

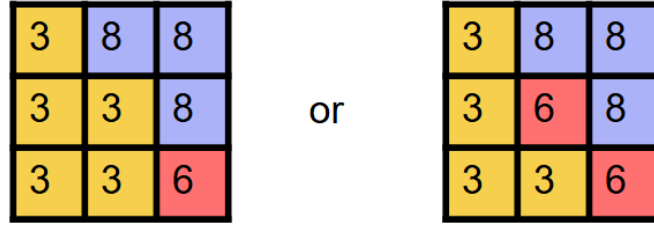


Figure 1.21: Trial flips to grain 3 (left) or grain 6 (right), with energies E^* indicated.

On the left: $E^* = 4\gamma$; on the right: $E^* = 7\gamma$. At $T = 0$ we accept the move to the configuration with lower energy (left) and reject the higher-energy move.

1.4.2 Definition of energy

Energy depends on all grain identities $G' = G(j', i')$ in the neighbourhood of the central cell $G(j, i)$. If a neighbouring pixel has a different identity, there is a grain boundary, with boundary energy $\gamma(G', G)$.

The energy of a central pixel is:

$$E = H = \sum_{(j', i') \in \text{neighbourhood}} \gamma(G(j', i'), G(j, i)).$$

Often one simplifies to identical boundary energies:

$$\gamma(G', G) = \gamma,$$

so we only count the number of unlike neighbours. Then

$$E = \sum_{(j', i') \in \text{neighbourhood}} \Gamma_{G'G},$$

with a look-up matrix Γ defined as:

$$\Gamma_{G'G} = \gamma(1 - \delta_{G'G}) = \begin{cases} \gamma, & G' \neq G, \\ 0, & G' = G, \end{cases}$$

i.e.

$$\Gamma = \gamma \begin{pmatrix} 0 & 1 & 1 & \dots & 1 \\ 1 & 0 & 1 & \dots & 1 \\ 1 & 1 & 0 & \dots & 1 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & 1 & \dots & 0 \end{pmatrix}.$$

In MATLAB:

```

1 %% Creation of energy matrix GGamma
2 % look up table
3
4 % same energy for all boundaries
5 GGamma = ones(NP);
6
7 % no boundary within grain, no boundary energy
8 for i=1:NP
9     GGamma(i,i)=0;
10 end

```

1.4.3 Energy of grain boundary: MATLAB function

Energy of the boundary around a given pixel can be computed with:

```

1 function Egb = Egrainboundary8only(yhere,xhere,AAtest,AA,GGamma)
2 % energy of configuration is calculated from boundaries
3 % between the central cell and the neighbourhood
4 % using the Gamma matrix as look-up table
5
6 nmax = numel(AA(1,:));
7
8 % calculate positions to account for periodic boundary conditions
9 yN = mod(yhere-2,nmax)+1; % y-1
10 xW = mod(xhere-2,nmax)+1; % x-1
11 yS = mod(yhere,nmax)+1; % y+1
12 xE = mod(xhere,nmax)+1; % x+1
13
14 Egb = GGamma(AA(yN,xhere),AAtest) + GGamma(AA(yhere,xE),AAtest) + ...
15 GGamma(AA(yS,xhere),AAtest) + GGamma(AA(yhere,xW),AAtest) + ...
16 GGamma(AA(yN,xW),AAtest) + GGamma(AA(yS,xE),AAtest) + ...
17 GGamma(AA(yN,xE),AAtest) + GGamma(AA(yS,xW),AAtest);

```

1.4.4 Monte Carlo simulation (8-neighbourhood)

If we simulate grain growth with only 4 neighbours, growth may saturate. Using all 8 neighbours gives realistic curvature-driven growth. The mean grain area then often shows approximately linear behaviour:

$$\bar{A} = kt + \bar{A}_0.$$

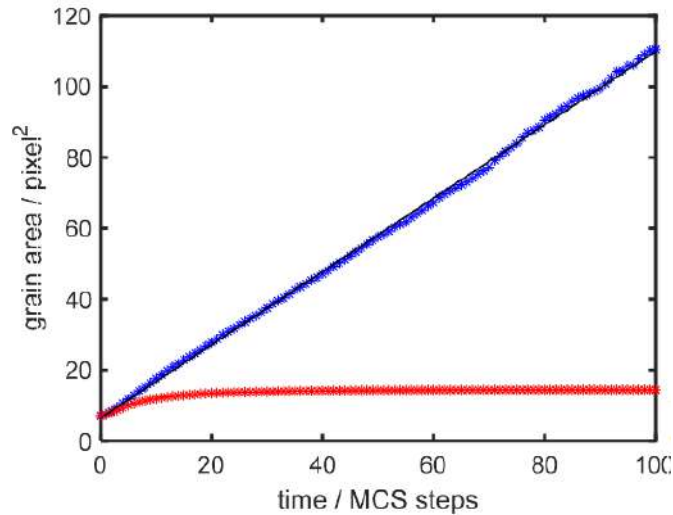


Figure 1.22: Comparison of grain growth with 4 vs 8 neighbours in the Monte Carlo model.

Connection to curvature-driven growth. In 2D grain growth, theory predicts $R^2 - R_0^2 \propto t$ for the mean grain radius. Since $A \propto R^2$, this implies $\langle A \rangle \propto t$, which matches the linear increase observed when using the full 8-neighbourhood energy in the Potts model. The 4-neighbour model fails to reproduce this because it does not approximate curvature correctly.

Example MATLAB Monte Carlo code:

```

1 %% Monte Carlo steps (version with energy matrix, slower)
2 disp('start MCS');
3 tic
4
5 % AA tessellation from above or any tessellation to start with
6 AA = AAstart;
7
8 % seed random generator
9 randseed = 05062019; % first course day F19
10 %randseed = 04062021; % first course day F21
11 rng(randseed);
12 Ntotal = nmax*nmax;

```

```

13
14 % definition of neighborhoods [dx dy]
15 % remember indices AA(y,x)
16 neighborhoodsize = 8;
17 neighborlist = [0 -1; 1 0; 0 1; -1 0; -1 -1; 1 1; 1 -1; -1 1]; % N E S W NW SE NE SW
18
19 % predefined colors for the different neighborhoods
20 colorstr ={'k','k','k','r','k','g','k','b'};
21
22 % times and temperatures
23 t=0;
24 tend = 100;
25 kT = 0.1; % Temperature scaled to boundary energy
26 % used only for T > 0
27 ngrains = zeros(1,tend);
28 gsizes = zeros(NP,tend);
29 nnb = zeros(NP,tend);
30
31 while t<tend % individual MCS steps
32
33     t = t+1; % increase time step
34
35     % list of positions for attempted changes
36     % all Ntotal positions created at once to make simulations reproducible
37     % same position might be attempted several times
38     xL = floor(rand(Ntotal,2)*nmax)+1;
39
40     Nchanges = 0;
41
42     for iL = 1:Ntotal % work through all the predefined attempts
43
44         AApos = AA(xL(iL,2),xL(iL,1)); % store grain index of selected position
45
46         % calculate energy of present configuration
47         Einit = Egrainboundary8only(xL(iL,2),xL(iL,1),AApos,AA,GGamma);
48
49         if Einit > 0 % otherwise in same grain as entire surrounding and flip
impossible!
50
51             indexlist = 1:neighborhoodsize;
52             neighbor = indexlist*0;
53
54             for iNB = indexlist
55                 xpos = xL(iL,:) + neighborlist(iNB,:); % position of neighbor
56                 xpos = mod(xpos-1,nmax)+1; % avoids pos below 1 and above
nmax
57                 neighbor(iNB) = AA(xpos(2),xpos(1)); % list with grain index of
neighbors
58             end
59
60             shortlist = neighbor(neighbor ~= AApos); % reduce to possible changes
61             indexlist = indexlist(neighbor ~= AApos); % reduce to possible changes
62             Ncand = numel(indexlist); % number of candidates for
flips
63
64             indexcand = 1+floor(Ncand*rand(1)); % select a candidate for
flipping
65
66             AAnew = shortlist(indexcand); % get grain index to be tested
67             % calculate energy of configuration flipped configuration
68             Etest = Egrainboundary8only(xL(iL,2),xL(iL,1),AAnew,AA,GGamma);
69
70             % Temperature = 0
71             if Etest <= Einit
72                 % accept change
73                 AA(xL(iL,2),xL(iL,1)) = AAnew;
74                 Nchanges = Nchanges + 1;

```

```

75         end
76
77         % Temperature > 0
78         % to be defined as exercise
79     end
80 end
81
82 % simple number of grains (fast)
83 ngrain = numberofgrains(AA);
84 ngrains(t) = ngrain;
85
86 % % if gsizes have to be calculated (slower)
87 gsize = grainsizenew(AA,1,NP);
88 gsizes(:,t) = gsize;
89 ngrains(t) = nnz(gsize);
90 % % if neighbours are relevant (slower)
91 % % Create connect matrix MM
92 % CreateConnectmatrixnew;
93 % % Number of neighbors
94 % nnb(:,t) = sum(MM);
95
96 disp(['done with MC step ',num2str(t,'%4.0d'),' ',num2str(Nchanges,'%4.0d'),' ',
97      'changes done']);
98 disp([num2str(ngrains(t),'%4.0d'),' grains left']);
99 end
100 toc

```

During the simulation we can monitor grain size distributions:

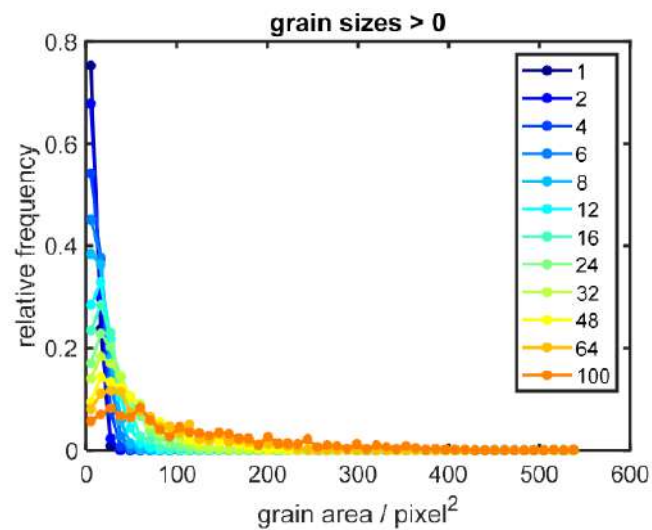


Figure 1.23: Evolution of grain size distribution during Monte Carlo simulation.

And also the neighbour number distribution at a given time (e.g. after 100 steps):

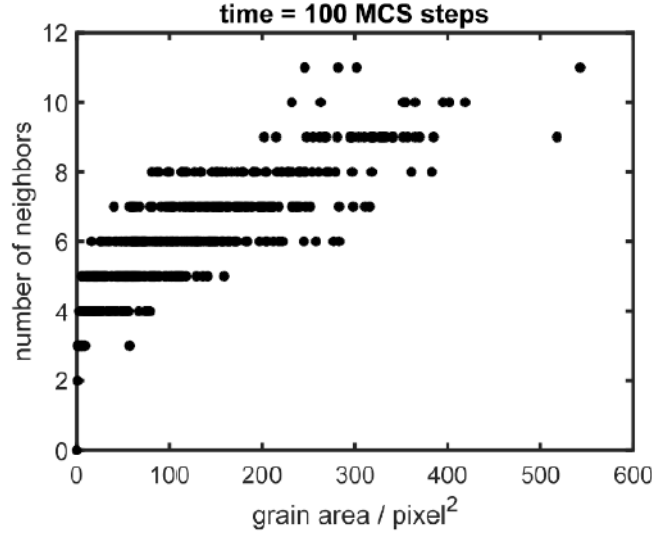


Figure 1.24: Number of neighbours per grain after 100 Monte Carlo steps.

Individual grain histories (size and neighbour number vs time) can also be traced:

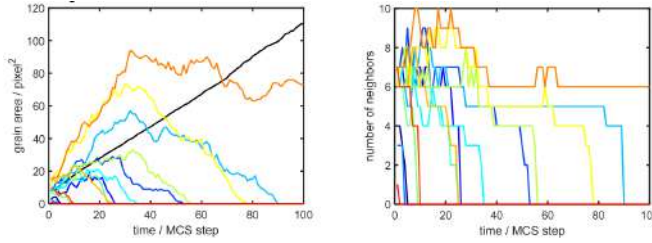


Figure 1.25: Example of individual grain size and neighbour evolution vs Monte Carlo time.

1.4.5 Potts' model and temperature

In the $T = 0$ recipe above, the energy of the configuration never increases; the system always moves downhill in energy. This corresponds to zero temperature and is not fully realistic.

To incorporate finite temperature, we allow energy-increasing moves with probability

$$w = \exp\left(-\frac{\Delta E}{kT}\right),$$

where k is Boltzmann's constant and T is temperature (often used in the scaled form kT/γ).

We have:

$$\begin{aligned} w &\approx 1 && \text{if } \Delta E \ll kT, \\ w &\approx 0 && \text{if } \Delta E \gg kT, \end{aligned}$$

so low-cost changes are likely to be accepted.

Comparing a uniform random number $z \in (0, 1)$ with w , we accept the change if

$$z < w = \exp\left(-\frac{\Delta E}{kT}\right) \iff \Delta E < -kT \ln z.$$

Recipe (finite temperature, $T > 0$)

- Array with N_{total} pixels ($N_{\text{total}} = N_{\text{max}} \times N_{\text{max}}$).
- Each pixel contains grain identity G (number, orientation).
- Single Monte Carlo step (Metropolis algorithm):
 1. Select randomly pixel (j, i) with grain identity $G(j, i)$.

2. Check neighbours: same or different grains?
3. Select randomly one neighbour grain identity G^* .
4. Change grain identity of pixel (j, i) to G^* (trial step).
5. Calculate E^* of new configuration and compare to old E .
6. Decide based on $\Delta E = E^* - E$:
 - 6.1. if $\Delta E \leq 0$: accept change,
 - 6.2. if $\Delta E > 0$: accept with probability

$$w = \exp\left(-\frac{\Delta E}{kT}\right).$$

7. Repeat for N_{total} attempts.

- Repeat Monte Carlo steps until the desired evolution is reached.

1.4.6 Transition probabilities and detailed balance

Transition rules in Monte Carlo simulations must be consistent with thermodynamics. Consider two states A and B (e.g. before and after a flip):

- Probabilities of being in each state: P_A, P_B .
- Transition probabilities:
 - w_{BA} : probability of transition $A \rightarrow B$,
 - w_{AB} : probability of transition $B \rightarrow A$.

Detailed balance condition:

$$w_{BA}P_A = w_{AB}P_B.$$

This ensures that in equilibrium, the net flux between states vanishes.

Key implications:

- **Equilibrium:** When detailed balance holds, the system's probability distribution is stationary (no net changes in time).
- **Entropy:** For a system sampling the Boltzmann distribution, detailed balance ensures the simulation tends towards the maximum-entropy equilibrium state.
- **Monte Carlo:** Transition rules (like the Metropolis algorithm) are designed to obey detailed balance; otherwise, simulations may produce unphysical steady states.
- **Irreversibility:** In processes far from equilibrium (e.g. recrystallisation, grain growth, plastic deformation), the system drifts in a preferred direction, and detailed balance may not strictly apply during the transient.

At thermodynamic equilibrium, the probability of a state with energy E is the Boltzmann factor:

$$P_A = \exp\left(-\frac{E_A}{kT}\right), \quad P_B = \exp\left(-\frac{E_B}{kT}\right).$$

Assume $E_A < E_B$ and define the Metropolis transition rules:

- From $B \rightarrow A$ (downhill): always allowed $\Rightarrow w_{AB} = 1$.
- From $A \rightarrow B$ (uphill): probability

$$w_{BA} = \exp\left(-\frac{\Delta E}{kT}\right) = \exp\left(-\frac{E_B - E_A}{kT}\right).$$

Check detailed balance:

$$w_{BA}P_A = \exp\left(-\frac{E_B - E_A}{kT}\right) \exp\left(-\frac{E_A}{kT}\right) = \exp\left(-\frac{E_B}{kT}\right) = P_B = w_{AB}P_B,$$

so the Metropolis algorithm satisfies detailed balance.

Note: During strongly non-equilibrium processes (like early stages of grain growth or recrystallisation), the system is not at equilibrium, and the instantaneous distribution does not satisfy detailed balance. However, the Monte Carlo algorithm itself is still constructed so that the *equilibrium* distribution, if reached, obeys detailed balance and matches the Boltzmann distribution.

Chapter 2

Module 2

Variable	Units	Notes
V	V	Electrical potential
R	Ω	Electrical resistance
I	A	Electrical current
ρ	Ωm	Resistivity
σ	S/m	Electrical conductivity (inverse of resistivity)
J	A/m^2	Current density
$Q_{j,v}$	A/m^3	Current source (default 0)
J_e	A/m^2	External current density (default 0)
E	V/m	Electrical field

$$R = \frac{U}{I}$$

$$R = \frac{L}{\sigma A}$$

Series (end-to-end)

$$I_1 = I_2 = \dots = I,$$

$$V_{eq} = V_1 + V_2 + \dots,$$

$$R_{eq} = R_1 + R_2 + \dots,$$

(voltage division) $V_k = \frac{R_k}{\sum_i R_i} V_{eq}.$

Parallel (share both nodes)

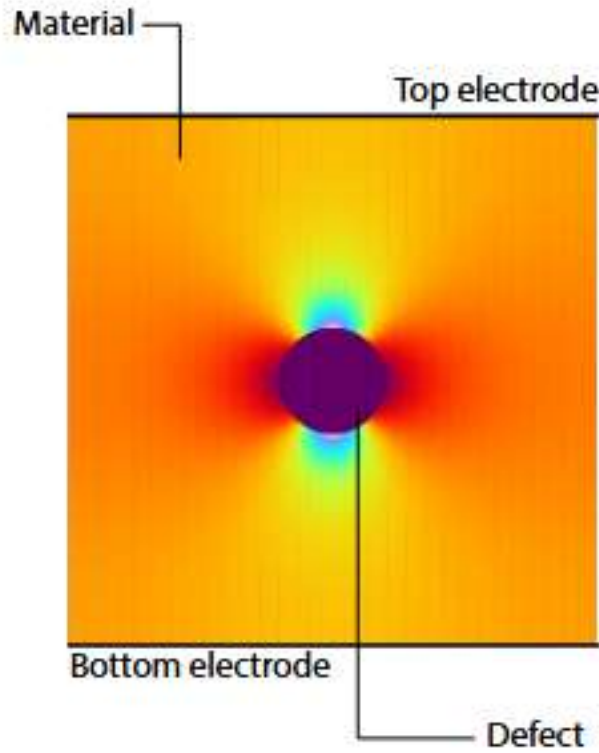
$$V_1 = V_2 = \dots = V_{eq},$$

$$I_{eq} = I_1 + I_2 + \dots,$$

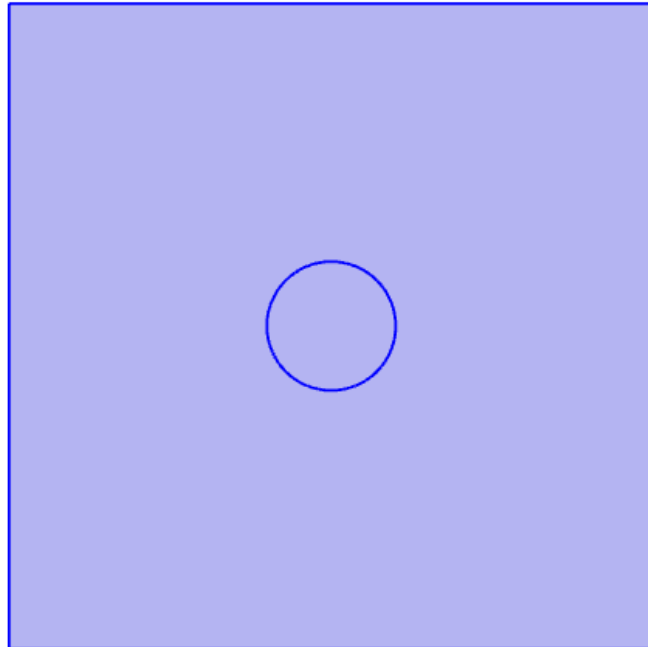
$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \dots$$

$$V_k = I_k R_k.$$

2.1 Parallel plates electrodes - Simulation



In COMSOL I started by defining the applied voltage $V_{app} = 1V$, a square centered at the center with a side $L_{square} = 1cm$, a circle also centered at the center with a radius of $R_{circle} = 0.1cm$, and the applied electric field $E_{app} = V_{app}/L_{square}$. The square and circle look like,



I then defined the electrical conductivity of the circle to $\sigma = 1S/m$ and the electrical conductivity of the rest of the square to $\sigma = 2$. I also set the electrical permittivity of the entire area to 1.

Looking at the physics, the equations that are applied to every point of the geometry we see,

$$\nabla J = Q_{j,v} \quad (2.1)$$

$$J = \sigma E + J_e \Leftrightarrow E = \frac{J - J_e}{\sigma} \quad (2.2)$$

$$E = -\nabla V \quad (2.3)$$

Substituting E from equation 2.2 in equation 2.3 we get,

$$\begin{aligned} \frac{J - J_e}{\sigma} &= -\nabla V \Leftrightarrow \\ \Leftrightarrow -\sigma \nabla V &= J - J_e \end{aligned}$$

We then get,

$$\nabla(-\sigma \nabla V) = \nabla J - \nabla J_e \quad (2.4)$$

Substituting ∇J in 2.4 from equation 2.1 we get,

$$\nabla(-\sigma \nabla V) = Q_{j,v} - \nabla J_e \quad (2.5)$$

The boundary conditions for the edges of the square were imposed by default in COMSOL with the expression $n \cdot J = 0$, which means that no current can run across the external boundary of the square, this boundary condition describes a material (the square) surrounded by an insulator.

After this I replaced some boundary conditions so that it would simulate our system with corresponding electrodes. I assigned the top side of the square to have an electrical potential $V = -V_{app}/2$ and the bottom side to have an electrical potential $V = V_{app}/2$. This way I applied a potential difference of V_{app} between the two electrodes.

When I neglected the defect and tried computing the Total current I using Ohms law, first computing the resistance with the cross sectional area $A = 1m \times 1cm = 10^{-2}m^2$, the length the current passes $L = 1cm = 10^{-2}m$,

$$R = \frac{L}{\sigma A} = \frac{10^{-2}}{2 \times 10^{-2}} = 0.5\Omega$$

since our Electrical potential is $V = 0.5 - (-0.5) = 1$ we compute the total current I as,

$$I = \frac{U}{R} = \frac{1}{0.5} = 2A$$

From COMSOL we got the total current to be $\approx 1.96A$ the error from the computed by hand can be explained by the defect and we can minimize it by doing the average electrical conductivity as,

$$\sigma_{average} = \frac{V_1\sigma_1 + V_2\sigma_2}{V_1 + V_2}$$

Where we get a ≈ 1.97 not the correct answer but very close.

2.2 Continuum physics and partial differential equations (PDE)

2.2.1 How the temperature T evolves inside a solid material

assuming that the temperature T continuous function,

$$T = T(x, y, z, t)$$

The time evolution is governed by,

$$\frac{\partial T}{\partial t} = \alpha \nabla^2 T$$

Where α is the diffusivity.

This equation relates to the first-order time-derivative of T with its second-order space-derivative.

The Laplace operator ∇^2 is defined as,

$$\nabla^2 \equiv \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$$

Deconstructing the heat equation

Fourier's law of heat conduction,

$$q = -k\nabla T$$

Where the heat flux density q is directed opposite to the temperature gradients, and its units are W/m^2 . The proportionality factor is the thermal conductivity k of the material, and its units are $W/(mK)$.

Energy conservation,

$$\frac{\partial u}{\partial t} = -\nabla \cdot q$$

The time derivative of the internal energy density, u , in a point is the amount of heat flowing "into that point", i.e., the opposite of the divergence of the heat flux density.

Heat capacity,

$$\frac{\partial T}{\partial t} = \frac{1}{c\rho} \frac{\partial u}{\partial t}$$

The temperature variation is related to the variation of the internal energy via the specific heat capacity c and the mass density ρ .

$$\begin{aligned} q &= -k\nabla T \\ \frac{\partial u}{\partial t} &= -\nabla \cdot q \\ \frac{\partial T}{\partial t} &= \frac{1}{c\rho} \frac{\partial u}{\partial t} \end{aligned}$$

From the bottom two equations, we get,

$$\frac{\partial T}{\partial t} = -\frac{1}{c\rho} \nabla \cdot q$$

from the equation we just achieved and Fourier's law of heat conduction, we get,

$$\frac{\partial T}{\partial t} = \frac{1}{c\rho} \nabla \cdot (k\nabla T)$$

For a uniform k , where $\alpha = \frac{k}{c\rho}$, we get,

$$\frac{\partial T}{\partial t} = \alpha \nabla^2 T$$

This equation describes/predicts what's going on inside a region of space.

Boundary conditions

To predict the evolution of T we need to know what happens at the boundaries (aka the boundary conditions).

- Prescribed temperature T , e.g. heat reservoir at constant T . Known as the Dirichlet boundary condition.
- Prescribed normal heat flux $\hat{n} \cdot (q = -k\nabla T)$, given amount of W/m^2 . A particular case is thermal insulation, i.e., zero heat flux. Known as the Neumann boundary condition.

Initial condition

For time-dependent equations we also need to know the initial state,

$$T = T(x, y, z, t = 0)$$

With this we can calculate how the temperature will evolve with time.

Equilibrium solution

Eventually, we will always reach the same equilibrium conditions, where the heat flux density and temperature are not evolving. and the governing equation becomes time-independent,

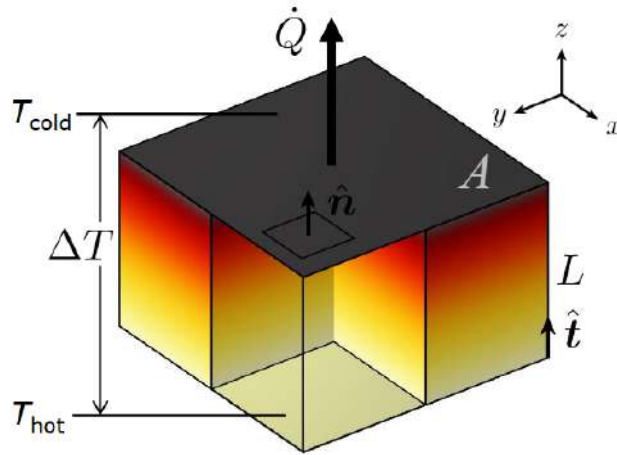
$$\frac{\partial T}{\partial t} = \alpha \nabla^2 T = 0$$

The lumped-element model: Circuits

This situation is the most essential. we can express it in terms of the net amount:

$\|q\| = \dot{Q}/A$, the total heat transfer rate $\dot{Q} \Leftrightarrow$ heat flux density q , with cross section area A .

$\|\nabla T\| = \Delta T/L$, temperature difference $\Delta T \Leftrightarrow$ temperature gradient ∇T , with length L .



$$\dot{Q} = \iint \hat{n} \cdot q \, dS$$

$$\dot{Q} = \iint \sum_k n_k q_k \, dS$$

↓ in z direction

$$\dot{Q} = \iint n_z q_z \, dx dy$$

$$\Delta T = \int \hat{t} \cdot (\nabla T) \, d\ell$$

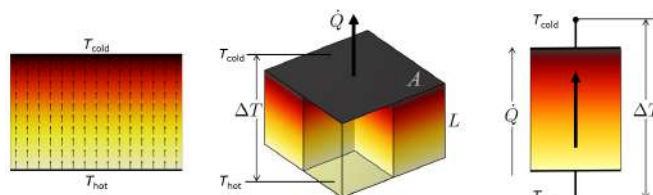
$$\Delta T = \int \sum_k t_k \frac{\partial T}{\partial x_k} \, d\ell$$

↓ in z direction

$$\Delta T = \int t_z \frac{\partial T}{\partial z} \, dz$$

Where the thermal resistance $R = \frac{L}{kS}$

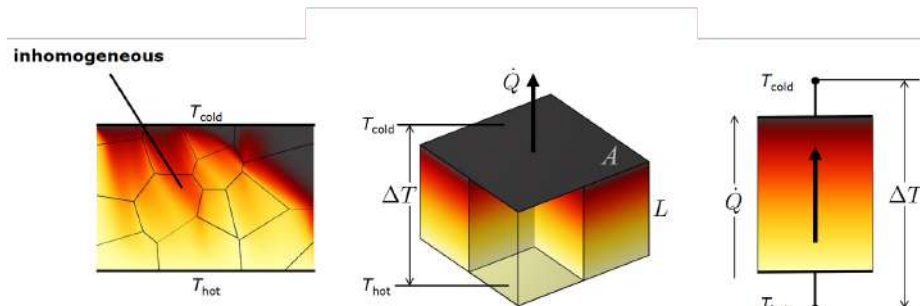
$$\begin{array}{c} \boxed{\begin{array}{l} \mathbf{q} = -k \nabla T \\ \dot{Q} = \iint \hat{n} \cdot \mathbf{q} \, dS \end{array}} \quad \boxed{\begin{array}{l} \Delta T = \int \hat{t} \cdot (\nabla T) \, d\ell \\ R \dot{Q} = \Delta T \end{array}} \end{array}$$



The basic idea of homogenization

$$\begin{array}{c} \overbrace{\qquad\qquad\qquad}^{\mathbf{q} = -k_{\text{eff}} \nabla T} \\ \left\| \mathbf{q} \right\| = \dot{Q}/A \qquad \left\| \nabla T \right\| = \Delta T/L \\ \underbrace{\qquad\qquad\qquad}_{R\dot{Q} = \Delta T} \end{array}$$

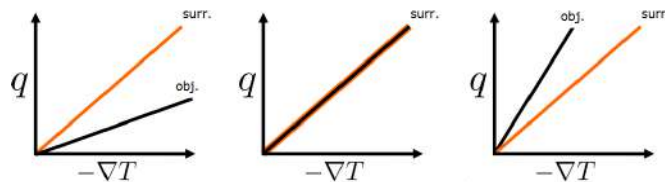
Where the thermal conductivity $k_{\text{eff}} = \frac{\dot{Q}/S}{\Delta T/L}$



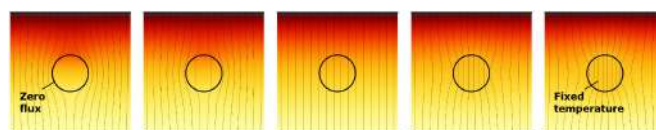
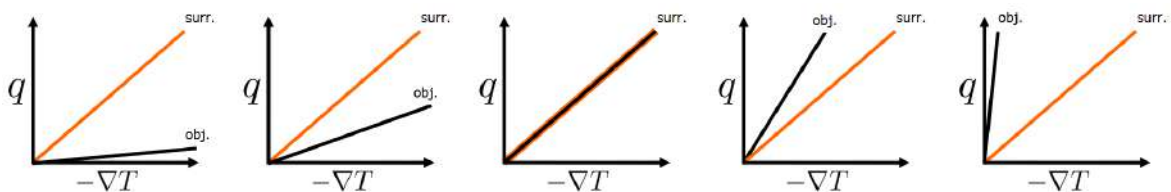
Conductors and Insulators

Constitutive relation of the object,

$$\mathbf{q} = -k \nabla T$$



Thermal Insulator	Same property as the surroundings	Thermal Conductor
$k < k_{\text{surr}}$	$k = k_{\text{surr}}$	$k > k_{\text{surr}}$
Smaller q for same ∇T		Larger q for same ∇T

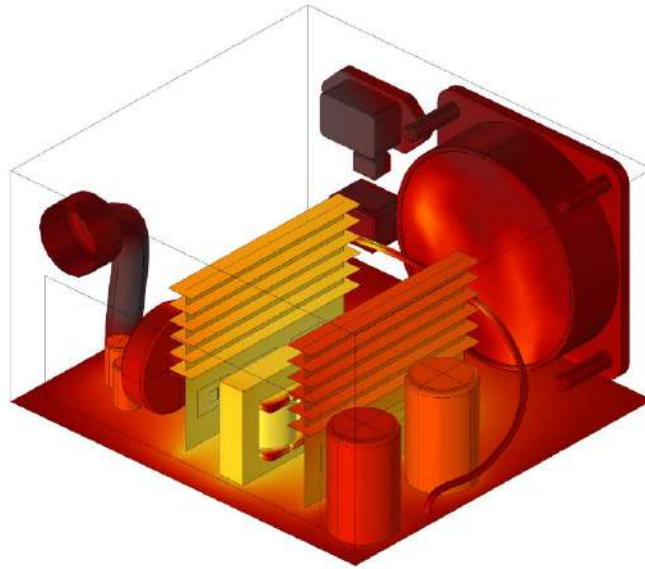


Perfect Insulator	←	Same properties as the surroundings	→	Perfect Conductor
$k \rightarrow 0$		$k = k_{\text{surr}}$		$k \rightarrow \infty$
$q = 0$				$\nabla T = 0$

The source terms - Heat source

- We may have a heat source Q_T inside the region governed by the PDE
- The units are W/m^3
- The heat equation then becomes,

$$\frac{\partial T}{\partial t} = \alpha \nabla^2 T + \frac{1}{c\rho} Q_T$$



Electrical Conduction example

- At steady state, from $\nabla \times E = 0$, we get $E = -\nabla V$
- Charge conservation equation $\nabla \cdot J = 0$
- Constitutive relation $J = \sigma E$ (Ohm's law)
- At steady state and for uniform conductivity:

$$\sigma \nabla^2 V = 0$$

- Steady state heat equation with uniform diffusivity:

$$\alpha \nabla^2 T = 0$$

- Boundary conditions:
 - Dirichlet: prescribed electrical potential
 - Neumann: prescribed electrical field or current (particular case: electrical insulation)

2.2.2 Analogies between PDE for different phenomena

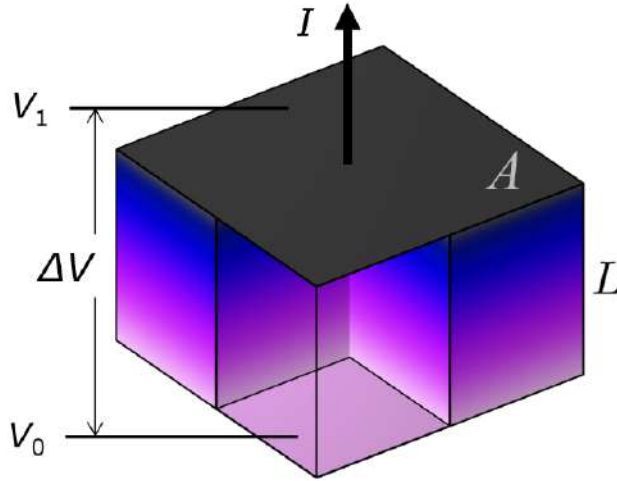
Potential	Gradient	Constitutive relation	Flux density	Conservation law
T [K] <i>Temperature</i>	$p_T = -\nabla T$ <i>Temp. grad.</i>	$q = k p_T$ <i>Heat flux relation</i>	q [W/m ²] <i>Heat flux dens.</i>	$\nabla \cdot q = Q_T$ <i>Energy cons.</i>
V_e [V] <i>Elec. potential</i>	$E = -\nabla V_e$ <i>Elec. field</i>	$J = \sigma E$ <i>Ohm's law</i>	J [A/m ²] <i>Elec. curr. dens.</i>	$\nabla \cdot J = Q_e$ <i>Charge cons.</i>
V_e [V] <i>Elec. potential</i>	$E = -\nabla V_e$ <i>Elec. field</i>	$D = \varepsilon E$ <i>Permittivity rel.</i>	D [C/m ²] <i>Elec. disp. field</i>	$\nabla \cdot D = \rho$ <i>Gauss's law</i>
V_m [A] <i>Magn. scalar pot.</i>	$H = -\nabla V_m$ <i>Magn. field</i>	$B = \mu H$ <i>Magnetic rel.</i>	B [Wb/m ²] <i>Magn. flux dens.</i>	$\nabla \cdot B = 0$ <i>Gauss (magn.)</i>
c [mol/m ³] <i>Molar conc.</i>	$p_c = -\nabla c$ <i>Conc. grad.</i>	$N = D p_c$ <i>Diffusion rel.</i>	N [mol/m ²] <i>Molar flux dens.</i>	$\nabla \cdot N = R$ <i>Matter cons.</i>

$$\|\nabla T\| = \Delta T / L$$

$$\begin{pmatrix} J_x \\ J_y \\ J_z \end{pmatrix} = \begin{pmatrix} \sigma_{xx} & \sigma_{xy} & \sigma_{xz} \\ \sigma_{yx} & \sigma_{yy} & \sigma_{yz} \\ \sigma_{zx} & \sigma_{zy} & \sigma_{zz} \end{pmatrix} \begin{pmatrix} E_x \\ E_y \\ E_z \end{pmatrix} = E_x \begin{pmatrix} \sigma_{xx} \\ \sigma_{yx} \\ \sigma_{zx} \end{pmatrix} + E_y \begin{pmatrix} \sigma_{xy} \\ \sigma_{yy} \\ \sigma_{zy} \end{pmatrix} + E_z \begin{pmatrix} \sigma_{xz} \\ \sigma_{yz} \\ \sigma_{zz} \end{pmatrix}$$

2.3 Parallel plates electrodes - Intro

2.3.1 Part 1 (1 homogeneous material)



In this exercise, a block of electrically conductive material was placed between two parallel electrodes, as shown in the figure above. The goal was to employ simple back-of-the-envelope calculation to predict the conductive behavior based on the material's properties and geometry.

Assumed parameters

$$V_0 = 0 \text{ V}, V_1 = 1 \text{ V}, L = 1 \text{ cm} = 0.01 \text{ m}, A = 0.01 \text{ m}^2, \sigma = 2 \text{ S/m}$$

Computing the magnitude of the electric field E :

From Subsection 2.2.2 we know the formula of the electric field to be,

$$E = -\nabla V_e$$

Where $\|\nabla V_e\| = \Delta V_e/L$. We calculate ∇V_e as,

$$\|\nabla V_e\| = \frac{\Delta V_e}{L} = \frac{V_1 - V_0}{L} = \frac{1 - 0}{0.01} = 100 \text{ V/m}$$

therefore we calculate the magnitude of the electrical field $\|E\|$ to be,

$$\|E\| = 100 \text{ V/m}$$

Estimating the electrical resistance R :

We know the formula of the electrical resistance to be,

$$R = \frac{L}{\sigma A}$$

With the values given, we compute the resistance R as,

$$R = \frac{0.01}{2 \cdot 0.01} = 0.5 \text{ } \Omega$$

Computing the expected current density J using the constitutive relationship:

From Subsection 2.2.2,

$$J = \sigma E$$

From the values given we compute J as,

$$J = \sigma E = 2 \cdot 100 = 200 \text{ A/m}^2$$

Computing the total current I :

From the known formula,

$$I = \frac{U}{R}$$

And from the given values of V and the computed value of the electrical resistance R , we compute I as,

$$I = \frac{U}{R} = \frac{1 - 0}{0.5} = 2 \text{ A}$$

Computing the current density J :

From the value of the total current I and the cross sectional area A we can compute the current density J as,

$$J = \frac{I}{A} = \frac{2}{0.01} = 200 \text{ A/m}^2$$

Which is equal to the one we calculate using the constitutive relationship.

2.3.2 Part 2 (2 heterogeneous materials)

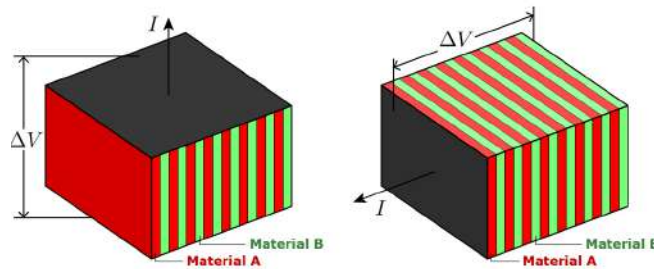


Figure 2.1: Stacks

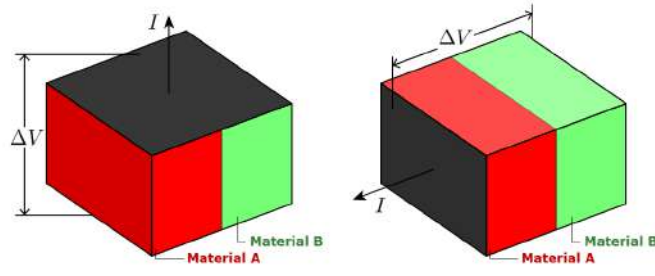
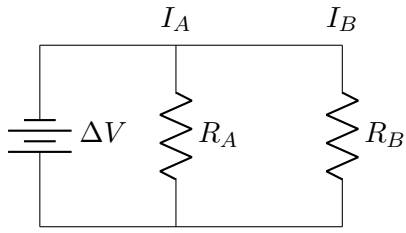


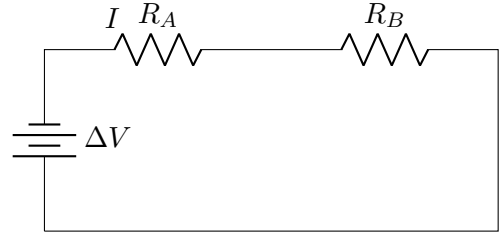
Figure 2.2: Equivalent arrangement of materials, when analyzed with the lumped element model

Now considering layered materials such as the ones in Figure 2.1, where the total volume is subdivided between the two materials. When analyzed with the lumped element model, these are equivalent to the ones shown in Figure 2.2.

The equivalent electric circuits: Parallel electric circuit:



(a) Parallel



(b) Series

Figure 2.3: Equivalent circuits for layered materials (parallel left, series right).

$$I = I_A + I_B$$

$$V = V_A = V_B$$

$$\frac{1}{R} = \frac{1}{R_A} + \frac{1}{R_B}$$

Series electric circuit:

$$I = I_A = I_B$$

$$V = V_A + V_B$$

$$R = R_A + R_B$$

Writing the expressions of electrical resistance R in terms of σ_A and σ_B :

Parallel electric circuit:

Individual resistances R_A and R_B are,

$$R_A = \frac{L}{\sigma_A A/2}$$

$$R_B = \frac{L}{\sigma_B A/2}$$

The total resistance will then be,

$$\begin{aligned} \frac{1}{R} &= \frac{1}{R_A} + \frac{1}{R_B} \\ &= \frac{1}{\frac{L}{\sigma_A A/2}} + \frac{1}{\frac{L}{\sigma_B A/2}} \\ &= \frac{\sigma_A A}{2L} + \frac{\sigma_B A}{2L} \\ &= \frac{A}{2L} (\sigma_A + \sigma_B) \\ R &= \frac{2L}{A(\sigma_A + \sigma_B)} \end{aligned} \tag{2.6}$$

Series electric circuit:

Individual resistances R_A and R_B are,

$$R_A = \frac{L/2}{\sigma_A A}$$

$$R_B = \frac{L/2}{\sigma_B A}$$

The total resistance will then be,

$$\begin{aligned} R &= R_A + R_B \\ &= \frac{L/2}{\sigma_A A} + \frac{L/2}{\sigma_B A} \\ &= \frac{L}{2A} \left(\frac{1}{\sigma_A} + \frac{1}{\sigma_B} \right) \end{aligned} \tag{2.7}$$

Expression of the effective conductivities σ_{eff} :

Using the following formula,

$$R = \frac{L}{\sigma_{eff} A}$$

We achieve the expression of the effective conductivity σ_{eff} as,

For parallel electric circuit,

$$\begin{aligned} \frac{L}{\sigma_{eff} A} &= R \\ \frac{L}{\sigma_{eff} A} &= \frac{2L}{A(\sigma_A + \sigma_B)} \\ \frac{1}{\sigma_{eff}} &= \frac{2}{\sigma_A + \sigma_B} \\ \sigma_{eff} &= \frac{\sigma_A + \sigma_B}{2} \end{aligned} \tag{2.8}$$

For series electric circuit,

$$\begin{aligned} \frac{L}{\sigma_{eff} A} &= R \\ \frac{L}{\sigma_{eff} A} &= \frac{L}{2A} \left(\frac{1}{\sigma_A} + \frac{1}{\sigma_B} \right) \\ \frac{1}{\sigma_{eff}} &= \frac{1}{2} \left(\frac{1}{\sigma_A} + \frac{1}{\sigma_B} \right) \\ \sigma_{eff} &= \frac{2}{\frac{1}{\sigma_A} + \frac{1}{\sigma_B}} \end{aligned} \tag{2.9}$$

Calculating numerical values of the electrical resistances R :

For the given values,

$$\sigma_A = 2 \text{ S/m}, \sigma_B = 1 \text{ S/m}, L = 1 \text{ cm} = 0.01 \text{ m}, A = 0.01 \text{ m}^2$$

For the parallel electrical circuit,

$$R = \frac{2 \cdot 0.01}{0.01 \cdot (2 + 1)} = \frac{2}{3} \approx 0.667 \text{ } \Omega$$

For the series electrical circuit,

$$R = \frac{0.01}{2 \cdot 0.01} \left(\frac{1}{2} + \frac{1}{1} \right) = \frac{1}{2} \cdot \frac{3}{2} = \frac{3}{4} = 0.75 \text{ } \Omega$$

Calculating numerical values of the effective conductivities σ_{eff} :

For the parallel electric circuit,

$$\sigma_{eff} = \frac{2+1}{2} = \frac{3}{2} = 1.5 \text{ S/m}$$

For the series electric circuit,

$$\sigma_{eff} = \frac{2}{\frac{1}{2} + \frac{1}{1}} = \frac{4}{3} \approx 1.33 \text{ S/m}$$

Considering the limit $\sigma_B \rightarrow 0$:

For the case where the conductivity of material B tends to 0. The following changes happen.

For the parallel electric circuit,

$$\begin{aligned} R &\rightarrow \frac{2L}{A\sigma_A} = \frac{2 \cdot 0.01}{0.01 \cdot 2} = 1 \text{ } \Omega \\ \sigma_{eff} &\rightarrow \frac{\sigma_A}{2} = \frac{2}{2} = 1 \text{ S/m} \end{aligned} \tag{2.10}$$

For the series electric circuit,

$$\begin{aligned} R &\rightarrow \infty \\ \sigma_{eff} &\rightarrow 0 \end{aligned}$$

Considering the limit $\sigma_B \rightarrow \infty$:

For the case where the conductivity of material B tends to ∞ . The following changes happen.

For the parallel electric circuit,

$$\begin{aligned} R &\rightarrow 0 \Omega \\ \sigma_{eff} &\rightarrow \infty \end{aligned}$$

For the series electric circuit,

$$\begin{aligned} R &\rightarrow \frac{L}{2A\sigma_A} = \frac{0.01}{2 \cdot 0.01 \cdot 2} = \frac{1}{4} = 0.25 \Omega \\ \sigma_{eff} &\rightarrow 2\sigma_A = 4 \text{ S/m} \end{aligned} \tag{2.11}$$

2.4 Anisotropic material

Anisotropic means having a physical property which has different values when measured in different directions.

Now considering the behavior of an isotropic material using the same example as in the previous Section 2.1. The purpose of this exercise is to understand the anisotropic behavior and to compute the effective conductivity tensor. The idea is to simulate a situation that is similar to the experimental characterization of the conductive response of a material sample.

Since we want to consider a homogeneous material, we delete the circle from the previous model.

In the previous model was also insulated on the left and right side, which won't work for our new model, as a square that is insulated on the left and right side wouldn't be able to generate a net flux along the x -direction, therefore, we replace the boundary condition with a Periodic Condition. By creating and assigning it to the corresponding boundaries, it will automatically override the default insulation boundary condition. This Periodic Condition corresponds to having each point on each point on the left the same potential as it is in the corresponding point of the right side.

We also switch change the boundary condition of the top and bottom sides from $\pm V_{app}/2$ to both having the same $-sign(y) * V_{app}/2$.

In the previous model we had the integration operator $intop1(ec.Jy)$ which would return the current density in the top side (AKA the current density on the y -direction). Now we add a second one

`intop2(ec.Jx)` that returns the current density on the right side (AKA the current density on the x -direction).

Note! that we are looking at a square prism with 1 m out-of-plane height and to normalize it we must divide it by the area of that side (with h out of plane height and L the actual J_x actual = $J_x/(h \cdot L)$).

To change the constitutive relation to an Anisotropic one, we change the Electrical conductivity σ property to User Defined, and change it from isotropic to diagonal. This then shows us a 3×3 matrix as,

	x	y	z
x	σ_{xx}	σ_{xy}	σ_{xz}
y	σ_{yx}	σ_{yy}	σ_{yz}
z	σ_{zx}	σ_{zy}	σ_{zz}

We then assign the following values $\sigma_{xx} = 1 \text{ S/m}$ and $\sigma_{yy} = 2 \text{ S/m}$, keeping the rest at 0.

After running the simulation, the following values for the current density are returned,

$$J_x = 3.9185 \cdot 10^{-14} \text{ A/m}$$

$$J_y = 2.0000 \text{ A/m}$$

These values are not the "true" current density values as they need to be normalized by dividing with the out-of-plane height's product with the side as,

$$\tilde{J}_x = \frac{J_x}{1 \cdot 0.01} = 3.9185 \cdot 10^{-12} \text{ A/m}$$

$$\tilde{J}_y = \frac{J_y}{1 \cdot 0.01} = 200 \text{ A/m}$$

Computing the electrical fields E_x and E_y ,

$$E_x = \frac{\Delta V_x}{L} = \frac{0}{0.01} = 0$$

$$E_y = \frac{\Delta V_y}{L} = \frac{1}{0.01} = 100$$

We can then estimate the yy and xy coefficients of effective conductivity,

$$\begin{pmatrix} J_x \\ J_y \end{pmatrix} = \begin{pmatrix} \sigma_{xx} & \sigma_{xy} \\ \sigma_{yx} & \sigma_{yy} \end{pmatrix} \begin{pmatrix} E_x \\ E_y \end{pmatrix} = E_x \begin{pmatrix} \sigma_{xx} \\ \sigma_{yx} \end{pmatrix} + E_y \begin{pmatrix} \sigma_{xy} \\ \sigma_{yy} \end{pmatrix}$$

$$\begin{cases} 3.9185 \cdot 10^{-12} = 0 \cdot \sigma_{xx} + 100 \cdot \sigma_{xy} \\ 200 = 0 \cdot \sigma_{yx} + 100 \cdot \sigma_{yy} \end{cases} \Leftrightarrow \begin{cases} \sigma_{xy} = 3.9185 \cdot 10^{-14} \text{ S/m} \\ \sigma_{yy} = 2 \text{ S/m} \end{cases}$$

Which are almost identical to the ones we input.

To fully characterize (in 2D) the response of the material, we want to apply the field along the x -direction.

After running the simulation, the following values for the current density are returned,

$$J_x = 1.0000 \text{ A/m}$$

$$J_y = 3.6633 \cdot 10^{-14} \text{ A/m}$$

These values are not the "true" current density values as they need to be normalized by dividing with the out-of-plane height's product with the side as,

$$\tilde{J}_x = \frac{J_x}{1 \cdot 0.01} = 100 \text{ A/m}$$

$$\tilde{J}_y = \frac{J_y}{1 \cdot 0.01} = 3.6633 \cdot 10^{-12} \text{ A/m}$$

Computing the electrical fields E_x and E_y ,

$$E_x = \frac{\Delta V_x}{L} = \frac{1}{0.01} = 100$$

$$E_y = \frac{\Delta V_y}{L} = \frac{0}{0.01} = 0$$

We can then estimate the xx and yx coefficients of effective conductivity,

$$\begin{pmatrix} J_x \\ J_y \end{pmatrix} = \begin{pmatrix} \sigma_{xx} & \sigma_{xy} \\ \sigma_{yx} & \sigma_{yy} \end{pmatrix} \begin{pmatrix} E_x \\ E_y \end{pmatrix} = E_x \begin{pmatrix} \sigma_{xx} \\ \sigma_{yx} \end{pmatrix} + E_y \begin{pmatrix} \sigma_{xy} \\ \sigma_{yy} \end{pmatrix}$$

$$\begin{cases} 3.6633 \cdot 10^{-12} = 100 \cdot \sigma_{xx} + 0 \cdot \sigma_{xy} \\ 100 = 100 \cdot \sigma_{yx} + 0 \cdot \sigma_{yy} \end{cases} \Leftrightarrow \begin{cases} \sigma_{xx} = 1 \text{ S/m} \\ \sigma_{yx} = 3.6633 \cdot 10^{-14} \text{ S/m} \end{cases}$$

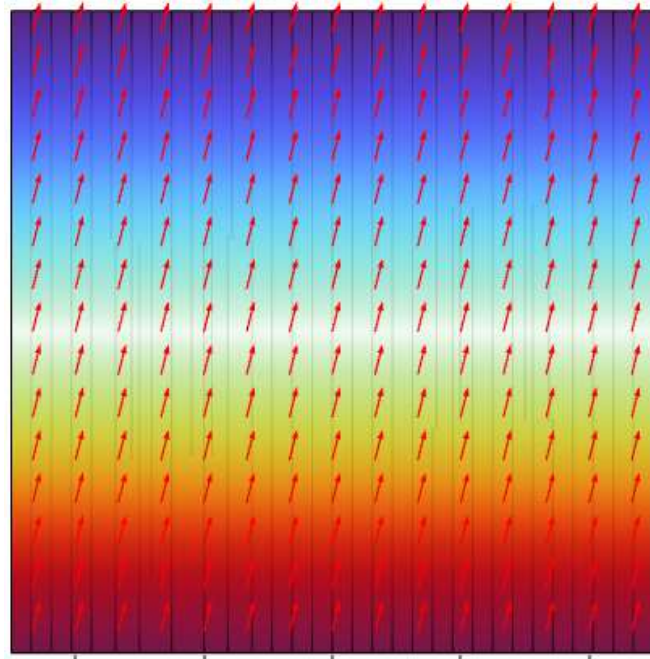
Which are almost identical to the ones we input.

Adding an off-diagonal entry in the conductivity tensor:

We switch the conductivity tensor σ from Diagonal to Symmetric, and enter 0.5 S/m for the xy component.

After running and computing the normalized current density $\tilde{J}_x$ and $\tilde{J}_y$ for both a vertical and horizontal electrical field, we can then compute the xx , xy , yx , and yy components of the conductivity, and we achieve the same ones we input as expected.

After running the simulations, we also get plots such as,



Why rotated coordinate systems matter. Many crystalline and anisotropic materials have conductivity (or stiffness) properties defined relative to crystallographic axes, which generally do not align with the geometry of a sample or the coordinate system used in simulations. Rotating the tensor therefore models the experimentally relevant situation where the material's intrinsic symmetry directions differ from the applied-field direction. This explains why the rotated conductivity tensor develops off-diagonal terms and why the current density need not align with the imposed electric field.

Physical interpretation of anisotropy. In anisotropic materials the current density $\mathbf{J}$ is not generally parallel to the electric field $\mathbf{E}$. Off-diagonal terms σ_{xy} , σ_{yx} indicate coupling between directions: an

electric field in y may produce current in x , and vice-versa. This behaviour represents microstructural alignment, texture, or layered arrangements that break rotational symmetry.

Here we can notice that the current density J represented by the arrows, run along a different direction than the applied field, indicated by the streamline. For example in the Figure ?? where we apply a field vertically, we also observe a horizontal component of the current. This is associated with the anisotropic response of the material that we include in the model by changing the conductivity tensor. It is a way to approximate phenomena happening at a much smaller scale, a scale that is not resolved in our model. Typically, this could be the scale of the crystal lattice, or the scale of the microstructure.

When the local coordinate system differs from the coordinate system aligned with the side of the sample:

In the case the local coordinate system describing anisotropy (e.g. the crystal coordinate system) does not match the coordinate system aligned with the sides of our sample. We can simulate this situation to get a grasp of the corresponding homogenization or experimental characterization. We start by creating a Rotated System, finding the property Rotation about the out-of-the-plane axis, and using $\pi/3$. And then in Current Conservation 1 node, we switch the coordinate system property to the rotated system.

Setting the off-diagonal components of the conductivity tensor back to zero since we assume that 1 S/m and 2 S/m are the conductivities along the x and y directions of the rotated system.

After running the simulation and calculating the coefficients of the effective conductivity tensor, we obtain,

$$\begin{pmatrix} \sigma_{xx} & \sigma_{xy} \\ \sigma_{yx} & \sigma_{yy} \end{pmatrix} = \begin{pmatrix} 1.75 & -0.43 \\ -0.43 & 1.25 \end{pmatrix} S/m$$

Interpreting results:

If we didn't know the conductivity tensor and we know that there is a coordinate system along which the tensor would be diagonal, meaning that in those direction we have that J and E are aligned. To find the intrinsic behavior, we need to diagonalize the effective tensor using Python as,

```
1 import numpy as np
2
3 # Where eff_coeff is the effective coefficient matrix
4 eff_coeff = np.array([[1.75, -0.43], [-0.43, 1.25]])
5
6 # Calculate the eigenvalues and eigenvectors
7 eigenvalues, eigenvectors = np.linalg.eig(eff_coeff)
8
9 # Create a diagonal matrix from the eigenvalues
10 D = np.diag(eigenvalues)
11
12 # Calculate the diagonalizing matrix
13 S = np.linalg.inv(eigenvectors) @ eff_coeff @ eigenvectors
14
15 # Calculate the angle of rotation
16 theta = np.arctan2(eigenvectors[1, 0], eigenvectors[0, 0]) * (180 / np.pi)
17 theta = np.round(theta, 2)
18
19
20 # Output the results
21 print("Effective Coefficient Matrix:\n", eff_coeff)
22 print("Diagonal Matrix D:\n", D)
23 print("diagonalizing Matrix S:\n", S)
24 print("Angle of Rotation (degrees):", theta)
```

2.5 Anisotropic arrangement of isotropic materials

Considering a layered material composed of a stack of alternating layers of two different materials. Each of the two composing material is expected to have a isotropic behavior, meaning that is the sample was

homogeneously composed by either one of them the $J - E$ would simply be given by $J = \sigma E$ where σ is a scalar (one single value). However, the fact that the two materials are geometrically arranged in an anisotropic way, will cause the overall behavior to be that of an anisometric composite.



Starting with the model from 2.4 I created a new parameter $Nstack = 10$ denoting the number of layers in the stack of each material. Followed by creating the geometry of the stack.

I then defined the electrical conductivity of each material as,

$$\sigma_A = 10 \text{ S/m}$$

$$\sigma_B = 1 \text{ S/m}$$

After running the simulation, the following values of the current density were achieved,
For a horizontal applied field,

$$J_x = 1.8182 \text{ A/m}$$

$$J_y = -7.256 \cdot 10^{-14}$$

For a vertical applied field,

$$J_x = 1.1762 \cdot 10^{-14} \text{ A/m}$$

$$J_y = 5.5000 \text{ A/m}$$

After solving the system we achieved the following coefficients of effective conductivity,

$$\begin{pmatrix} \sigma_{xx} & \sigma_{xy} \\ \sigma_{yx} & \sigma_{yy} \end{pmatrix} \approx \begin{pmatrix} 1.82 & 0 \\ 0 & 5.5 \end{pmatrix} \text{ S/m}$$

We can make sense of this result by considering the circuit model from Section 2.3. For the case of the field being applied in the y direction the two resistances are added in parallel as in Equation 2.6, while for the case of the field applied in the x direction the resistances are added in series as in Equation 2.7. The effective conductivities are computed using the formulas in Equation 2.8 for the field applied in the y direction and using the formula in Equation 2.9 for the field being applied in the x direction. Computing them as,

For the vertical field (parallel),

$$R = \frac{2 \cdot 0.01}{0.01^2 \cdot (10 + 1)} = \frac{200}{11} = 18.1818 \text{ } \Omega$$

$$\sigma_{yy} = \frac{(10 + 1)}{2} = 5.5 \text{ S/m}$$

For the horizontal field (series),

$$R = \frac{0.01}{2 \cdot 0.01^2} \left(\frac{1}{10} + \frac{1}{1} \right) = 50 \frac{11}{20} = 55 \text{ } \Omega$$

$$\sigma_{xx} = \frac{2}{\frac{1}{10} + \frac{1}{1}} = \frac{20}{11} = 1.8182 \text{ S/m}$$

These values are identical to the ones computed in the simulation.

Since one of the two materials is much more conductive than the other one, we can use our intuition to make an even simpler estimation for the two cases. When the field is applied parallel to the stack, the more conductive material will dominate, but its cross sectional area will be 50% (so it will be half as conductive). When the field is applied perpendicularly to the stack, the more resistive material will dominate, but its length along the field direction will be 50% (so it will be twice as conductive). Therefore, using these considerations, we can estimate the effectivity xx and yy coefficients. And we can estimate them using the formulas in Equation 2.10 for the yy and the formulas in Equation 2.11 for the xx , as, For the vertical field,

$$R = \frac{2 \cdot 0.01}{0.01^2 \cdot 10} = 20 \text{ } \Omega$$

$$\sigma_{yy} = \frac{10}{2} = 5 \text{ S/m}$$

For the horizontal field,

$$R = \frac{0.01}{2 \cdot 0.01^2 \cdot 1} = \frac{1}{0.02} = 50 \text{ } \Omega$$

$$\sigma_{xx} = 2 \cdot 1 = 2$$

These values are not perfect, however, they are a good approximation.

In the case where we couldn't run a simulation, we can still estimate the effective conductivity matrix if the stack was oriented along a different direction (not parallel to the x or y axis), we can use the following code,

```

1 # Setting base parameters
2 L = 0.01
3 A = 0.01**2
4
5 # For a coordinate system rotated by theta degrees
6 theta = 30 * (np.pi / 180) # Convert degrees to radians
7
8 # Electrical conductivity values for odd and even layers
9 sigma_odd = 10
10 sigma_even = 1
11
12 # Calculate the effective conductivity considering the lumped element model
13 # For parallel Electric Field
14 sigma_eff_parallel = (sigma_odd + sigma_even) / 2
15
16 # For series Electric Field
17 sigma_eff_series = 2 / ((1/sigma_odd) + (1/sigma_even))
18
19 # Construct the effective conductivity matrix and rotate it
20 sigma_eff = np.array([[sigma_eff_series, 0], [0, sigma_eff_parallel]])
21 sigma_rotated = np.array([[np.cos(theta), -np.sin(theta)], [np.sin(theta), np.cos(theta)
22     ]]) @ sigma_eff @ np.array([[np.cos(theta), np.sin(theta)], [-np.sin(theta), np.cos
23     (theta)]])
24
25 print("Effective Conductivity Matrix (Rotated):\n", sigma_rotated)

```

2.6 Linear elasticity

2.6.1 Hooke's law

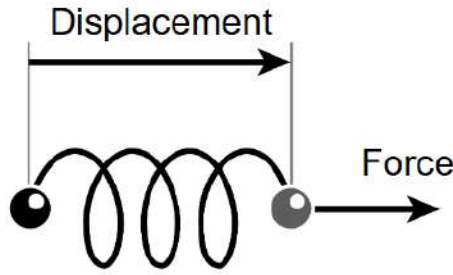
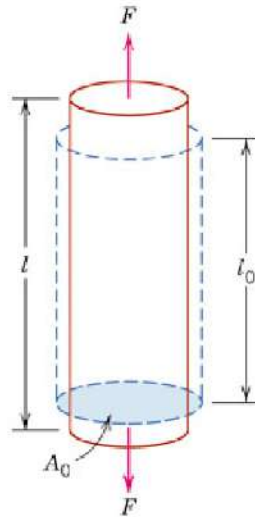


Figure 2.4: Hooke's Law

$$F = k\Delta x \quad (2.12)$$

Where variables in Equation 2.12 refer to: F the Force in Newtons N from the spring, Δx the displacement in meters m both seen in Figure 2.4, and the spring constant k in N/m , which measures the spring's stiffness.

Normalization of force and displacement



We normalize the Force F by dividing it by the area A_0 over which it acts, this is called the stress σ ,

$$\sigma = \frac{F}{A_0}$$

Which is given in Pascal Pa (N/m^2).

We normalize the displacement Δx or Δl by dividing it with the initial sample size x_0 or l_0 , this is called the strain ε ,

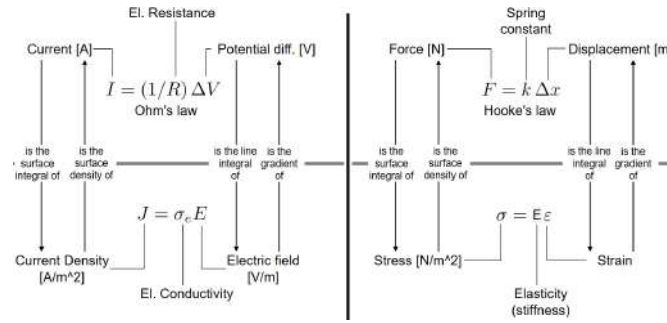
$$\varepsilon = \frac{\Delta l}{l_0} = \frac{l - l_0}{l_0}$$

Which is dimensionless either in m/m or %.

Then we use the Elasticity modulus (also known as Young's modulus, E-modulus) E to substitute the spring constant k to achieve the normalized Hooke's Law as,

$$\sigma = E \cdot \varepsilon$$

Analogy with electrical conduction,



2.6.2 Elastic deformation

In an undeformed material, it has a specific volume. After deforming it (like stretching), the bonds inside the material are stretched, and therefore the volume increases. The part of the deformation that is considered elastic is the deformation that is reversible. So in a stress-strain graph such as the one in Figure 2.5, the elastic region (AKA the reversible deformation) is that with the initial constant slope, E-modulus E .

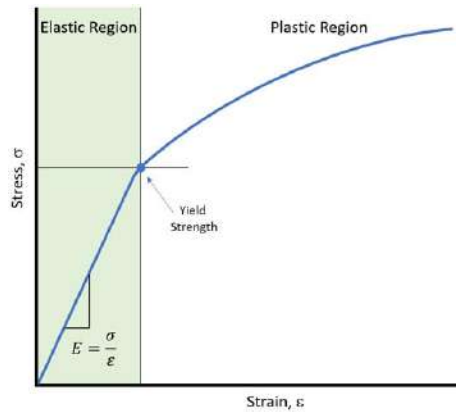
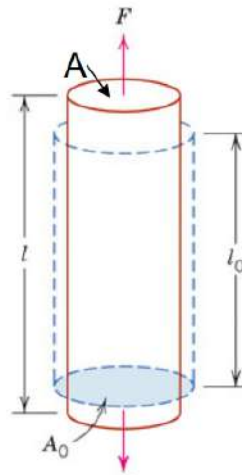


Figure 2.5

Volume changes in elastic deformation



$$A_0 \cdot l_0 \neq A \cdot l$$

Poisson's ratio ν is then defined as,

$$\nu = -\frac{\varepsilon_x}{\varepsilon_z} = -\frac{\varepsilon_y}{\varepsilon_z}$$

For uniaxial deformation along the z axis and isotropic materials, the following Poisson's ratio are known,

Metals	$\nu \approx 0.33$
Ceramics	$\nu \approx 0.25$
Polymers	$\nu \approx 0.40$

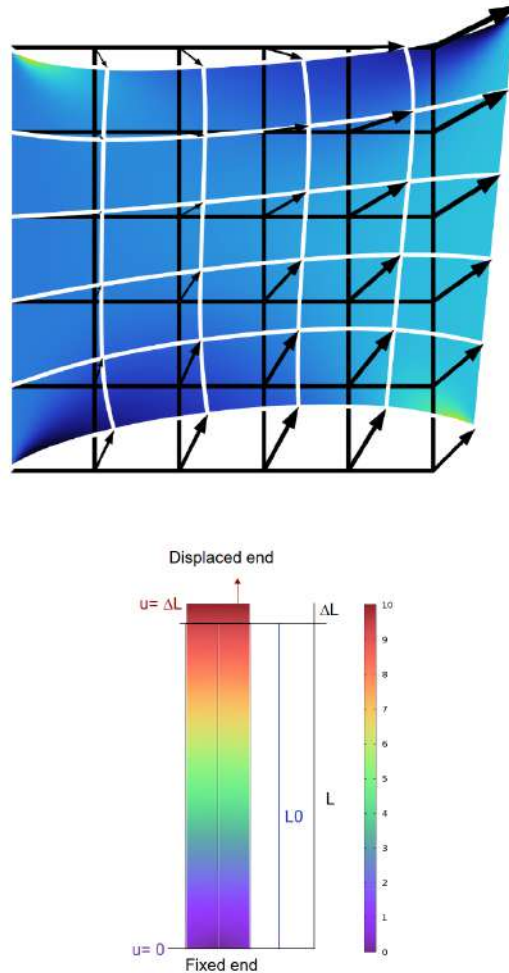
2.6.3 Isotropic linear elasticity

For isotropic materials, there are two constants the E-modulus E and Poisson's ratio ν .

2.7 Continuum elasticity in 3D

When considering the 3D case, things are more complicated for elasticity than they are for electrical or heat conductivity. For each physical quantity, we need more components than for conductivity, scalars become vectors, and vectors become matrices, with $9 = 3 \times 3$ coefficients. However the operators do not change, the gradient, divergence are defined normally.

The displacement field u



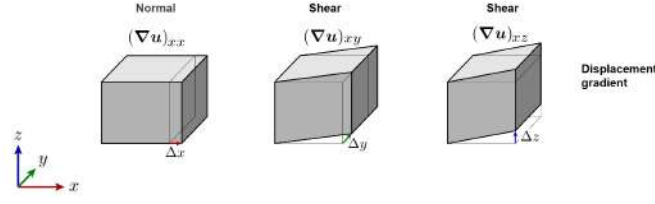
The displacement gradient,

$$\nabla u = \begin{pmatrix} (\nabla u)_{xx} & (\nabla u)_{xy} & (\nabla u)_{xz} \\ (\nabla u)_{yx} & (\nabla u)_{yy} & (\nabla u)_{yz} \\ (\nabla u)_{zx} & (\nabla u)_{zy} & (\nabla u)_{zz} \end{pmatrix} = \begin{pmatrix} \frac{\partial u_x}{\partial x} & \frac{\partial u_y}{\partial x} & \frac{\partial u_z}{\partial x} \\ \frac{\partial u_x}{\partial y} & \frac{\partial u_y}{\partial y} & \frac{\partial u_z}{\partial y} \\ \frac{\partial u_x}{\partial z} & \frac{\partial u_y}{\partial z} & \frac{\partial u_z}{\partial z} \end{pmatrix}$$

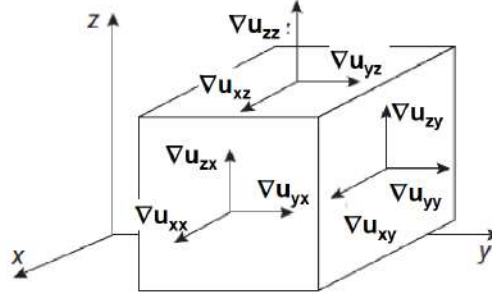
2.7.1 3D, x direction

As seen in the figure below we can see the normal component $(\nabla u)_{xx} = \partial u_x / \partial x$, which tells us how much the x displacement (u_x) changes as we move along the x direction. We can then also see the shear

components $(\nabla u)_{xy} = \partial u_x / \partial y$ and $(\nabla u)_{xz} = \partial u_x / \partial z$ which tells us how much the x displacement changes as we move along the y and z direction, this is shear deformation in the xy and xz planes.



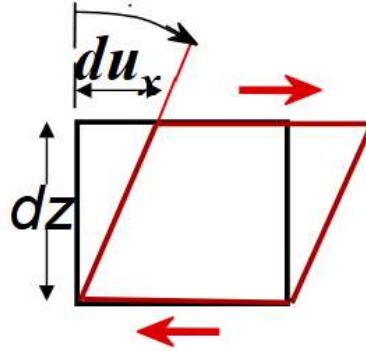
This applied to all faces as,



Displacement gradient to strain

$$\nabla u = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ \gamma & 0 & 0 \end{pmatrix}$$

Corresponds to the following distortion,



Where $\gamma = \partial u_x / \partial z$. It is split into a symmetric strain ε and a skew rotation ω , as,

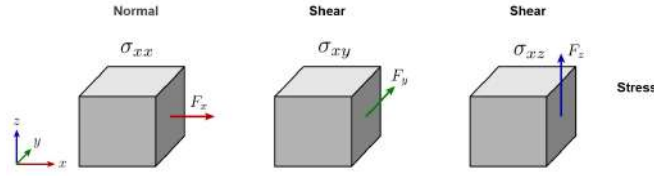
$$u = d\varepsilon + d\omega = \begin{pmatrix} 0 & 0 & \gamma/2 \\ 0 & 0 & 0 \\ \gamma/2 & 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 & -\gamma/2 \\ 0 & 0 & 0 \\ \gamma/2 & 0 & 0 \end{pmatrix}$$

ε represents pure shear strain, which changes shape but not orientation, and its formula is,

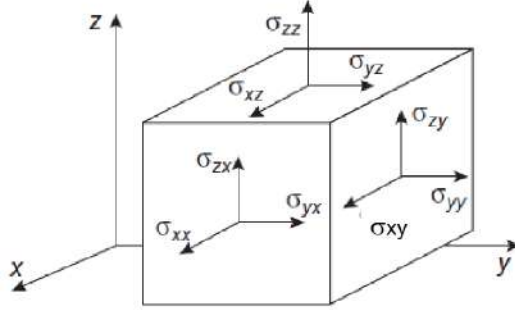
$$\varepsilon = \frac{1}{2}(\nabla u + \nabla u^T)$$

ω represents a small rigid body rotation (no shape change) and its formula is,

$$\omega = \frac{1}{2}(\nabla u - \nabla u^T)$$



This is applied to all faces as,



The strain as,

$$\underline{\underline{\varepsilon}} = \begin{pmatrix} \varepsilon_{xx} & \varepsilon_{xy} & \varepsilon_{xz} \\ \varepsilon_{yx} & \varepsilon_{yy} & \varepsilon_{yz} \\ \varepsilon_{zx} & \varepsilon_{zy} & \varepsilon_{zz} \end{pmatrix} = \begin{pmatrix} \varepsilon_{xx} & \varepsilon_{xy} & \varepsilon_{xz} \\ \varepsilon_{xy} & \varepsilon_{yy} & \varepsilon_{yz} \\ \varepsilon_{xz} & \varepsilon_{yz} & \varepsilon_{zz} \end{pmatrix} \text{ Symmetric per definition}$$

The stress as,

$$\underline{\underline{\sigma}} = \begin{pmatrix} \sigma_{xx} & \sigma_{xy} & \sigma_{xz} \\ \sigma_{yx} & \sigma_{yy} & \sigma_{yz} \\ \sigma_{zx} & \sigma_{zy} & \sigma_{zz} \end{pmatrix} = \begin{pmatrix} \sigma_{xx} & \sigma_{xy} & \sigma_{xz} \\ \sigma_{xy} & \sigma_{yy} & \sigma_{yz} \\ \sigma_{xz} & \sigma_{yz} & \sigma_{zz} \end{pmatrix} \text{ Symmetric per definition}$$

2.7.2 Von Mises stress

Wrapping all stress components into one parameter,

$$\sigma_v^2 = \frac{1}{2} \left((\sigma_{xx} - \sigma_{yy})^2 + (\sigma_{yy} - \sigma_{zz})^2 + (\sigma_{zz} - \sigma_{xx})^2 + 6(\sigma_{yz}^2 + \sigma_{zx}^2 + \sigma_{xy}^2) \right)$$

From the 4th-order elasticity tensor to Voigt notation

The constitutive relation of linear elasticity can be written in tensor form as

$$\sigma_{ij} = C_{ijkl} \varepsilon_{kl},$$

where C_{ijkl} is the fourth-order stiffness tensor. The symbol “:” in expressions such as $C : \varepsilon$ denotes a *double tensor contraction*, meaning

$$(C : \varepsilon)_{ij} = \sum_{k=1}^3 \sum_{l=1}^3 C_{ijkl} \varepsilon_{kl}.$$

Because the stress and strain tensors are symmetric, and because the stiffness tensor obeys the minor and major symmetries

$$C_{ijkl} = C_{jikl} = C_{ijlk} = C_{klij},$$

the 81 possible components reduce to 21 independent ones. To write Hooke’s law in matrix form, we group components as

$$\begin{aligned} \{\sigma\} &= (\sigma_{xx}, \sigma_{yy}, \sigma_{zz}, \sigma_{yz}, \sigma_{xz}, \sigma_{xy})^T, \\ \{\varepsilon\} &= (\varepsilon_{xx}, \varepsilon_{yy}, \varepsilon_{zz}, 2\varepsilon_{yz}, 2\varepsilon_{xz}, 2\varepsilon_{xy})^T. \end{aligned}$$

The factors of 2 on shear strains correspond to *engineering shear strain*, which simplifies the matrix representation. In this form, the constitutive law reduces to the standard 6×6 matrix shown next.

2.7.3 Isotropic elasticity in 3D

$$\begin{bmatrix} \sigma_{xx} \\ \sigma_{yy} \\ \sigma_{zz} \\ \sigma_{yz} \\ \sigma_{xz} \\ \sigma_{xy} \end{bmatrix} = \frac{E}{(1+\nu)(1-2\nu)} \begin{bmatrix} 1-\nu & \nu & \nu & 0 & 0 & 0 \\ \nu & 1-\nu & \nu & 0 & 0 & 0 \\ \nu & \nu & 1-\nu & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1-2\nu}{2} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1-2\nu}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1-2\nu}{2} \end{bmatrix} \begin{bmatrix} \varepsilon_{xx} \\ \varepsilon_{yy} \\ \varepsilon_{zz} \\ 2\varepsilon_{yz} \\ 2\varepsilon_{xz} \\ 2\varepsilon_{xy} \end{bmatrix}$$

Poisson's ratio. The parameter ν quantifies lateral contraction: a material stretched along z shrinks in x and y by $\varepsilon_{xx} = \varepsilon_{yy} = -\nu\varepsilon_{zz}$.

Due to symmetry we do not need to write all nine components of the stress strain. **Example** Uniaxial tension along z .

$$\begin{aligned} \sigma_{zz} &= E \cdot \varepsilon_{zz} \\ \vartheta &= -\frac{\varepsilon_{xx}}{\varepsilon_{zz}} = -\frac{\varepsilon_{yy}}{\varepsilon_{zz}} \\ \varepsilon_{yz} &= \varepsilon_{xz} = \varepsilon_{xy} = 0 \\ \sigma_v^2 &= \frac{1}{2} \left((\sigma_{xx} - \sigma_{yy})^2 + (\sigma_{yy} - \sigma_{zz})^2 + (\sigma_{zz} - \sigma_{xx})^2 + 6(\sigma_{yz}^2 + \sigma_{zx}^2 + \sigma_{xy}^2) \right) \end{aligned}$$

Example for uniaxial tension along z

$$\begin{bmatrix} 0 \\ 0 \\ \sigma_{zz} \\ 0 \\ 0 \\ 0 \end{bmatrix} = \frac{E}{(1+\nu)(1-2\nu)} \begin{bmatrix} 1-\nu & \nu & \nu & 0 & 0 & 0 \\ \nu & 1-\nu & \nu & 0 & 0 & 0 \\ \nu & \nu & 1-\nu & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1-2\nu}{2} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1-2\nu}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1-2\nu}{2} \end{bmatrix} \begin{bmatrix} -\vartheta\varepsilon_{zz} \\ -\vartheta\varepsilon_{zz} \\ \varepsilon_{zz} \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

2.7.4 Stress and strain transformation

When the coordinate system is rotated, the stress tensor transforms as

$$\sigma' = R \sigma R^T,$$

where R is the rotation matrix. Principal stresses are obtained by solving

$$\det(\sigma - \lambda I) = 0,$$

so the eigenvalues of σ give the principal stresses, and the eigenvectors give their directions. The same relations hold for the strain tensor.

2.7.5 Practical elasticity calculations used in exercises

Some of the elasticity exercises required manual evaluation of stress and strain states. The main steps are:

- **Principal stresses:** Solve the eigenvalue problem of the stress tensor σ ; the eigenvalues are the principal stresses.
- **Principal strains:** Apply the same procedure to the strain tensor ε or use Hooke's law if the stress state is known.
- **Von Mises stress:** Compute

$$\sigma_{vM} = \sqrt{\frac{1}{2}[(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2]}.$$

- **Mohr's circle (2D):** Graphical method to obtain principal stresses and maximum shear stress for plane stress problems.

- **Plane stress vs. plane strain:** Plane stress: $\sigma_{zz} = 0$. Plane strain: $\varepsilon_{zz} = 0$. Used depending on geometry (thin sheet vs. long constrained body).
- **Uniaxial loading check:** For pure tension in z , the relations reduce to $\sigma_{zz} = E\varepsilon_{zz}$ $\varepsilon_{xx} = \varepsilon_{yy} = -\nu\varepsilon_{zz}$.

These tools complement the COMSOL results by showing how stress states relate to the tensor formulation discussed earlier.

2.7.6 Equations in COMSOL

$$0 = \nabla \cdot S + F_v$$

This is the static equilibrium condition: the divergence of the stress tensor represents the net internal force per unit volume, while F_v is the applied body force density (e.g. gravity). In static simulations acceleration is zero, so internal forces must balance external ones. Each component expands as

$$(\nabla \cdot S)_i = \sum_{j=1}^3 \frac{\partial S_{ij}}{\partial x_j}.$$

Newton's 2nd law, with no acceleration, change in stress state balances external forces. Second Piola-Kirchhoff stress, for small deformations, the same as σ .

$$S = S_{inel} + S_{el}, \quad \varepsilon_{el} = \varepsilon - \varepsilon_{inel}$$

Constitutive symmetry. For linear elastic materials the stiffness tensor satisfies

$$C_{ijkl} = C_{jikl} = C_{ijlk} = C_{klij},$$

which expresses both minor and major symmetry. These constraints reduce the number of independent elastic constants and ensure that the elastic energy is well-defined.

Elastic strain.

$$\varepsilon_{inel} = \varepsilon_0 + \varepsilon_{ext} + \varepsilon_{th} + \varepsilon_{hs} + \varepsilon_{pl} + \varepsilon_{cr} + \varepsilon_{vp} + \varepsilon_{ve}$$

Meaning of the inelastic strain components. In COMSOL, the total strain may include several contributions:

- ε_0 : initial (predefined) strain,
- ε_{ext} : prescribed external strain,
- ε_{th} : thermal strain from temperature change,
- ε_{hs} : hygroscopic or swelling strain,
- ε_{pl} : plastic strain (permanent deformation),
- ε_{cr} : creep strain (time-dependent),
- ε_{vp} : viscoplastic strain,
- ε_{ve} : viscoelastic strain.

In this course we consider purely elastic materials, so all inelastic components vanish.

All other contributions to strain other than elastic deformation. For us, this is 0.

$$S_{el} = C : \varepsilon_{el}$$

Hooke's law Here C is the fourth-order elasticity (stiffness) tensor and the symbol “:” denotes a double tensor contraction. In index notation this means

$$(S_{el})_{ij} = \sum_{k=1}^3 \sum_{l=1}^3 C_{ijkl} (\varepsilon_{el})_{kl},$$

so we multiply C_{ijkl} with the elastic strain components and sum over the two repeated indices k and l to obtain the elastic stress tensor. For an isotropic material, this general relation reduces to the 6×6 matrix form of Hooke's law given in the *Isotropic elasticity in 3D* subsection.

$$S_{inel} = S_0 + S_{ext} + S_q$$

All other contributions to stress other than elastic deformation. For us, this is 0.

$$\varepsilon = \frac{1}{2} \left[(\nabla u)^T + \nabla u \right]$$

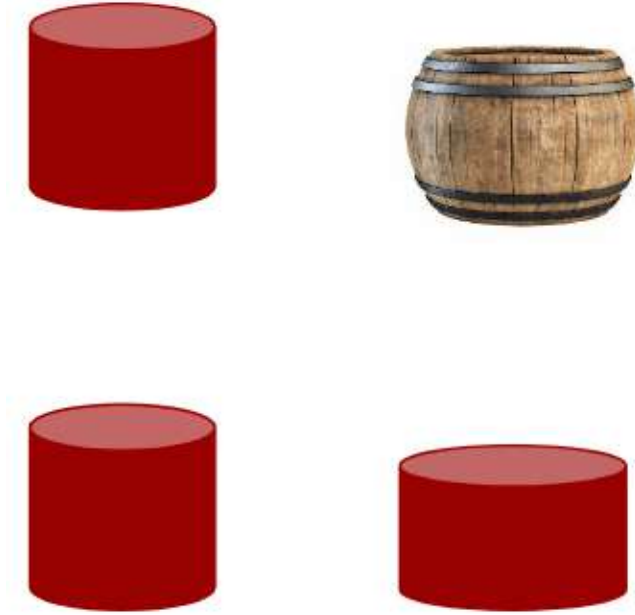
Symmetric part of the displacement gradient.

$$C = C(E, \nu)$$

Stiffness matrix.

2.7.7 Boundary conditions

Boundary conditions are defined as either a prescribed displacement or an applied force. Displacement control is when a machine grips the sample at the ends and moves the grips apart, measuring the force response. The force control is, for example, when a sample is attached to the ceiling on one end and on the other a load is applied, it then measures the resulting displacement in the sample. In cases of compression, if there is strong friction between the sample and the tool, then the sample can start looking like a barrel. If there is no friction between the sample and the tool, then it will simply flatten the cylinder as seen in the figure below.



2.7.8 Newton's 1st law

An object at rest remains at rest, and an object in motion remains at constant speed and in a straight line unless acted on by an unbalanced force. Since we do not want objects to fly during the simulation,

- **For 1D:** we need to fix a point in space.
- **For 2D:** we must restrict one other point from motion along the normal line to the line connecting this point to the fully fixed one, otherwise the sample may rotate around the fixed point. We do this by setting a fixed point with the attributes $u_x = u_y = 0$ and another with either $u_x = 0$ or $u_y = 0$ to prevent motion along the x or y direction, respectively.

- **For 3D:** we must also prevent rotation around another axis. We do this by setting the fixed point with the attributes $u_x = u_y = u_z = 0$, another with the attributes $u_x = u_y = 0$ (or $u_x = u_z = 0$ or $u_y = u_z = 0$), and a third with the attributes $u_y = 0$ (or $u_x = 0$ or $u_z = 0$), the first option makes it so there's no rotation around x and y axes, i.e. keeping the direction of the z axis. Making it so the point can still move along z . There are many ways of doing this, and the best way depends on the application.

2.7.9 Numerical example: tensile test of an aluminium cube

To connect the continuum elasticity theory with a concrete simulation, we built a 3D COMSOL model of a homogeneous aluminium cube subjected to uniaxial tension in the z -direction.

We defined a cube with edge length $L = 0.05$ m and applied a prescribed displacement $u_z = 0.005$ m on the top face, while the bottom face was constrained to have zero displacement in z and free contraction in x and y . One corner point on the bottom face was fixed in x and y to remove rigid-body motion. The material was set to aluminium with Young's modulus $E \approx 70$ GPa and Poisson's ratio $\nu \approx 0.33$, in agreement with the isotropic elastic constants introduced earlier.

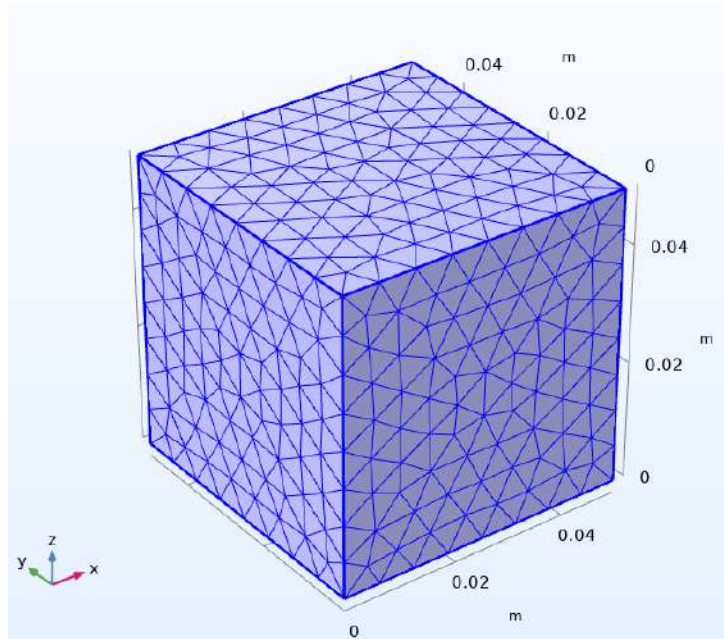


Figure 2.6: Finite element mesh of the aluminium cube used in the COMSOL simulation.

After meshing the cube and solving the stationary solid mechanics problem, COMSOL directly provided the von Mises stress distribution σ_{vM} , the displacement field $\mathbf{u}$ and the strain tensor components, such as the normal tensile strain ε_{zz} .

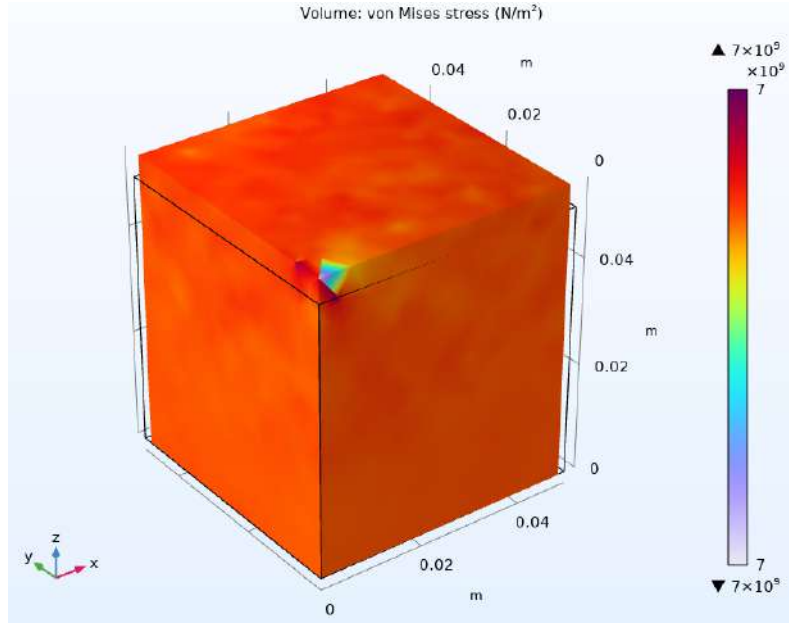


Figure 2.7: Von Mises stress distribution in the aluminium cube under uniaxial tension. The stress is nearly uniform, with only minor numerical fluctuations.

The displacement field increased linearly along z , while ε_{zz} was essentially uniform throughout the specimen, apart from small numerical noise.

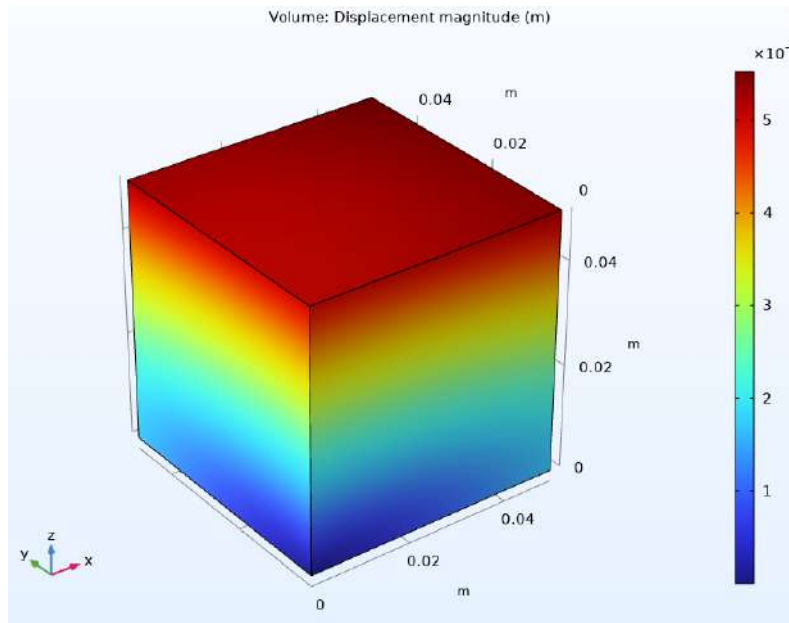


Figure 2.8: Displacement magnitude field. The linear gradient along the loading direction confirms the expected uniaxial deformation.

We then evaluated the average strain in the volume, $\langle \varepsilon_{zz} \rangle$, and the average normal stress on the loaded top surface, $\langle \sigma_{zz} \rangle$.

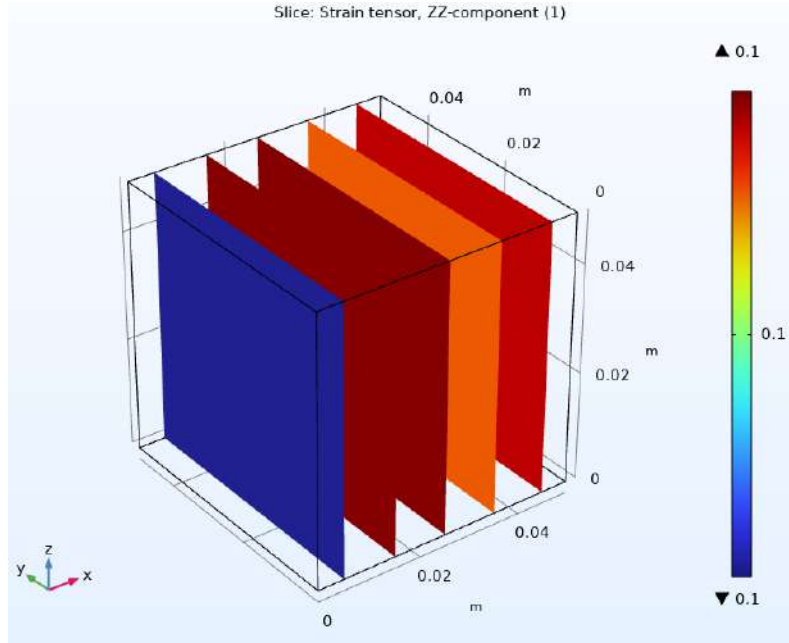


Figure 2.9: Slice plot of the tensile strain component ε_{zz} . The strain is almost perfectly uniform, consistent with a homogeneous uniaxial tension state.

In the simulation, COMSOL returned the following averaged quantities:

$$\langle \varepsilon_{zz} \rangle_{\text{COMSOL}} = 0.10000, \quad \langle \sigma_{zz} \rangle_{\text{COMSOL}} = 7.0000 \times 10^9 \text{ Pa}.$$

For comparison, the theoretical predictions based on the applied displacement and Hooke's law are:

$$\frac{u_z}{L} = \frac{0.005}{0.05} = 0.10, \quad E \frac{u_z}{L} = 70 \times 10^9 \cdot 0.10 = 7.0 \times 10^9 \text{ Pa}.$$

The simulation therefore reproduces the theoretical strain and stress with effectively zero deviation ($< 0.01\%$), confirming the correctness of the boundary conditions, material model, and overall setup.

These quantities satisfied, to a very good approximation,

$$\langle \varepsilon_{zz} \rangle \simeq \frac{u_z}{L}, \quad \langle \sigma_{zz} \rangle \simeq E \langle \varepsilon_{zz} \rangle,$$

which is exactly the uniaxial version of the 3D Hooke's law derived in Sections 3.6 and 3.7.3. This exercise therefore confirmed that the tensorial constitutive law implemented in COMSOL reproduces the simple one-dimensional stress-strain relation when the loading and boundary conditions correspond to a pure uniaxial tensile test.

Chapter 3

Module 3

3.1 Introduction to CALPHAD and phase diagrams

3.1.1 Phase diagrams and phases

Phase diagrams

A phase diagram in physical chemistry, engineering, mineralogy, and materials science is a chart that shows the conditions (e.g. temperature, composition, pressure) at which thermodynamically distinct phases occur at equilibrium. It represents the *equilibrium* state of lowest Gibbs energy.

Phases

In the physical sciences, a phase is a region of space (a thermodynamic system) throughout which all physical properties of a material are essentially uniform. Examples of physical properties include density, index of refraction, magnetization and chemical composition.

- Gas $\rightarrow$ Liquid $\rightarrow$ Crystalline or amorphous solid
- Single solid phase materials (isomorphism)
- Several solid phase materials (allotropy, polymorphism)

Material properties determined by microstructure

- Number of phases
- Chemical composition and crystal structure of each phase
- Phase fraction (mass or volume fraction)
- Size and spatial arrangement of phases

Characteristics of equilibrium phase diagrams

- Represent the state of *lowest* Gibbs energy G .
- Show the phases that are in thermodynamic equilibrium.
- Do not show how long it takes to reach equilibrium.
- If equilibrium is not attained in a realistic time, the system may be in a *metastable* (non-equilibrium) state.

3.1.2 Binary systems

Mass balance and lever rule

For a two-phase region (e.g. solid α and liquid L) in a binary system:

Mass fractions

$$1 = W_\alpha + W_L.$$

Average composition (mass conservation)

$$c_0 = c_\alpha W_\alpha + c_L W_L.$$

Lever rule Define the “arms” of the tie-line:

$$R = c_0 - c_L, \quad S = c_\alpha - c_0.$$

Then

$$W_\alpha = \frac{c_0 - c_L}{c_\alpha - c_L} = \frac{R}{S + R}, \quad W_L = \frac{c_\alpha - c_0}{c_\alpha - c_L} = \frac{S}{S + R},$$

and

$$W_L R = W_\alpha S.$$

Example: isomorphous binary system (complete solubility)

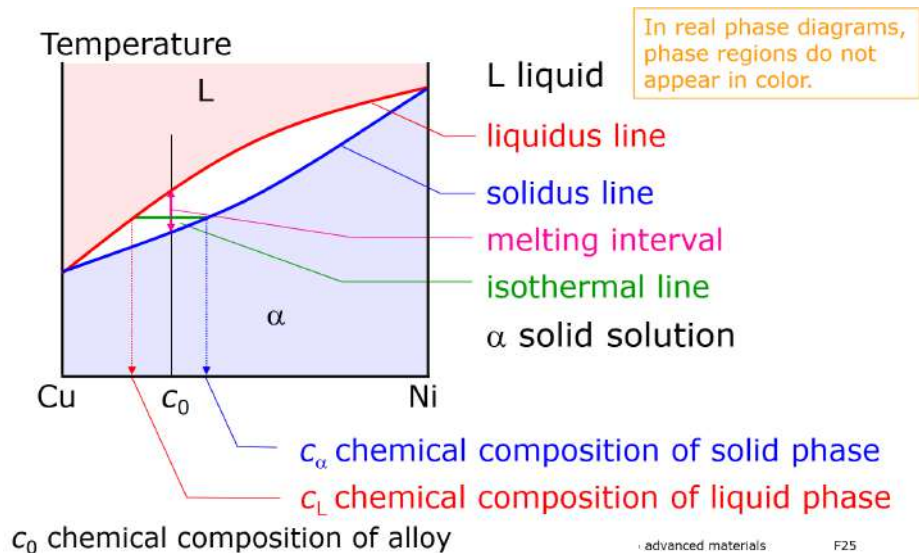


Figure 3.1: Example of an isomorphous binary phase diagram (complete solid solubility).

Microstructure for slow solidification

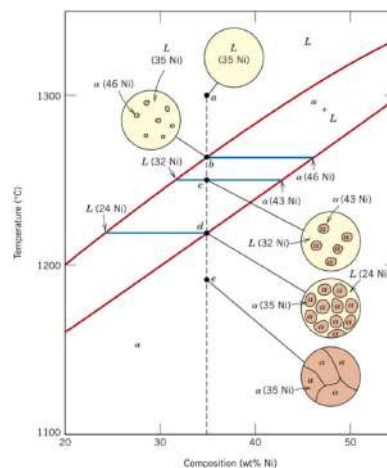


Figure 3.2: Typical solid-solution microstructure after slow cooling.

- Extremely slow cooling allows diffusion to redistribute solute during solidification.
- The composition of phases changes during solidification via atomic motion (diffusion).
- Diffusion depends strongly on time and temperature (faster at higher T).

Example: binary eutectic system (limited solubility in the solid state)

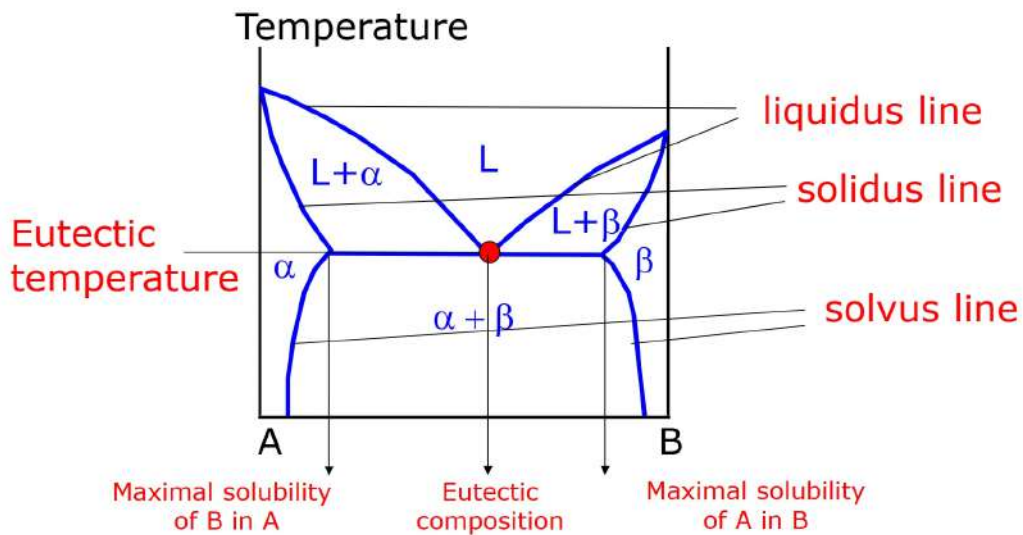


Figure 3.3: Example of a binary eutectic phase diagram with limited solid solubility.

Microstructures

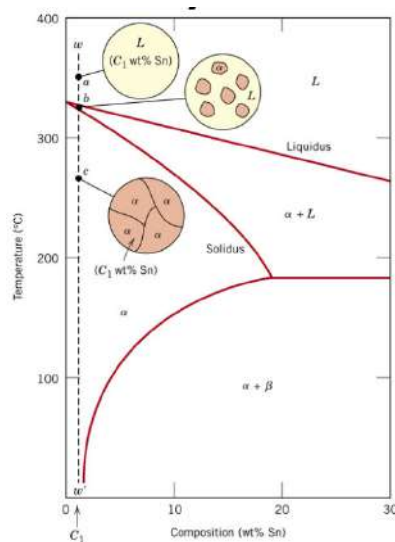


Figure 3.4: Microstructure where the entire content of β is soluble in α .

- The entire content of β is soluble in α .
- Single-phase α as in an isomorphous system.

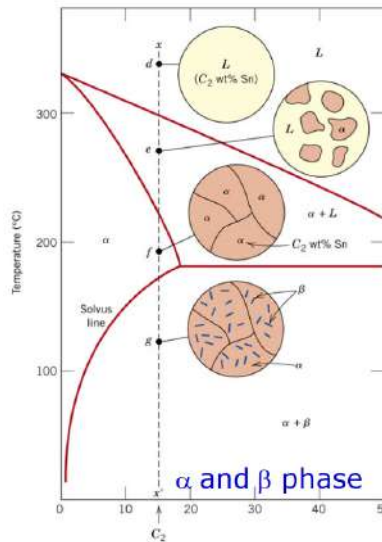


Figure 3.5: Microstructure with precipitates of β in an α matrix.

- At higher temperature, all β is soluble in α .
- At room temperature, solubility is lower: precipitates of β form in an α matrix.

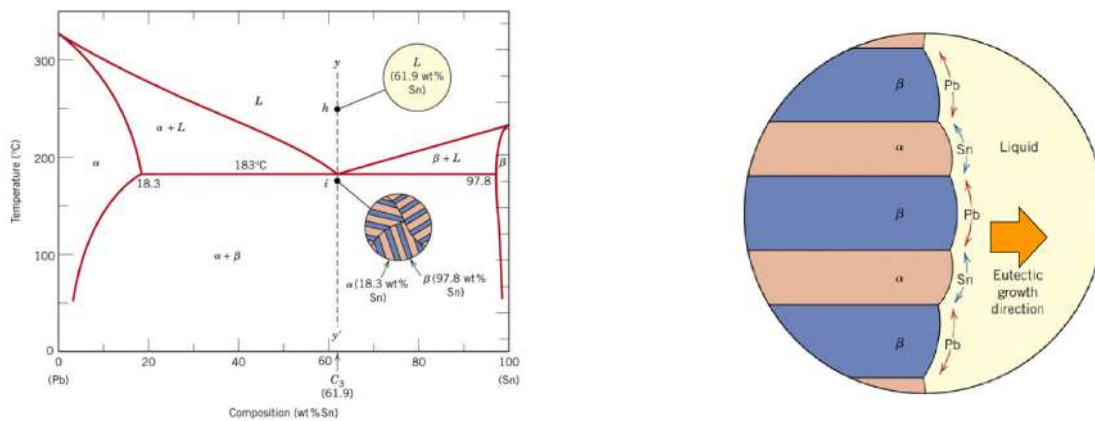


Figure 3.6: Lamellar eutectic microstructures: alternating layers of α and β .

- Lamellar structure: alternating layers of α and β .
- Lamellar spacing depends on cooling rate:
 - Slow solidification $\rightarrow$ wide lamellae.
 - Fast solidification $\rightarrow$ thin lamellae.
- Mechanical properties depend on lamellar spacing.

3.1.3 Ternary systems

Ternary $Zr-Y-O$ phase diagram (isothermal section)

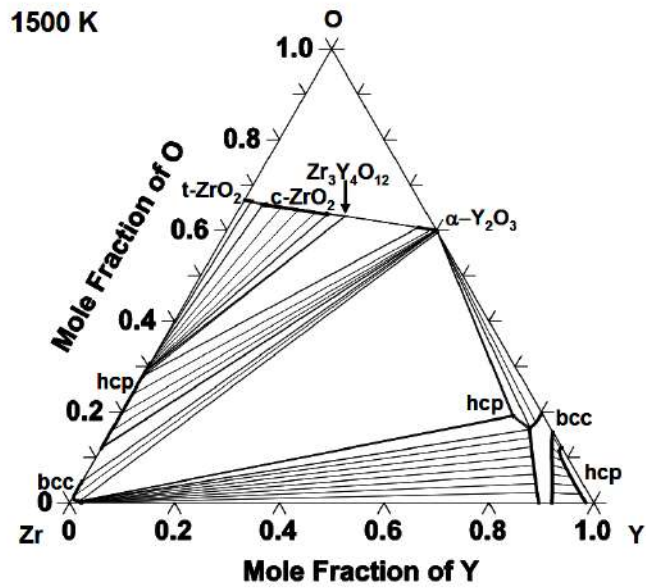


Figure 3.7: Isothermal section of a ternary $Zr-Y-O$ phase diagram.

How to read a ternary phase diagram

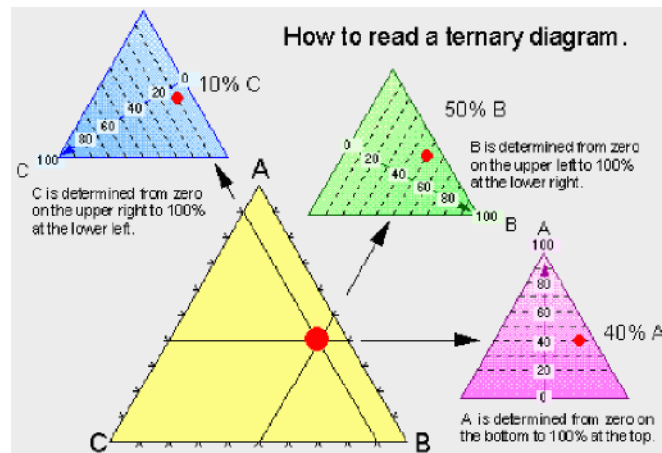


Figure 3.8: Composition reading in a ternary diagram.

- Each corner represents 100 % of one component.
- Any point inside the triangle represents a mixture of the three components.
- Composition is read by projecting parallel to triangle edges onto the axes.
- Two-phase regions are bounded by tie-lines; three-phase regions by tie-triangles.

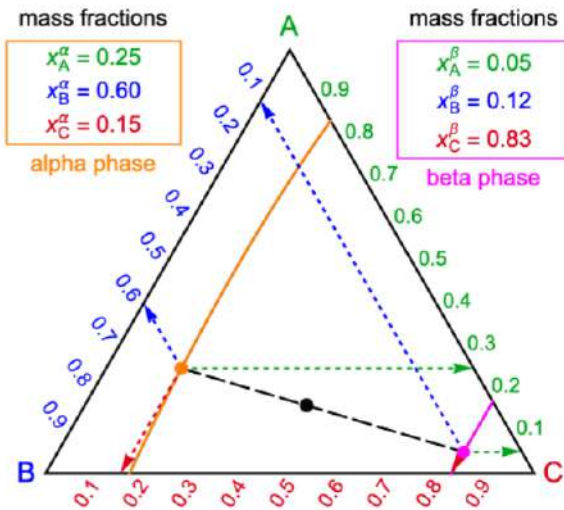


Figure 3.9: Example on how to read a 2 phase region of a Ternary phase diagram

3.1.4 Pseudo-binary and pseudo-ternary phase diagrams

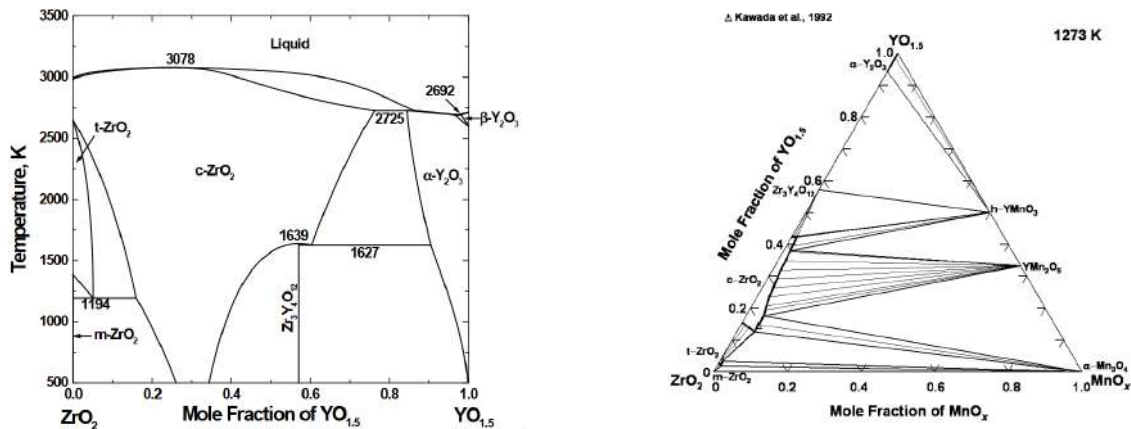


Figure 3.10: Examples of pseudo-binary and pseudo-ternary phase diagrams.

A pseudo-phase diagram is a simplified representation of a complex multi-component system, where the “components” are themselves compounds (like oxides) rather than pure elements.

- A **pseudo-binary** phase diagram shows phase relationships along a fixed compositional line between two composite end-members.
- A **pseudo-ternary** phase diagram extends this to three oxide components at a constant temperature.

These diagrams make it easier to visualise and understand phase stability in complex ceramic and oxide systems.

3.1.5 Different variables and projections in phase diagrams

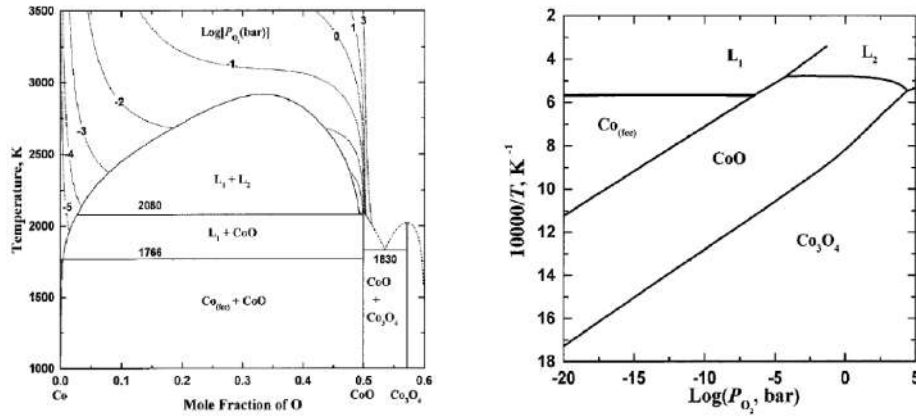


Figure 3.11: Two ways of plotting equilibria in the Co–O system: composition vs T (left) and T vs oxygen partial pressure (right).

The left diagram shows phase equilibria in the Co–O binary system at fixed bulk composition (closed system), while the right diagram expresses the same equilibria as functions of temperature and oxygen partial pressure (open system). Both are projections of the same thermodynamic relationships and are therefore consistent.

3.1.6 Gibbs phase rule

The Gibbs phase rule gives the number of degrees of freedom F (variance) of a chemical system at equilibrium:

$$F = 2 + C - P,$$

where

- C is the number of chemical components,
- P is the number of phases in the system,
- the number 2 corresponds to temperature T and pressure P as variables.

If only one of T or P is allowed to vary (e.g. phase diagrams at constant pressure), then the number of degrees of freedom is reduced by one.

3.1.7 Material design and property tailoring: Fe–C

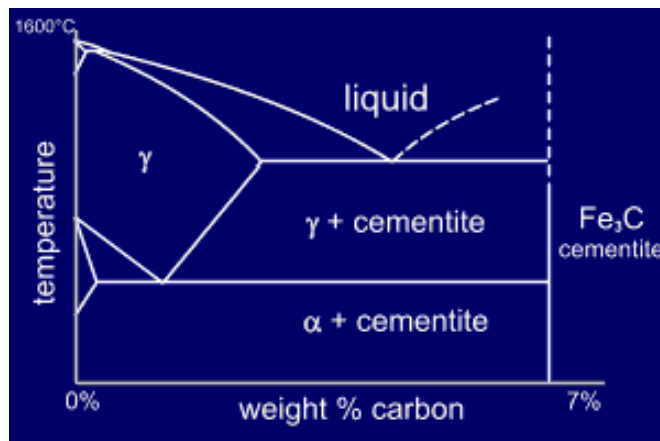


Figure 3.12: Fe–C phase diagram.

- γ phase is called **austenite**. Austenite is a high-temperature phase with FCC (face-centred cubic) structure.
- α phase is called **ferrite**. Ferrite is common in steels and has BCC (body-centred cubic) structure.
- Fe_3C is called **cementite**.
- The “eutectoid-like” mixture of α + cementite is called **pearlite**.

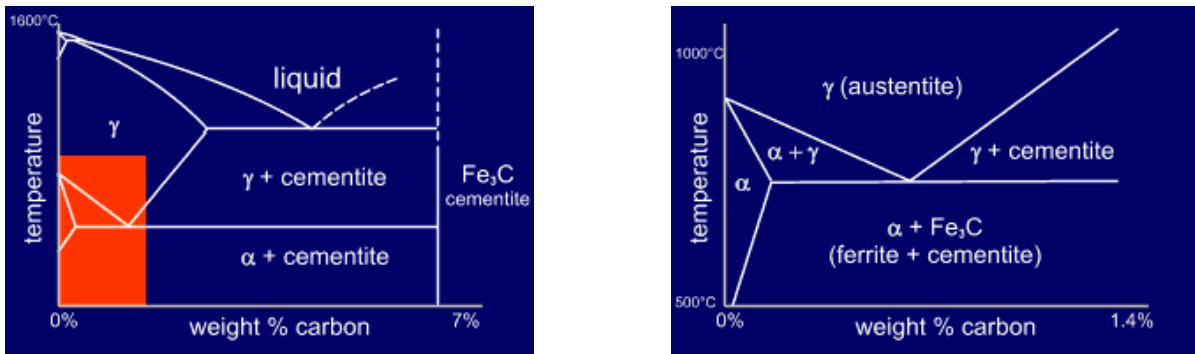


Figure 3.13: Solidification and transformation sequences in steels.

- The eutectoid composition is $Fe-0.83 \text{ wt\% C}$.
- At this composition, high-temperature austenite undergoes the eutectoid reaction at 723°C :



- Ferrite and cementite grow cooperatively as a lamellar mixture: **pearlite**.

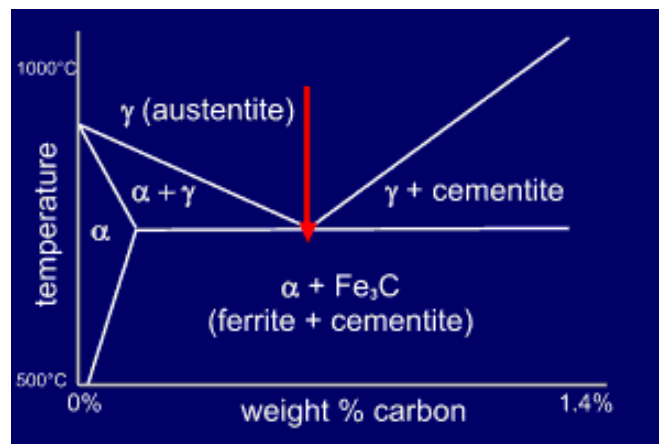


Figure 3.14: Pearlite and microstructural evolution in steels.

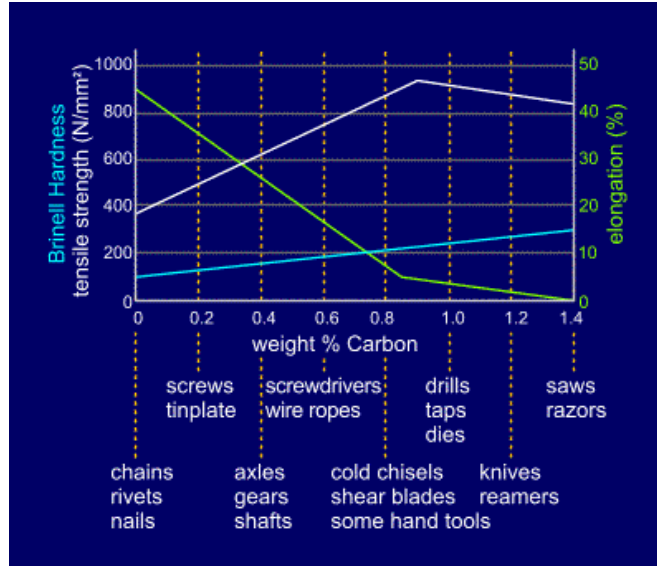


Figure 3.15: Mechanical properties vs carbon content in steels.

The microstructure of steels varies considerably with carbon content. Increasing carbon content increases the amount of hard, brittle cementite, which changes the engineering properties. For example, strength increases with carbon content up to the eutectoid composition, but then starts to drop as a grain-boundary network of brittle cementite forms.

3.2 ZrO_2 and Yttria-stabilised zirconia (YSZ)

3.2.1 Phases of zirconia

Phase	Structure	Stable Temperature Range	Notes
$m\text{-}ZrO_2$	Monoclinic	Room T $\approx$ 1273 K	Stable at low temperatures, large volume, brittle.
$t\text{-}ZrO_2$	Tetragonal	$\approx$ 1273–2650 K	Metastable at room temperature, transforms to monoclinic with volume change.
$c\text{-}ZrO_2$	Cubic (fluorite)	$>\approx$ 2650 K	Stable only at very high temperatures.

Table 3.1: Phases of zirconia and their stability ranges.

Pure ZrO_2 exists in monoclinic, tetragonal, and cubic forms depending on temperature. The cubic and tetragonal phases are only stable at high temperatures. By adding aliovalent oxides such as Y_2O_3 , CaO , or MgO , oxygen vacancies are introduced, lowering the free energy of the cubic phase and stabilising it at lower temperatures. This concept of stabilised zirconia is fundamental in advanced ceramics, as it allows the use of ZrO_2 -based materials at room temperature without phase transformation or cracking.

Why Y_2O_3 stabilises cubic ZrO_2 . Introducing Y^{3+} on Zr^{4+} sites creates oxygen vacancies for charge neutrality. The cubic fluorite structure accommodates vacancies with minimal elastic penalty, lowering its Gibbs free energy relative to the tetragonal and monoclinic forms. Thus vacancy formation shifts the stability range of the cubic phase down to room temperature.

3.2.2 Yttria-stabilised zirconia (ZrO_2) $_{1-x}$ ($YO_{1.5}$) $_x$

TZP – Tetragonal Zirconia Polycrystal

Submicron grains with tetragonal symmetry. High strength and fracture toughness:

$$\text{Bend strength: } 800\text{--}1200 \text{ MPa}, \quad K_{IC} = 6\text{--}8 \text{ MPa m}^{1/2}.$$

PSZ – Partially Stabilised Zirconia

Cubic matrix with fine tetragonal precipitates ($< 1 \mu m$). High fracture toughness:

$$K_{IC} = 6\text{--}12 \text{ MPa m}^{1/2}.$$

YSZ – Fully stabilised zirconia

Cubic fluorite structure. High oxygen ion conductivity:

$$\sigma \approx 0.1 \text{ S cm}^{-1} \text{ at } 1000^\circ\text{C}.$$

Applications: structural ceramics, tooth-bridges, oxygen sensors, solid oxide fuel cells.

Process of producing PSZ with about 8 mol% $YO_{1.5}$

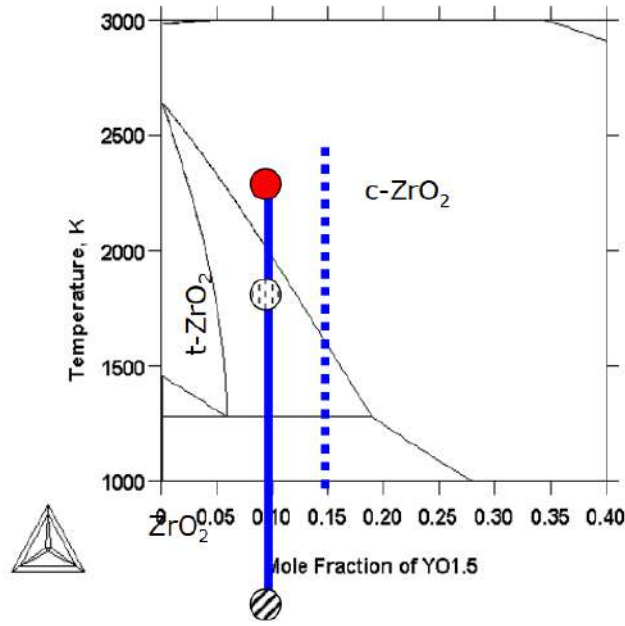


Figure 3.16: Schematic ZrO_2 - $YO_{1.5}$ phase diagram and PSZ processing path.

- At the red point, the material is sintered in the cubic single-phase field.
- Subsequent dissolution-annealing in the $t + c$ two-phase region produces tetragonal precipitates; precipitates remain below the critical size for $t \rightarrow m$ transformation.
- Cooling to room temperature: the remaining cubic phase does not transform.

3.2.3 Modelling stoichiometric compounds

Stoichiometric ZrO_2

$$G(ZrO_2) = \overset{\circ}{G}_{Zr^{4+};O^{2-}} - H_{Zr}^{SER} - 2H_O^{SER} = A + BT + CT \ln T + DT^2 + ET^{-1},$$

where

- $-H_{Zr}^{SER} - 2H_O^{SER}$: reference enthalpies for the selected element reference (SER),
- A : enthalpy term (enthalpy of formation / Gibbs energy of formation),
- $BT + CT \ln T + DT^2 + ET^{-1}$: entropy and heat capacity contributions,
- B is related to entropy, and C, D, E are heat capacity constants.

Stoichiometric compound with magnetic transition: CoO

Magnetic contribution to the Gibbs energy:

$$\Delta G_m^{\text{MAG}} = RT \ln(\beta + 1) f(\tau),$$

where

- $\tau = T/T_c$,
- β is a parameter related to the total magnetic entropy,
- T_c is the critical temperature for magnetic ordering.

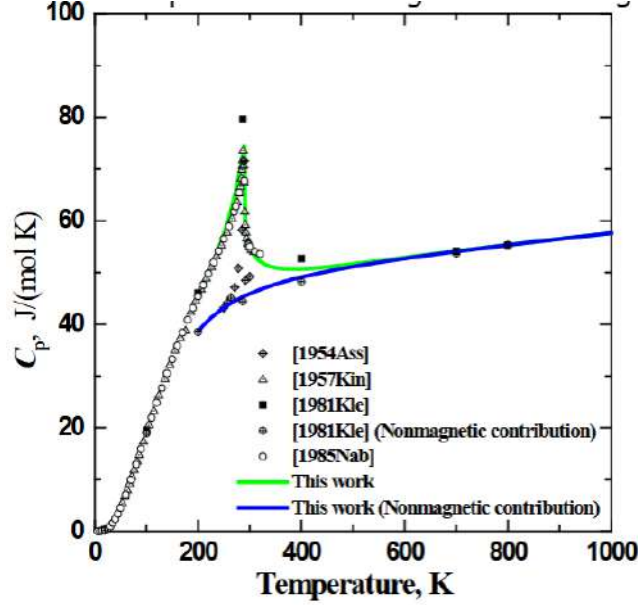


Figure 3.17: Magnetic contribution to G for CoO.

3.2.4 Modelling solid solutions: Y_2O_3 -stabilised cubic ZrO_2

Solution 1: substitutional model ($YO_{1.5}$, ZrO_2)

Gibbs energy of the c - ZrO_2 - $YO_{1.5}$ solution:

$$G_m = y_{YO_{1.5}} \hat{G}_{YO_{1.5}} + y_{ZrO_2} \hat{G}_{ZrO_2} + RT(y_{YO_{1.5}} \ln y_{YO_{1.5}} + y_{ZrO_2} \ln y_{ZrO_2}) + {}^E G_m,$$

with excess Gibbs energy

$${}^E G_m = y_{YO_{1.5}} y_{ZrO_2} \sum_{i=0}^n L_{YO_{1.5}, ZrO_2} (y_{YO_{1.5}} - y_{ZrO_2})^i.$$

Experimental data used for optimisation:

- Phase boundary data,
- ZrO_2 or $YO_{1.5}$ activity in c -YSZ.

Solution 2: sublattice model $(Y^{3+}, Zr^{4+})_1(O^{2-}, Va)_2$

$$G_m = y_{Y^{3+}} y_{O^{2-}}^2 {}^\circ G_{Y^{3+}, O^{2-}} + y_{Zr^{4+}} y_{O^{2-}}^2 {}^\circ G_{Zr^{4+}, O^{2-}} + y_{Y^{3+}} y_{Va} {}^\circ G_{Y^{3+}, Va} + y_{Zr^{4+}} y_{Va} {}^\circ G_{Zr^{4+}, Va} \\ + RT \left[y_{Y^{3+}} \ln y_{Y^{3+}} + y_{Zr^{4+}} \ln y_{Zr^{4+}} + 2 y_{O^{2-}} \ln y_{O^{2-}} + y_{Va} \ln y_{Va} \right] + {}^E G_m,$$

with

$${}^E G_m = y_{Y^{3+}} y_{Zr^{4+}} y_{O^{2-}} \sum_{i=0}^n {}^i L_{Y^{3+}, Zr^{4+}, O^{2-}} (y_{Y^{3+}} - y_{Zr^{4+}})^i \\ + y_{Y^{3+}} y_{Zr^{4+}} y_{Va} \sum_{i=0}^n {}^i L_{Y^{3+}, Zr^{4+}, Va} (y_{Y^{3+}} - y_{Zr^{4+}})^i.$$

Again, optimisation uses:

- Phase boundary data,
- ZrO_2 or $YO_{1.5}$ activity in c -YSZ.

3.3 Experimental determination of phase relations in oxide systems

Techniques: XRD, SEM, TEM, EDX, DTA, TG, dilatometry, electrical conductivity, infrared spectroscopy, etc.

Difficulties: sluggish reaction kinetics, metastable phases, inhomogeneity, etc.

3.3.1 Determination of the eutectic point

The eutectic point in the CeO_2 – CoO system was determined experimentally using **Differential Thermal Analysis (DTA)**. A sample with 20 mol% CeO_2 and 80 mol% CoO was heated and cooled at a constant rate while the heat flow was monitored. A sharp thermal peak on the DTA curve indicates the eutectic reaction, corresponding to the temperature where liquid transforms into two solid phases simultaneously.

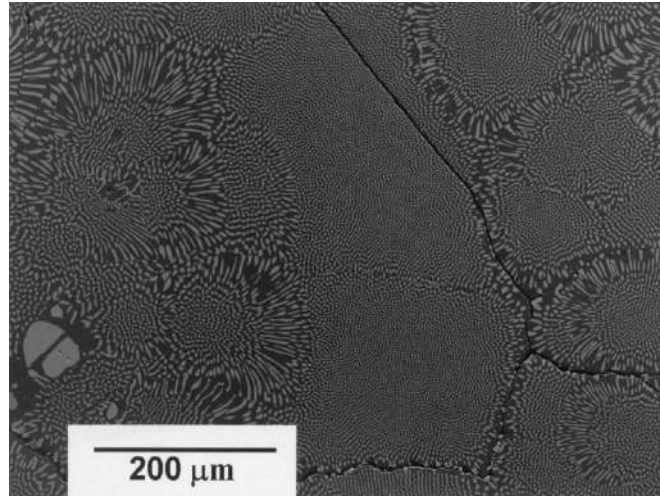


Figure 3.18: DTA signal and eutectic microstructure in CeO_2 – CoO .

The micrograph shows a fine, lamellar two-phase microstructure characteristic of eutectic solidification. Thus the eutectic temperature and composition are identified:

- by the distinct thermal event in the DTA curve;
- by confirmation from the lamellar eutectic microstructure at the eutectic composition.

3.3.2 Determination of the sub-solidus

The **sub-solidus region** is that part of the phase diagram below the solidus line, where only solid phases coexist (no liquid). It is determined not from melting behaviour but by analysing quenched microstructures and measuring compositions of coexisting solid phases.

In the CeO_2 – CoO example, a 20 mol% CeO_2 –80 mol% CoO sample sintered at 1580°C showed two distinct solid solutions: a CeO_2 -rich and a CoO -rich phase. A WDX line scan across the interface revealed

a sharp compositional change, defining the **solid solubility limits** and hence the sub-solidus phase boundary.

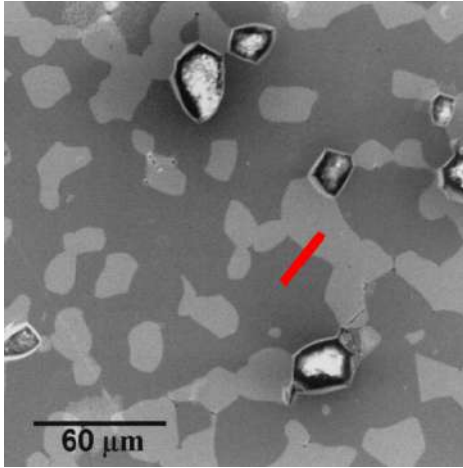


Figure 3.19: SEM micrograph of CeO₂–CoO (20 mol% CeO₂–80 mol% CoO) sintered at 1580°C for 40 h and quenched.

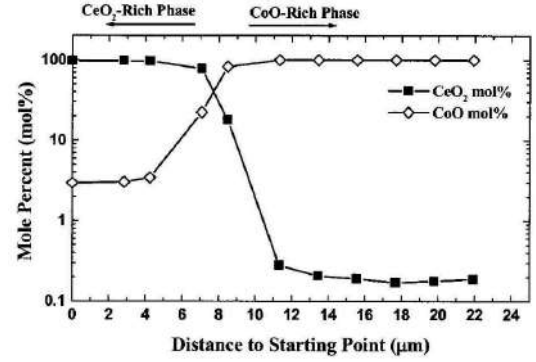


Figure 3.20: WDX line scan across the phase boundary giving the solubility limits.

3.4 CALPHAD: CALculation of PHase Diagrams

Phase diagrams correspond to the equilibrium state that minimises the total Gibbs free energy $G(T, P, x_i)$ at each T , P , and composition.

3.4.1 Principle in binary systems

For a binary A–B system, each phase has a Gibbs energy $G^\alpha(x_B, T)$, $G^\beta(x_B, T)$, $G^L(x_B, T)$, etc. The stable phase at a given x_B is the one with the lowest G .

When two phases coexist, their equilibrium compositions are given by the endpoints of a **common tangent** between their $G(x_B)$ curves:

- equal slopes $\Rightarrow$ equal chemical potentials:

$$\left(\frac{\partial G^\alpha}{\partial x_B} \right)_T = \left(\frac{\partial G^\beta}{\partial x_B} \right)_T ;$$

- the tangent touches both curves at their equilibrium compositions.

Repeating this at different temperatures yields the liquidus, solidus and other phase boundaries.

Why common tangents give equilibrium compositions. The slope of the Gibbs energy curve is the chemical potential:

$$\mu_B = \left. \frac{\partial G}{\partial x_B} \right|_T .$$

Two coexisting phases must satisfy $\mu_B^\alpha = \mu_B^\beta$ and $\mu_A^\alpha = \mu_A^\beta$. A common tangent has identical slopes at the two touching points, enforcing equality of chemical potentials and therefore thermodynamic equilibrium.

3.4.2 CALPHAD method

- Modern phase diagrams are computed using the **CALPHAD** approach.
- Each phase's $G(T, x_i)$ is modelled using experimental and thermodynamic data, typically with polynomial expansions (e.g. Redlich–Kister).

- The system's equilibrium is found by minimising the total Gibbs energy over all phases and compositions.
- Software such as **Thermo-Calc**, **FactSage**, **Pandat** performs these calculations using thermodynamic databases.

3.5 Thermo-Calc

Thermo-Calc is a CALPHAD-based software used to calculate thermodynamic and phase equilibria in multi-component systems.

3.5.1 How Thermo-Calc works

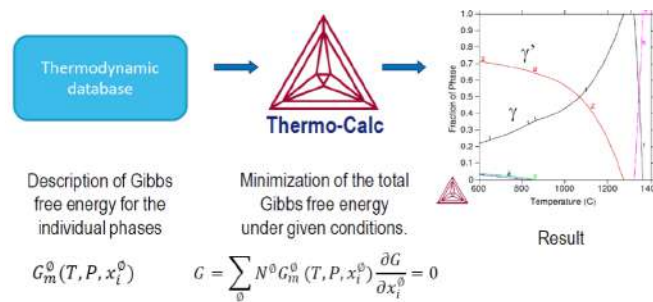


Figure 3.21: Thermo-Calc concept: databases + Gibbs energy models → equilibria.

Thermo-Calc uses thermodynamic databases that contain assessed Gibbs energy descriptions of phases. These are based on the CALPHAD approach.

3.5.2 What can be calculated using Thermo-Calc

Thermo-Calc is flexible and can be used to calculate, for example:

- Stable and metastable heterogeneous phase equilibria.
- Amount and composition of phases.
- Transformation temperatures: liquidus, solidus, solvus.
- Solubility limits and partition coefficients.
- Driving forces for phase transformations.
- Phase diagrams (binary, ternary, isothermal, isoplethal, etc.).
- Molar volume, density, thermal expansion.
- Scheil–Gulliver (non-equilibrium) solidification simulations.
- Thermochemical data:
 - Enthalpies,
 - Heat capacity,
 - Activities, etc.
- Thermodynamic properties of chemical reactions.

Different types of calculations allow variation of one or more system variables.

Single point equilibrium

Single-point calculations are for one specific set of conditions and output the phases present, their amounts and compositions. Information about unstable phases (e.g. driving force for precipitation) can also be obtained.

Output from POLY-3
Tue Aug 08 2006 11:41:39

Database: TCFe3

Conditions:
T=1000, P=1E5, N=1, W(C)=2E-2, W(CR)=0.1, W(MN)=3E-2, W(NI)=1E-2
DEGREES OF FREEDOM 0

Temperature 1000K (727C, 1340F), Pressure 1.000000E+05
Number of moles of components 1.00000E+00, Mass 5.16905E+01
Total Gibbs energy -4.42393E+04, Enthalpy 2.56582E+04, Volume 5.22429E-06

Component	Moles	W-Fraction	Activity	Potential	Ref.State
C	8.6072E-02	2.0000E-02	8.9724E-02	-2.0046E+04	SER
CR	9.9412E-02	1.0000E-01	4.4290E-04	-6.4206E+04	SER
FE	7.7748E-01	8.4000E-01	6.0347E-03	-4.2489E+04	SER
MN	2.8227E-02	3.0000E-02	4.6517E-05	-8.2943E+04	SER
NI	8.8074E-03	1.0000E-02	3.3138E-05	-8.5758E+04	SER

FCC_A1#1 STATUS ENTERED Driving force 0.0000E+00
Number of moles 7.5868E-01, Mass 4.1735E+01
Mass fractions:
FE 9.44835E-01 CR 1.41596E-02 C 3.93605E-03
MN 2.47653E-02 NI 1.23038E-02

FCC_A1#2 STATUS ENTERED Driving force 0.0000E+00
Number of moles 0.0000E+00, Mass 0.0000E+00
Mass fractions:
FE 9.44835E-01 CR 1.41596E-02 C 3.93605E-03
MN 2.47653E-02 NI 1.23038E-02

M7C3#1 STATUS ENTERED Driving force 0.0000E+00
Number of moles 2.4122E-01, Mass 9.9559E+00
Mass fractions:
CR 4.59839E-01 C 8.73393E-02 NI 3.42660E-04
FE 4.00536E-01 MN 5.19435E-02

Figure 3.22: Example of single-point equilibrium output.

Step calculations

Step calculations vary one system variable (e.g. temperature or composition) and show how phases change with that variable.

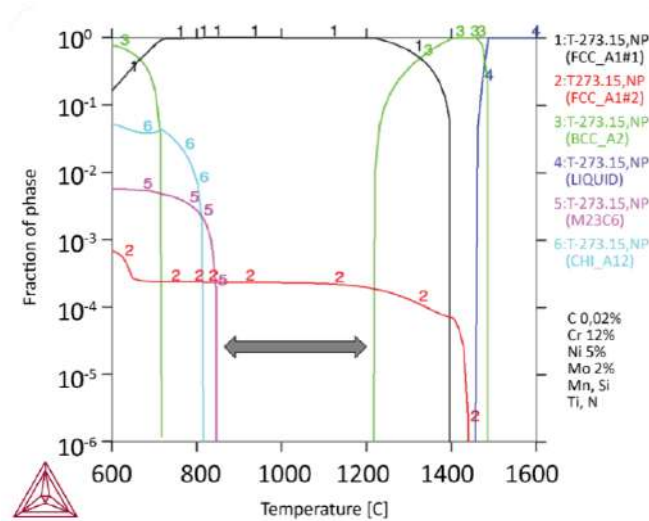


Figure 3.23: Step calculation with temperature as the varying variable.

Map calculations

Map calculations are used to compute binary and ternary phase diagrams and isoplethal/ isothermal sections.

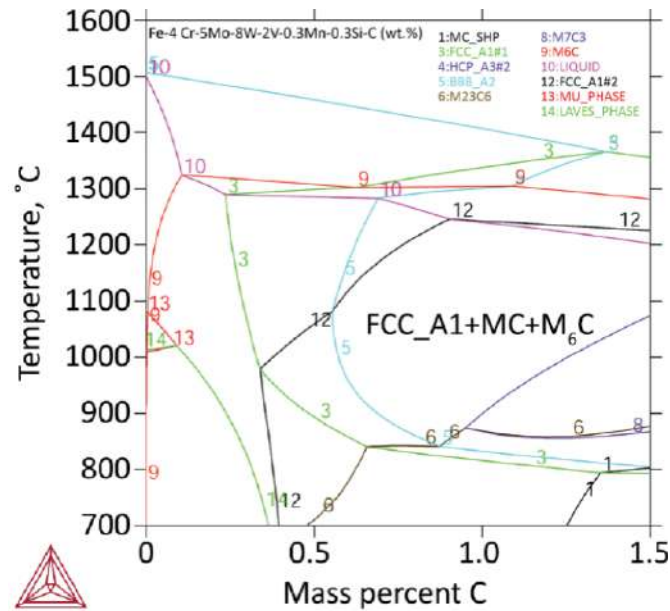


Figure 3.24: Example of an isopleth section for steel.

Scheil solidification simulations

Scheil calculations model non-equilibrium solidification, predicting fraction solid curves, incipient melting, microsegregation, etc.

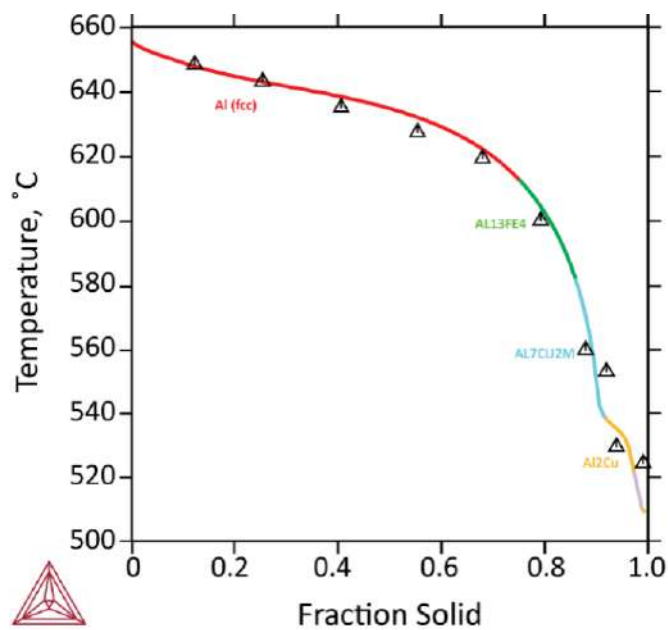


Figure 3.25: Scheil solidification simulation example.

Pourbaix diagrams

Using the console mode, Pourbaix (potential-pH) diagrams can also be calculated for metals and alloys in aqueous environments.

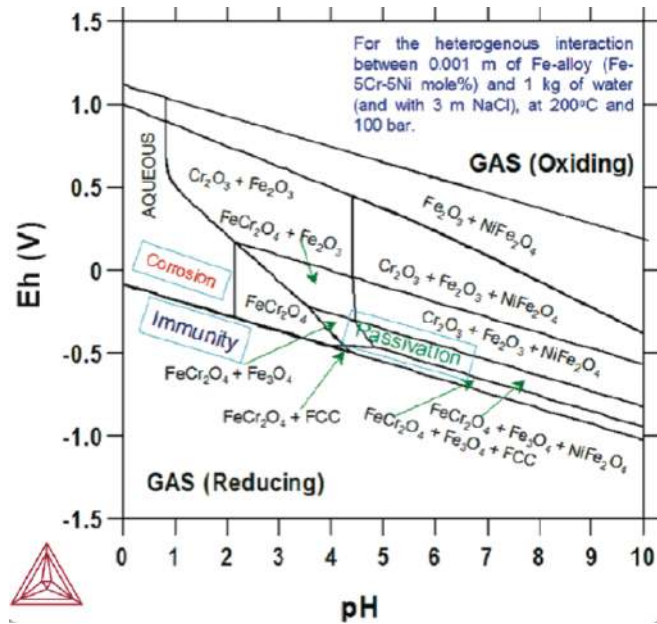


Figure 3.26: Example of a Pourbaix diagram.

3.5.3 Thermo-Calc product structure

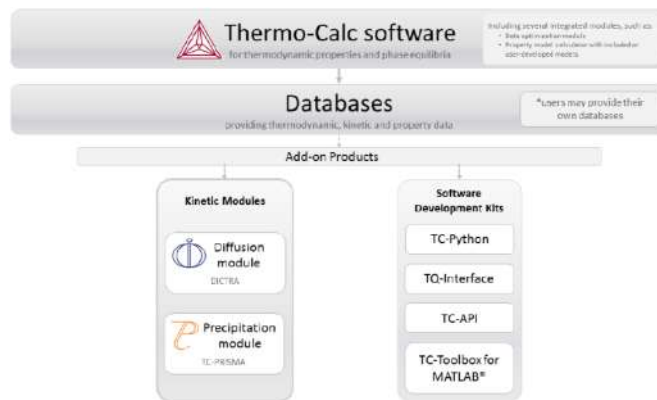


Figure 3.27: Thermo-Calc product structure and module overview.

Thermodynamic databases

Over 30 thermodynamic databases are available:

- Steels and Fe-alloys
- Nickel-base superalloys
- Aluminium/Titanium/Magnesium-base alloys
- Gases, inorganic/organic substances and general alloys
- Slags, metallic liquids, molten salts
- Ceramic systems and hard materials
- Semiconductors and solder alloys
- Noble metal alloys
- Materials processing and environmental aspects
- Aqueous solutions, corrosion, hydrometallurgical systems

- Minerals and geochemical/environmental processes
- Nuclear materials and nuclear fuel/waste processing

Diffusion module (DICTRA)

DICTRA is used for diffusion-controlled transformations in multicomponent systems:

- Homogenisation
- Carburising and decarburising
- Nitriding and carbonitriding
- Microsegregation during solidification
- Coarsening/dissolution of precipitates
- Diffusion-controlled phase transformation kinetics
- Inter-diffusion between coating and substrates
- Post-weld heat treatment
- Mobility databases for Fe-, Al-, Ni-, Si-, Ti-based alloys

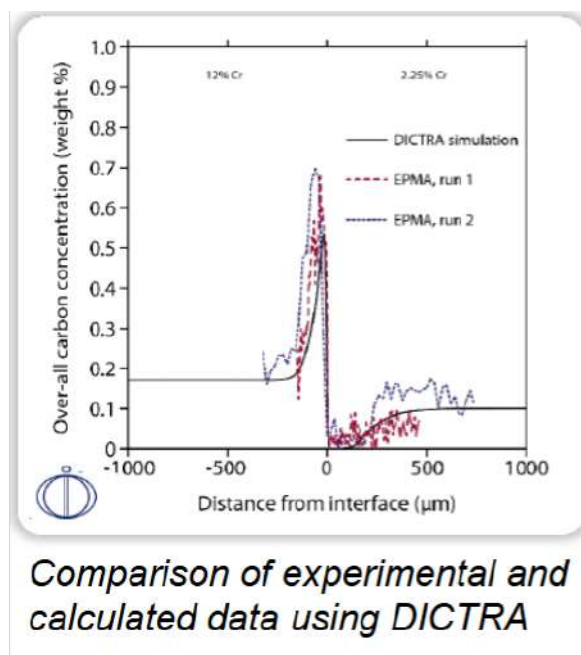


Figure 3.28: Example of a DICTRA diffusion simulation.

Precipitation module (TC-PRISMA)

TC-PRISMA simulates precipitation kinetics in multicomponent systems:

- Concurrent nucleation, growth/dissolution and coarsening of precipitates
- Temporal evolution of particle size distribution
- Average particle radius and number density
- Volume fraction and composition of precipitates
- Nucleation and coarsening rates

- Time–Temperature–Precipitation (TTP) diagrams
- Uses thermodynamic and kinetic data from Thermo-Calc and DICTRA

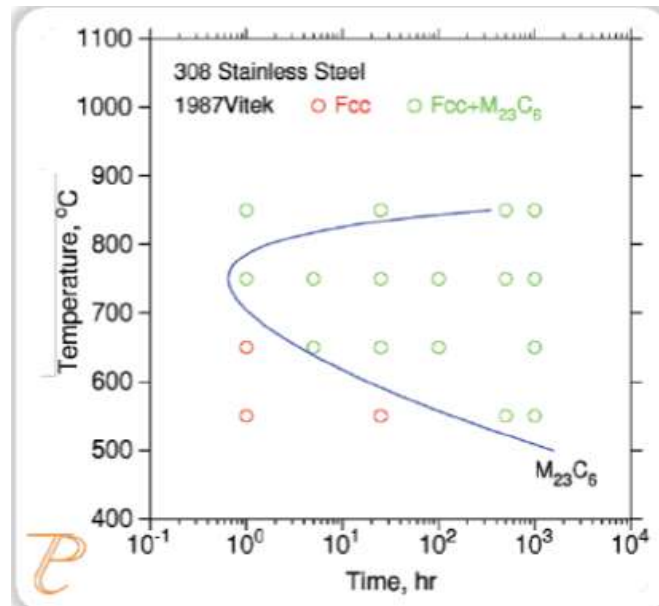


Figure 3.29: Precipitation simulation example with TC-PRISMA.

Software development kits

Interfaces to couple in-house software with Thermo-Calc:

- **TC-Python:** Python-based interface to Thermo-Calc.
- **TQ-Interface:** Fortran library for computationally intensive tasks.
- **TC-API:** C library suitable for C/C++/Java/VB applications.
- **TC-Toolbox for MATLAB:** Access to Thermo-Calc from MATLAB.

3.5.4 What can Thermo-Calc be used for (in the materials life cycle)?

Thermodynamics drives many processes during the materials life cycle: alloy design, solidification, heat treatment, precipitation, corrosion, etc. Thermo-Calc tools can be used to:

- Design alloys and optimise precipitate fractions.
- Avoid deleterious phases.
- Understand microsegregation during solidification.
- Identify precipitate-free forging windows and optimal heat treatments.

3.5.5 Practical Thermo-Calc workflow

- Select database (e.g. TCFE, TCOX).
- Define components (elements or compound components).
- Choose calculator (equilibrium, binary/ternary, etc.).
- Set conditions (T, P, composition; axis variables for diagrams).
- Run calculation (Gibbs energy minimisation).

- Inspect results: phase fractions, compositions, boundaries.
- Adjust settings (metastable phases, disabled phases, axis ranges) if needed.

3.5.6 Example calculations performed in the course

Examples included:

- Single-point equilibrium in Fe–C (e.g. Fe–0.1 mass% C at $T = 1000$ K).
- Temperature-stepping at fixed composition (phase-fraction vs T curves).
- Full Fe–C phase diagram with the Binary Calculator and via Equilibrium Calculator.
- A pseudo-binary oxide diagram treating Al_2O_3 and MgO as components.

These illustrate how CALPHAD concepts are implemented in Thermo-Calc.

3.6 Solid oxide cells (SOCs): materials and modelling

3.6.1 Introduction to SOCs

Solid oxide fuel cells (SOFCs) and solid oxide electrolysis cells (SOECs) operate with solid ceramic electrolytes at high temperature.

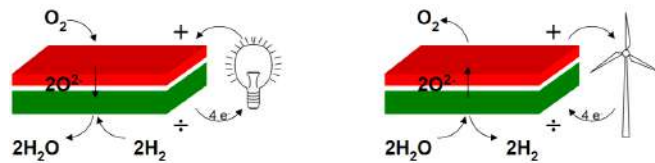


Figure 3.30: Principle of SOFC (left) and SOEC (right) with $\text{H}_2/\text{H}_2\text{O}$.

- SOFC: consumes H_2 and produces electricity.
- SOEC: consumes H_2O and electricity, produces H_2 .

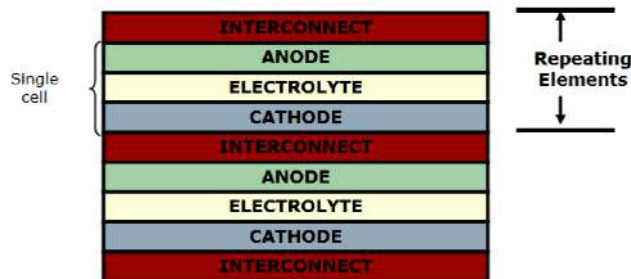


Figure 3.31: Stack of SOC cells.

- Single cell voltage under load < 1 V (H_2/O_2).
- Stack voltage determined by the number of cells.
- Stack current determined by current density and cell area.

3.6.2 Cell components (2G and 2.5G SOCs)

Example: anode-supported *Ni/YSZ*–*YSZ*–*LSM*/*YSZ* cell.

- Cathode current collector (LSM): $\approx 40 \mu\text{m}$.
- Active cathode (LSM/*YSZ*): $\approx 20 \mu\text{m}$.
- Electrolyte (*YSZ*): $\approx 10 \mu\text{m}$.
- Active anode (*NiO*/*YSZ*): $\approx 15 \mu\text{m}$.
- Anode support (*NiO*/*YSZ*): $\approx 250 \mu\text{m}$.

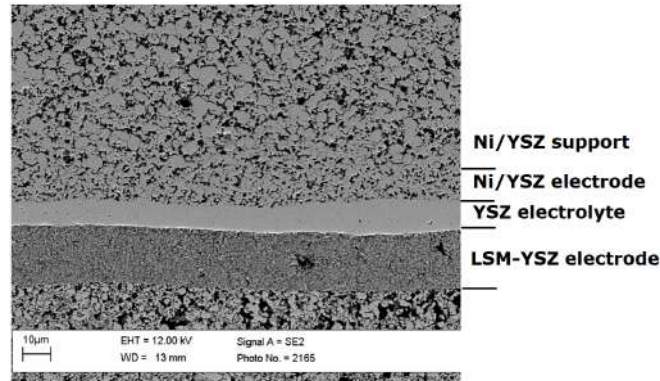


Figure 3.32: Cell components – 2G SOC.

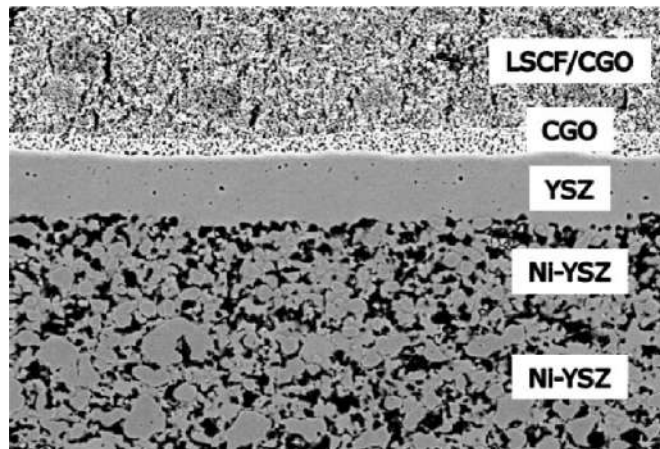


Figure 3.33: Cell components – 2.5G SOC.

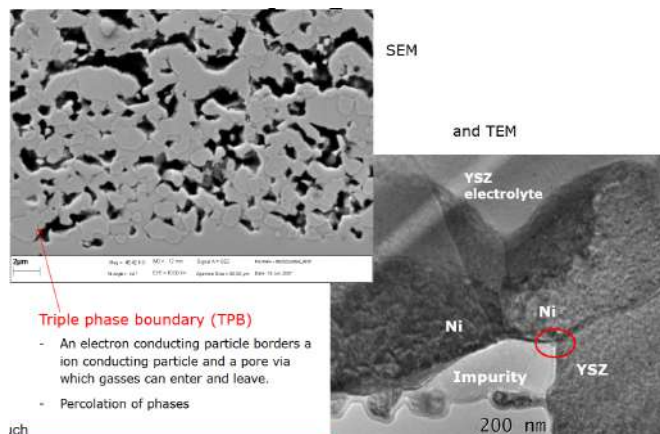


Figure 3.34: Zoom in on the Ni/*YSZ* hydrogen electrode.

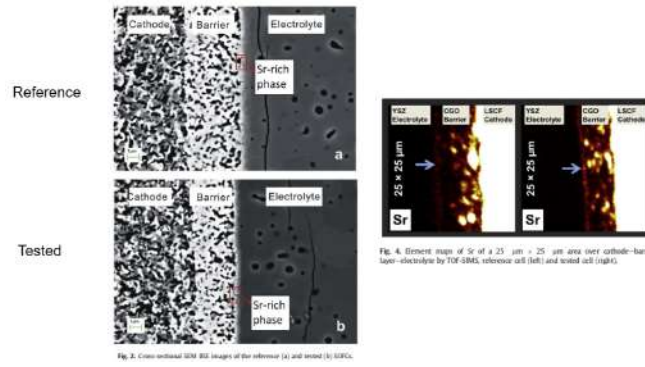


Figure 3.35: Zoom in on the LSCF/CGO oxygen electrode.

3.6.3 CALPHAD modelling of SOC materials

- Development of relevant thermodynamic databases.
- Modelling of oxygen electrode/electrolyte interfaces.
- Modelling of Ni/YSZ electrode degradation.
- Nano-phase diagrams.

Database development:

- ETH + DTU groups: $La-Sr-Mn-Cr-Co-Fe-O$, $Y-Zr-O$, $Co-Ce-Gd-O$.
- MPI group: $Zr-Ln-O$ (Ln = lanthanides), $La-Sr-Ga-Mg-Ni-O$.

Modelling the oxygen electrode / electrolyte interface

Zirconate formation at the LSM-YSZ interface decreases SOFC performance and is partly responsible for long-term degradation.

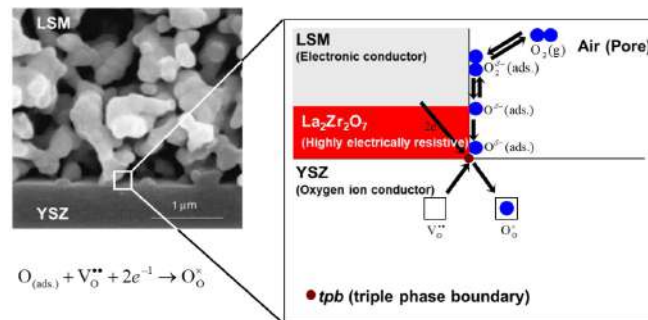


Figure 3.36: Phase formation and degradation at the LSM-YSZ interface.

Thermodynamic modelling of La-Mn-Sr-Y-Zr-O

Strategy: build the database and models gradually:

- binaries $\rightarrow$ ternaries $\rightarrow$ quaternaries $\rightarrow$ full La-Mn-Sr-Y-Zr-O system.

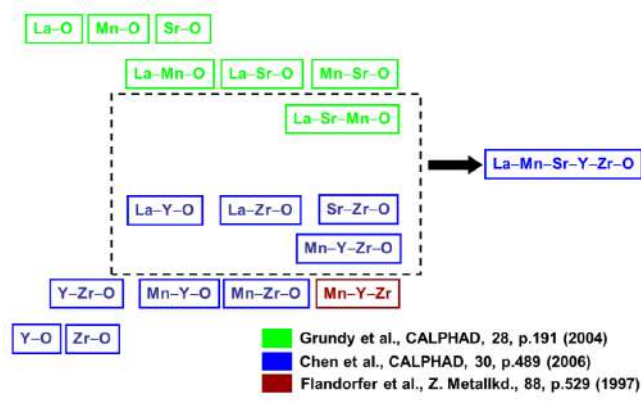
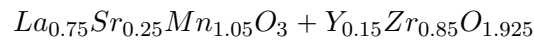


Figure 3.37: Building up a large oxide system from simpler subsystems.

Phase equilibrium example

Equilibrium phase fractions for



at 1273 K:

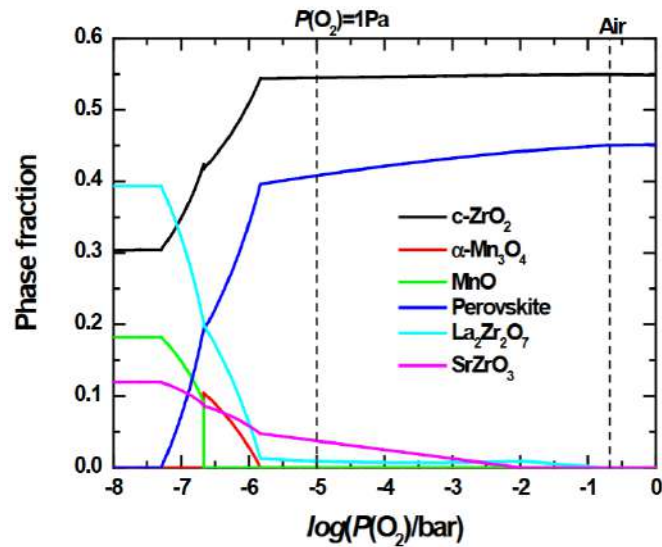


Figure 3.38: Equilibrium phases for a typical cathode/electrolyte mixture.

Correlating with operating conditions

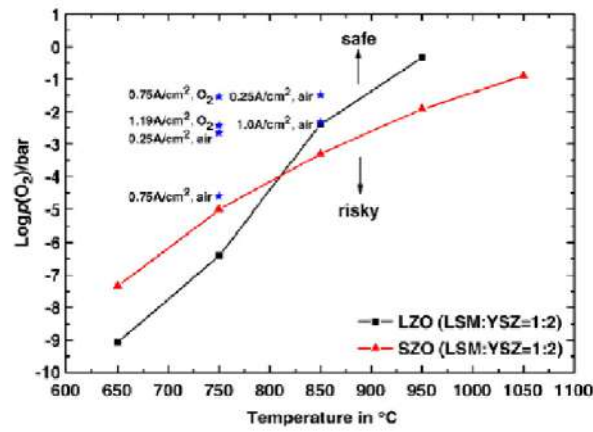
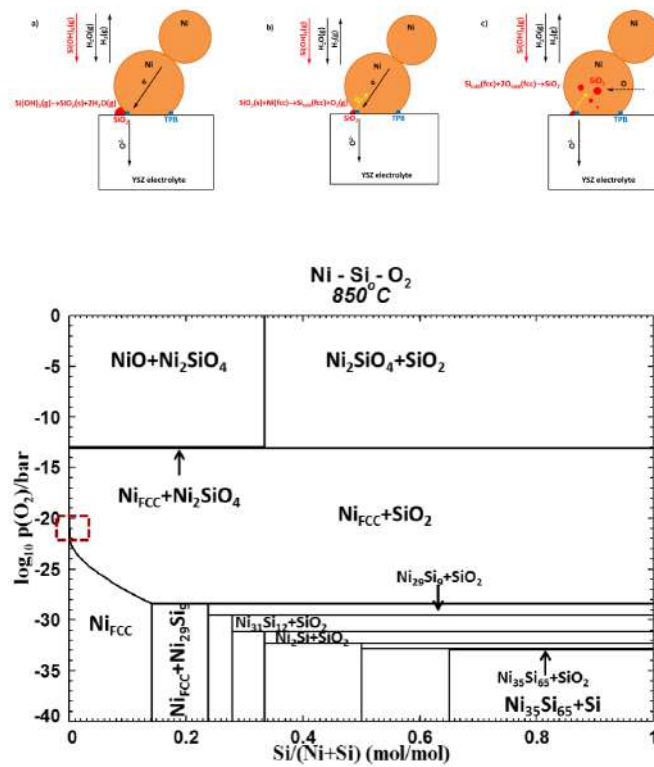


Fig. 6. Critical $p(\text{O}_2)$ for the zirconate formation as a function of temperature for the present LSM-YSZ system. Star symbols show the cathode interface ($T, p(\text{O}_2)$) conditions for the tested cells in Table 4.

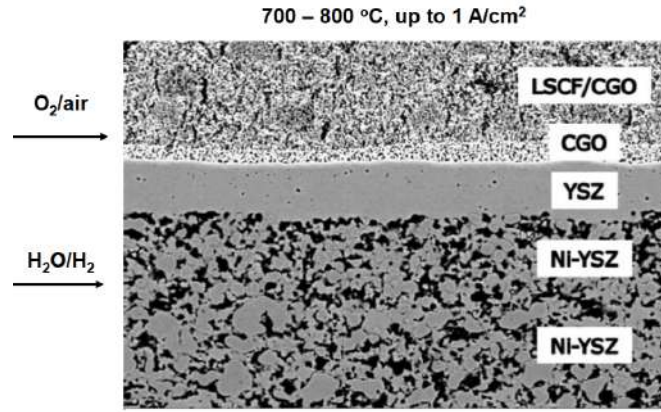
Figure 3.39: Relation between calculated phase equilibria and SOFC operating conditions.

Modelling Ni/YSZ electrode degradation

Schematic: SiO_2 deposition at Ni-YSZ triple-phase boundaries (TPBs) and formation of Si/SiO₂ inclusions in Ni particles:



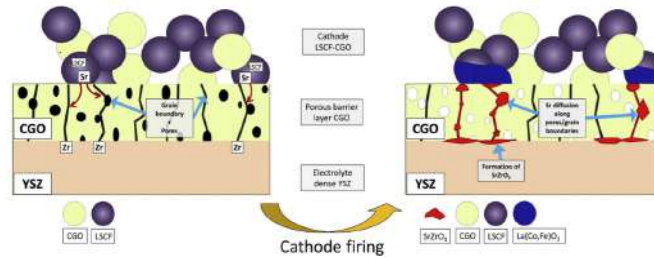
Challenges in 2.5G cells



Issues include material stability, chemical compatibility, inter-diffusion, interface reactions, coarsening, migration and poisoning.

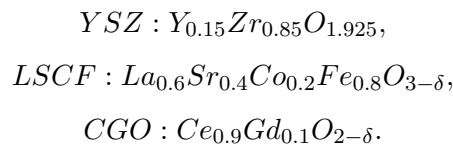
3.7 Project 3: CGO–YSZ interdiffusion in SOCs

3.7.1 Background

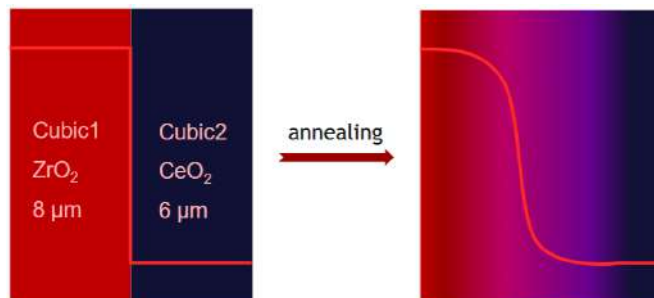


- Interdiffusion between YSZ electrolyte and CGO barrier layer.
- Interdiffusion between LSCF cathode and CGO barrier layer.
- Significant Sr diffusion to the CGO–YSZ interface.
- $SrZrO_3$ formation at the CGO–YSZ interface.

Compositions:



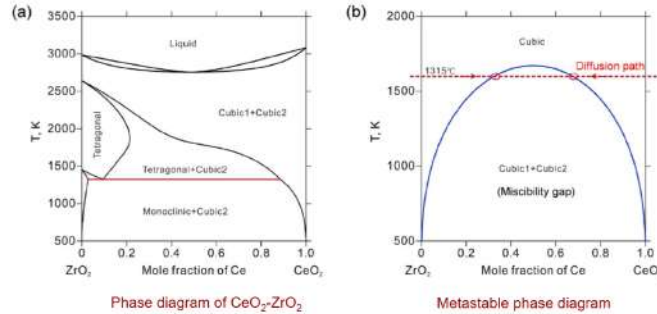
3.7.2 Model system



Simplifications

- Interdiffusion between cubic ZrO_2 and cubic CeO_2 only; the effect of Y and Gd is neglected.
- Consider *cation* diffusion only.
- Influence of pO_2 neglected.

3.7.3 Phase diagram information



Thermodynamics: solubility of CeO_2 in ZrO_2 is larger than that of ZrO_2 in CeO_2 .

Kinetics: diffusion depths are determined by mobility of Ce in ZrO_2 and Zr in CeO_2 .

3.7.4 Assessment of mobility data

To model diffusion in fluorite oxides with the CALPHAD mobility framework, cation mobilities are needed for each cation (Zr, Ce) in each fluorite phase.

Two fluorite phases are considered:

- **Cubic1:** ZrO_2 -rich fluorite phase.
- **Cubic2:** CeO_2 -rich fluorite phase.

Experimental diffusivity data exist only for Zr and Ce in ZrO_2 -rich fluorite (typically in YSZ). No measurements exist for cation diffusion in CeO_2 -rich fluorite. Thus the following assumption is made:

$$M(\text{Cubic1, Zr}) = M(\text{Cubic2, Zr}), \quad M(\text{Cubic1, Ce}) = M(\text{Cubic2, Ce}).$$

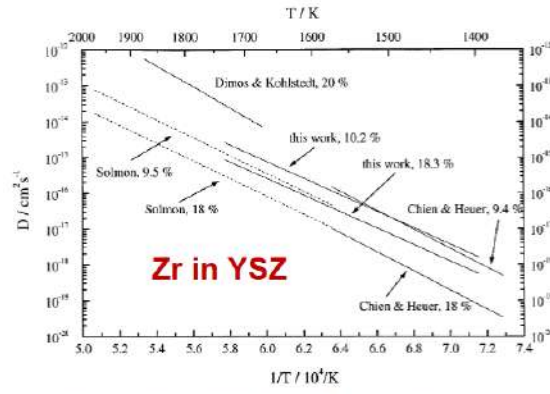
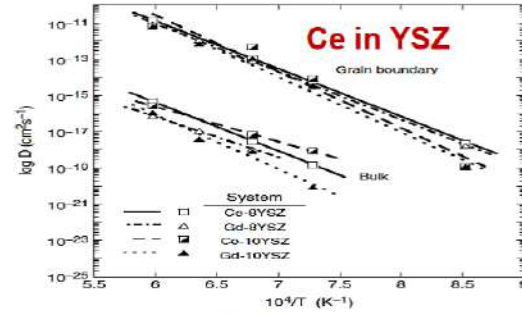


Fig. 1. Comparison of recent experimental data on cation diffusion in yttria stabilized zirconia (cf. Table I).



Comparison of diffusivities according to the yttria content in stabilized zirconia.

Figure 3.40: Arrhenius plots for cation diffusion used to parameterise mobility.

These Arrhenius-type diffusion data are used to parameterise cation mobilities in Cubic1; they are then assumed to apply also to Cubic2, enabling a complete mobility description.

3.7.5 Experimental validation

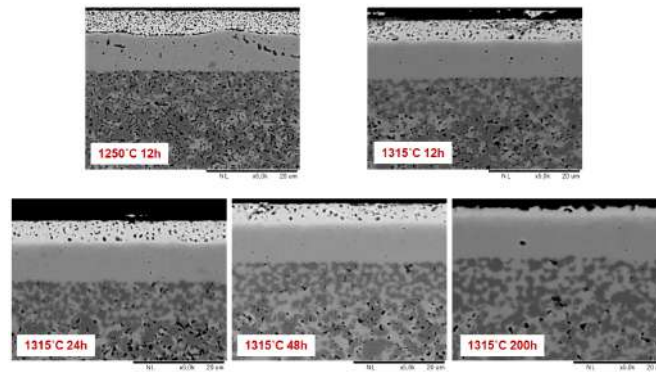


Figure 3.41: Cross-sectional SEM-BSE images of SOFC half-cells sintered at different conditions.

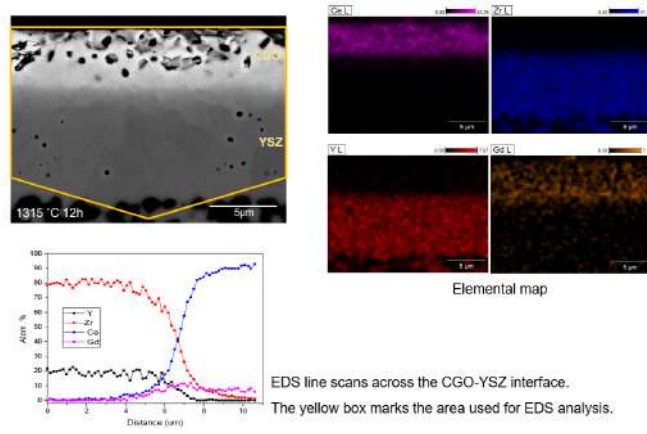


Figure 3.42: SEM-EDS analysis of tested half-SOFCs showing interdiffusion profiles.

3.7.6 Experiment vs simulation

Simulation of diffusion (composition) profiles:

- Grain size of CGO: 1 μm .
- Grain size of YSZ: 1 μm .

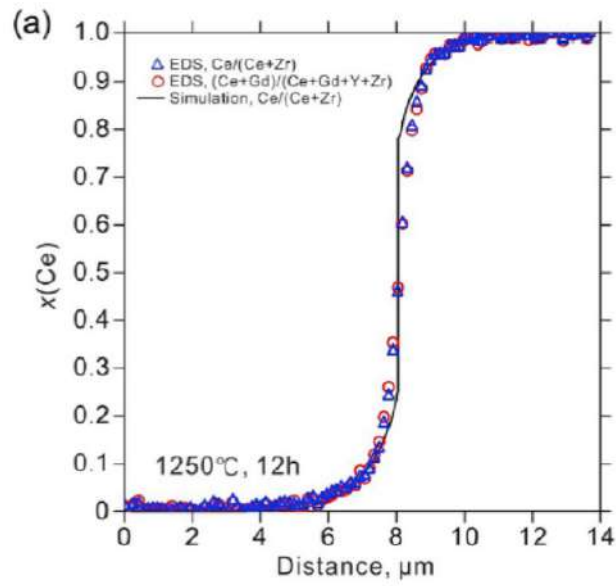


Figure 3.43: Simulated diffusion profiles vs experimental data.

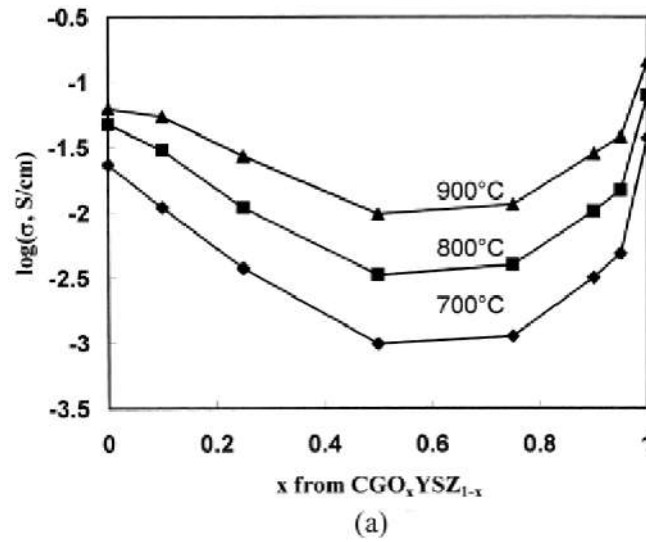


Figure 3.44: From composition profile to local conductivity.

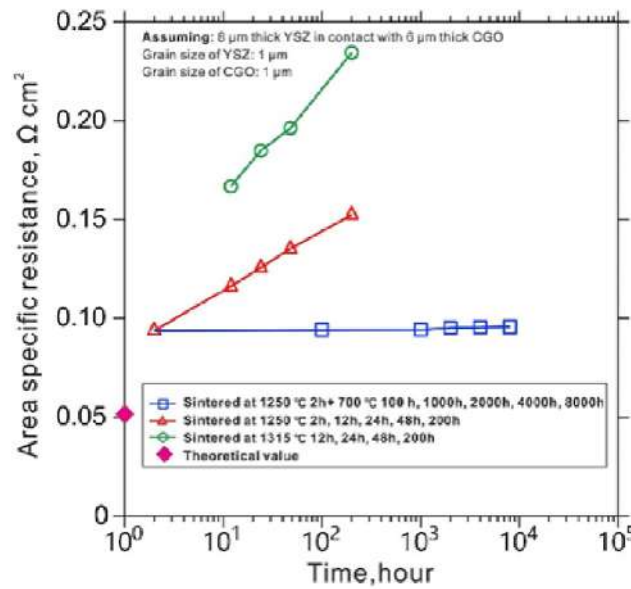
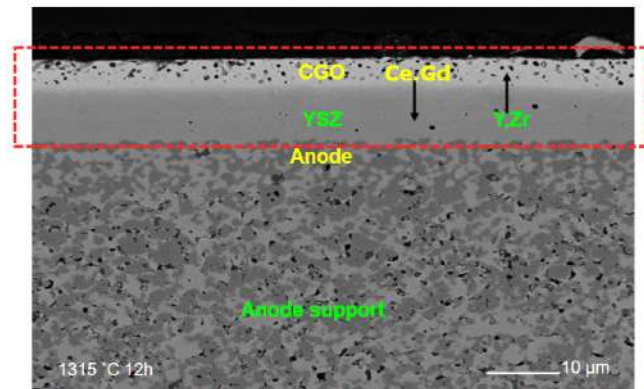


Figure 3.45: Prediction of area-specific resistance (ASR) from composition-dependent conductivity.



Cross-sectional SEM image of an as-fabricated SOC halfcell

Figure 3.46: CGO–YSZ interdiffusion and its impact on serial resistance.

Consequence: CGO–YSZ interdiffusion causes cell serial resistance (ASR) to increase.

Aim of the study: model CGO–YSZ interdiffusion during cell fabrication (1200–1300°C, 5–20 h) and during operation (700–800°C, up to years) and evaluate its influence on cell resistance increase.

Area Specific Resistance (ASR) of CGO–YSZ Bilayers

The local ionic conductivity of the fluorite phase depends strongly on the CeO_2 mole fraction x . An empirical fit (from 2019 Xu) relates the conductivity σ to x :

$$\log_{10} \sigma = 129.408x^6 - 343.667x^5 + 335.289x^4 - 142.568x^3 + 26.596x^2 - 4.893x - 1.629,$$

where σ is in S/cm.

Given a composition profile $x(l)$ across the electrolyte thickness L , the ohmic area-specific resistance is

$$\text{ASR}_{\text{ohm}} = \int_0^L \frac{1}{\sigma(l)} dl.$$

This expression is used together with diffusion simulations (e.g. DICTRA) to quantify how interdiffusion between CGO and YSZ affects cell performance.

3.8 Diffusion kinetics simulations

3.8.1 What is diffusion?

Diffusion is the transport of matter driven by gradients in chemical potential (Gibbs energy). It occurs:

- within a single phase (intra-phase diffusion),
- across phase boundaries (interface migration).

Diffusion controls the *rate* at which a system approaches equilibrium, even when the final equilibrium state is known.

3.8.2 Driving force for diffusion

- Diffusion always acts to reduce the Gibbs free energy G of the system.
- At equilibrium, G is minimised and no net diffusion occurs.

Example: in one case the Gibbs energy is already minimised; in another, Ni diffusion from FCC to an interface reduces G and causes interface motion.

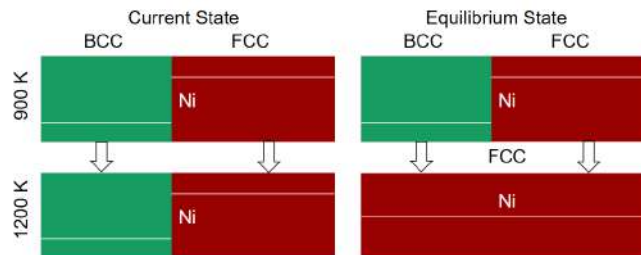


Figure 3.47: Driving force for diffusion illustrated by Gibbs energy change.

3.8.3 Analogy with heat transfer

Diffusion equations are analogous to heat conduction and charge transport:

$$\text{Heat flux: } q = -k\nabla T, \quad \text{Diffusion flux: } J = -D\nabla c.$$

The same structure appears: gradient of potential $\rightarrow$ flux $\rightarrow$ conservation law.

3.8.4 Fick's laws of diffusion

- Fick's 1st law (steady state):

$$J_B = -D_B \frac{\partial C_B}{\partial x}.$$

- Fick's 2nd law (transient):

$$\frac{\partial C_B}{\partial t} = D_B \frac{\partial^2 C_B}{\partial x^2}.$$

- If $D = D(C)$ depends on composition:

$$\frac{\partial C}{\partial t} = \frac{\partial}{\partial x} \left(D(C) \frac{\partial C}{\partial x} \right).$$

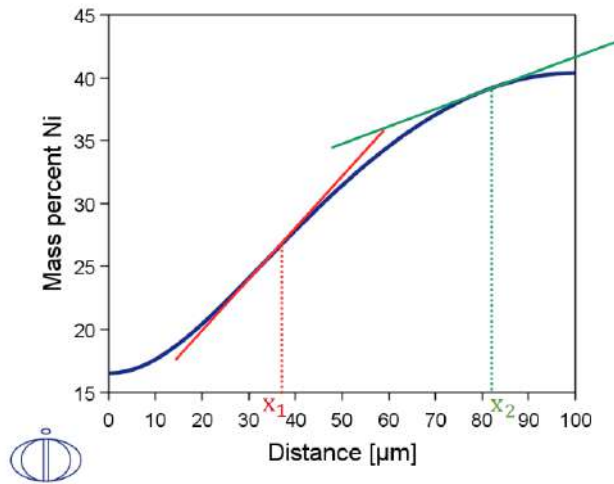


Figure 3.48: Concentration profile evolution according to Fick's 2nd law.

3.8.5 Atomistic mechanisms of diffusion

Diffusion coefficients often follow an Arrhenius relation:

$$D = D_0 \exp\left(-\frac{Q}{RT}\right),$$

where D_0 is a prefactor and Q the activation energy.

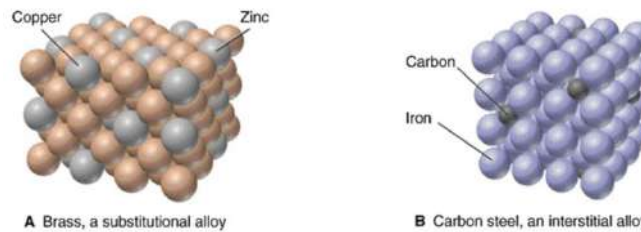


Figure 3.49: Substitutional vs interstitial diffusion mechanisms.

Atomistic Diffusion Mechanisms

Diffusion in solids occurs by atomic jumps, whose mechanism depends on the type of solute:

- **Substitutional diffusion (vacancy mechanism):** Atoms on lattice sites move by exchanging places with vacancies. This mechanism dominates in metallic alloys and requires thermal activation to create and move vacancies.

- **Interstitial diffusion:** Small atoms (e.g. H, C, N, O) diffuse through interstitial sites without needing vacancies. This mechanism is typically much faster than vacancy diffusion due to lower activation energies.

These microscopic mechanisms determine the diffusion coefficient D used in Fick's laws.

3.8.6 Numerical solution of Fick's 2nd law

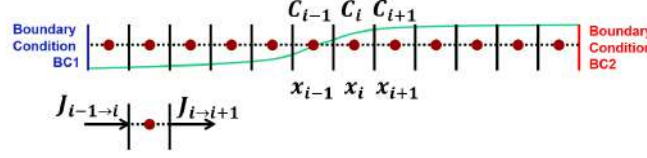


Figure 3.50: Discretisation of a 1D concentration profile.

We discretise concentration C_i^t on a 1D grid:

$$C_i^t \rightarrow C_{i-1}^t, C_i^t, C_{i+1}^t.$$

Finite-difference time stepping:

$$\frac{C_i^{t+\Delta t} - C_i^t}{\Delta t} = D_i^t \left(\frac{C_{i+1}^t - 2C_i^t + C_{i-1}^t}{(\Delta x)^2} \right) + \left(\frac{D_{i+1}^t - D_{i-1}^t}{2\Delta x} \right) \left(\frac{C_{i+1}^t - C_{i-1}^t}{2\Delta x} \right).$$

Boundary conditions

- **Dirichlet:** fixed composition $C(x, t) = C_0$.
- **Neumann:** fixed flux $J = -D\nabla C$ (e.g. zero-flux).

3.8.7 Multicomponent diffusion

For multicomponent systems:

$$\frac{\partial [c]}{\partial t} = \frac{\partial}{\partial x} \left([D] \frac{\partial [c]}{\partial x} \right),$$

with

$$[c] = \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_{n-1} \end{bmatrix}, \quad [D] = \begin{bmatrix} D_{1,1} & D_{1,2} & \cdots & D_{1,n-1} \\ D_{2,1} & D_{2,2} & \cdots & \vdots \\ \vdots & \vdots & \ddots & \vdots \\ D_{n-1,1} & \cdots & \cdots & D_{n-1,n-1} \end{bmatrix}.$$

Diffusion of one component can depend on gradients of all others.

3.8.8 Application: diffusion in $\text{ZrO}_2\text{--CeO}_2$

- Thermodynamics: CeO_2 has higher solubility in ZrO_2 than vice versa.
- Kinetics: diffusion depths governed by mobilities of Ce in ZrO_2 and Zr in CeO_2 .

Mobility assessment Two fluorite phases:

- Cubic1: ZrO_2 -rich fluorite,
- Cubic2: CeO_2 -rich fluorite.

Experimental diffusion data only exist for Cubic1 (e.g. YSZ). The assumption

$$M(\text{Cubic1}, \text{Zr}) = M(\text{Cubic2}, \text{Zr}), \quad M(\text{Cubic1}, \text{Ce}) = M(\text{Cubic2}, \text{Ce})$$

is used to approximate mobilities in Cubic2.

3.8.9 Validation and simulation (YSZ/CGO interdiffusion)

- SEM-EDS shows interdiffusion across CGO/YSZ interface.
- Simulated diffusion profiles agree with experimental data.
- Interdiffusion increases ASR via composition-dependent conductivity.

Key takeaway: Diffusion kinetics describe how fast systems approach equilibrium. Fick's laws relate fluxes to concentration gradients, atomistic mechanisms determine the diffusion coefficient, and CALPHAD-based mobility data enable quantitative simulations of interdiffusion in complex oxides (e.g. CGO-YSZ).

3.8.10 Practical DICTRA workflow

DICTRA simulations in the course used the following workflow:

- Define geometry and regions (1D domains representing phases).
- Assign phases to each region.
- Set initial compositions (sharp jumps for diffusion couples).
- Choose temperature schedule (constant or time-dependent).
- Apply boundary conditions (e.g. zero-flux).
- Select mobility database consistent with thermodynamic one.
- Run time integration (solve multicomponent diffusion with moving interfaces).
- Analyse concentration profiles, interface movement, phase fraction evolution.

This applies to cementite dissolution, pearlite growth, and the SOC diffusion couple.

3.8.11 Additional DICTRA example problems

Besides the SOC-related diffusion couple, built-in DICTRA console examples show classical diffusion-controlled transformations:

- **Cementite dissolution** in Fe-Cr-C: three initial particle sizes, DICTRA tracks carbon profiles and shrinking particles. Smaller particles dissolve faster; interface composition evolves roughly with $\sqrt{t}$.
- **Pearlite growth** in Fe-C-Mn: cooperative growth of ferrite/cementite lamellae; DICTRA solves coupled diffusion in austenite and ferrite while lamellae advance. Growth rate is controlled by thermodynamic driving forces and diffusion kinetics.

These examples demonstrate that the same diffusion equations and mobility models used for SOC diffusion couples can describe a wide range of microstructural evolution scenarios.